Berry–Esseen bounds for two summands and Bernoulli arrays

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Abstract

The Berry–Esseen constant for two arbitrary independent summands is strictly smaller than $49/120$. Independent centered two-point summands satisfy the bound with Esseen's constant $C_{\mathrm{E}} = (3+\sqrt{10})/(6\sqrt{2\pi})$ whenever there are at most three summands, or all nondegenerate summands have the same support diameter. In the latter class the constant is optimal. For at most two Bernoulli summands, the standardized normal-approximation error Δ and Lyapunov ratio ℓ obey $C_{\mathrm{E}}\ell - \Delta > 3/80$. These finite results imply the Esseen bound for an arbitrary independent array containing two Bernoulli coordinates with at least $22/23$ of its variance, or two arbitrary coordinates with at least $439/440$ of its variance. The two-summand proof uses moment optimization to reduce an attained maximum to two laws with at most three support points. Contact equations and second variations then exclude its possible threshold configurations, with interval certificates for the remaining parameter domains. The Bernoulli proofs use explicit deficit estimates and a reduction to binomial sums. The supplement supplies frozen partitions for the two-law and three-Bernoulli calculations, and the arithmetic and deterministic subdivision program for the common-diameter curvature inequality.

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1 Results

The universal Berry–Esseen problem asks for an estimate valid for every independent array. Restrictions on the number or form of the summands can give sharper estimates. We study two arbitrary summands and several classes of Bernoulli arrays, then use these bounds in arrays whose variance is concentrated in two coordinates. For two arbitrary laws, the main difficulty is to control their

support points and the location of a maximizing distribution-function jump. Moment optimization and the conditions for an attained maximum reduce this question to finitely many threshold configurations.

Let $X_1, \dots, X_n$ be independent centered real random variables with finite absolute third moments. Write

$$B^2 = \sum_i \mathbb{E}X_i^2 > 0, \qquad R = \sum_i \mathbb{E}|X_i|^3, \qquad \ell = R/B^3, \qquad \Delta = \sup_x \left|\mathbb{P}\{B^{-1}\textstyle\sum_i X_i \le x\} - \Phi(x)\right|. \tag{1.1}$$

Here Φ and ϕ denote the standard normal distribution function and density. Put

$$C_{\mathrm{E}} = \frac{3+\sqrt{10}}{6\sqrt{2\pi}}, \qquad K_{\mathrm{E}} = \sqrt{2\pi}C_{\mathrm{E}} = \frac{3+\sqrt{10}}{6}, \qquad c_2 = \frac{49}{120}, \qquad t_* = \sqrt{10} - 3. \tag{1.2}$$

Esseen proved that a universal Berry–Esseen constant cannot be smaller than C_{E} [4]. Equality with C_{E} remains open for general independent arrays.

Theorem 1.1 (Two arbitrary summands)**.** *Let C_2 be the supremum of Δ/ℓ over arrays with at most two independent summands. Then $C_2 < 49/120$.*

A *Bernoulli coordinate* means a centered random variable with at most two support points; its two values need not be 0 and 1. Deterministic zero coordinates are allowed.

Theorem 1.2 (Bernoulli arrays)**.** *The inequality $\Delta \le C_{\mathrm{E}}\ell$ holds if either*

(i) the array consists of at most three Bernoulli coordinates, or

(ii) every coordinate is Bernoulli and all nondegenerate coordinates have the same support diameter.

The constant in (ii) is optimal over that class.

Theorem 1.3 (A uniform two-Bernoulli deficit)**.** *For every independent array consisting of at most two Bernoulli coordinates and satisfying $B > 0$,*

$$C_{\mathrm{E}}\ell - \Delta > \frac{3}{80}.$$

Corollary 1.4 (Variance concentrated in two coordinates)**.** *An independent array satisfies $\Delta \le C_{\mathrm{E}}\ell$ if it contains either two Bernoulli coordinates with combined variance at least $22B^2/23$, or two arbitrary coordinates with combined variance at least $439B^2/440$.*

Theorem 1.1 is a bound on a supremum, so its strictness requires exclusion of equality at 49/120. Its proof first attains a hypothetical maximum at or above this value, replaces its two laws by laws with at most three atoms, and excludes every possible threshold configuration. The Bernoulli results use direct deficit inequalities.

For the common-diameter assertion we use Schulz's binomial theorem [10, Theorem 1]: if $N \sim \mathrm{Bin}(n,p)$, $q = 1-p$, and $0 < p < 1$,

$$\sup_x |\mathbb{P}\{(N-np)/\sqrt{npq} \le x\} - \Phi(x)| < C_{\mathrm{E}}\frac{p^2+q^2}{\sqrt{npq}}. \tag{1.3}$$

The proof below extends this theorem to unequal success probabilities. Optimality follows from Esseen's i.i.d. example: with $p_{\mathrm{E}} = (4-\sqrt{10})/2$ and $q_{\mathrm{E}} = 1-p_{\mathrm{E}}$, the law

$$\mu_{\mathrm{E}} = p_{\mathrm{E}}\delta_{\sqrt{q_{\mathrm{E}}/p_{\mathrm{E}}}} + q_{\mathrm{E}}\delta_{-\sqrt{p_{\mathrm{E}}/q_{\mathrm{E}}}} \tag{1.4}$$

has variance one, and its i.i.d. sums satisfy $\Delta/\ell \to C_{\mathrm{E}}$ [4].

The finite-support reduction uses classical moment-set methods [3, 12]. The transfer to a Gaussian remainder uses the zero-bias estimate obtained independently by Goldstein and Tyurin [5, 11]. We state these inputs where they enter the proofs. Interval arithmetic is performed with Arb [7].

For a centered sum S of variance B^2 and $\varepsilon \in \{-1, 1\}$, define

$$F_S^+(y) = \mathbb{P}(S \le y), \qquad F_S^-(y) = \mathbb{P}(S < y), \qquad H_S^\varepsilon(y) = C_{\mathrm{E}}\ell + \varepsilon\Phi(y/B) - \varepsilon F_S^\varepsilon(y).$$

Continuity of Φ gives

$$\Delta = \sup_y \max\{F_S^+(y) - \Phi(y/B),\ \Phi(y/B) - F_S^-(y)\}. \tag{1.5}$$

For finite support these extrema occur at atoms, with the indicated closed or strict convention. Reflection gives $H_S^-(y) = H_{-S}^+(-y)$.

Sections 2–4 prove the Bernoulli assertions and the uniform two-summand deficit. Sections 5–6 use that deficit and moment optimization to prove the bound for two arbitrary summands; Appendix B supplies the contact-case exclusions. Section 7 proves the variance-dominance corollaries.

2 Two Bernoulli summands with arbitrary diameters

For independent Bernoulli variables B_i with $\mathbb{P}(B_i = 1) = p_i \in (0, 1)$, write $q_i = 1 - p_i$ and

$$X_i = h_i(B_i - p_i), \quad a_i = p_i q_i, \quad r_i = p_i^2 + q_i^2, \quad V = \sum_i a_i h_i^2 = B^2, \quad R = \sum_i a_i r_i h_i^3 = B^3\ell.$$

We first record the estimate used when a compact calculation reaches a small variance.

Lemma 2.1 (Optimization over the diameters)**.** *Let $0 < p_i < 1$ and $h_i \ge 0$, with $\sum_i a_i h_i^2 > 0$. Then*

$$\frac{\sum_i a_i r_i h_i^3}{(\sum_i a_i h_i^2)^{3/2}} \ge \left(\sum_i \frac{a_i}{r_i^2}\right)^{-1/2}. \tag{2.1}$$

Equality holds precisely when $r_i h_i$ is constant in i.

Proof. Hölder's inequality gives

$$\sum_i a_i h_i^2 = \sum_i (a_i r_i h_i^3)^{2/3} (a_i/r_i^2)^{1/3} \le \left(\sum_i a_i r_i h_i^3\right)^{2/3} \left(\sum_i a_i/r_i^2\right)^{1/3}.$$

Its equality condition is $a_i r_i h_i^3$ proportional to a_i/r_i^2. □

The endpoint estimate below holds for any finite number of summands.

Proposition 2.2 (Support endpoints)**.** *For every finite Bernoulli array with $0 < p_i < 1$ and $h_i > 0$, both $H_S^+(y)$ and $H_S^-(y)$ are strictly positive at each endpoint of the support of $S = \sum_i h_i(B_i - p_i)$.*

The proof is in Appendix A. It reduces the two CDF values containing an endpoint atom to an explicit rational inequality for the Bernoulli odds. The other two values follow from an elementary normal-tail estimate.

The interior atoms of a two-summand law are covered by the following two compact inequalities.

Lemma 2.3 (Two compact inequalities)**.** *Let $0 < p_1, p_2 < 1$, $q_i = 1 - p_i$, $a_i = p_i q_i$, and $r_i = p_i^2 + q_i^2$. For $0 \le t \le 1$, put $V = a_1 t^2 + a_2$ and $R = a_1 r_1 t^3 + a_2 r_2$. Then*

$$C_{\mathrm{E}} \frac{R}{V^{3/2}} + \Phi\left(\frac{-p_1 t + q_2}{\sqrt{V}}\right) - (1 - p_1 p_2) > 3/80. \tag{2.2}$$

For $0 \le \theta \le \pi/2$, set $\alpha = \cos\theta$, $\beta = \sin\theta$. Then

$$C_{\mathrm{E}} \left(\alpha^3 \frac{r_1}{\sqrt{a_1}} + \beta^3 \frac{r_2}{\sqrt{a_2}}\right) + \Phi\left(\alpha\sqrt{\frac{q_1}{p_1}} - \beta\sqrt{\frac{p_2}{q_2}}\right) - q_2 > 3/80. \tag{2.3}$$

A separately formulated Arb interval calculation in the supplement proves these inequalities on the full two unit cubes. It substitutes intervals directly into the squared-variance-share formulas of Section 8, using the same Arb library as the other calculations. Its combined three-domain cover, including the endpoint square below, has 63,792 leaves at 320 bits; the least certified excess over $3/80$ is greater than $2.5361 \cdot 10^{-7}$. Rational probability-variance maxima and Hölder bounds handle singular faces; Section 8 gives the exact acceptance formulas. Inequality (2.3) is proved for every $\alpha, \beta \ge 0$ with $\alpha^2 + \beta^2 = 1$. The parameters obtained from ordered diameters $0 < h_1 \le h_2$ form a subset of this domain.

Lemma 2.4 (The uniform endpoint deficit)**.** *At either support endpoint of a one- or two-Bernoulli array, both signed sharp deficits exceed $3/80$.*

Proof. At the lower endpoint, let $A = \sum_i a_i/r_i^2$, $s = \sum_i p_i/q_i$, and $\Pi = \prod_i q_i$. After variance normalization, Hölder and Cauchy–Schwarz give $R \ge A^{-1/2}$ and $m = \sum_i p_i h_i \le \sqrt{s}$. The atom-bearing deficit is therefore at least

$$C_{\mathrm{E}}/\sqrt{A} + \Phi(-\sqrt{s}) - \Pi.$$

On $[0, 99/100]^2$ the independent endpoint cover proves this expression greater than $3/80$. Its all-zero corner has no positive-variance law. If some $p_i \ge 99/100$, the deficit is greater than $49/180 - 1/100 > 3/80$, since $R \ge 1/\sqrt{2} > 2/3$. For an atom-free endpoint, the proof of Lemma A.1 gives $R \ge c/x$, where $c = 2(\sqrt{2} - 1) > 49/60$. The elementary bound $x\Phi(-x) < 1/4$ then gives deficit greater than

$$(C_{\mathrm{E}} - 1/(4c))R > (49/120 - 15/49)(2/3) = 601/8820 > 3/80.$$

For completeness, $x\Phi(-x) < 1/4$ follows from $\Phi(-x) < 1/2$ for $x \le 1/2$, from $\Phi(-1/2) < 1/3$ and $\Phi(-3/4) < 1/4$ on the next two quarter intervals, and from Mills' bound $x\Phi(-x) < \phi(x) \le \phi(1) < 1/4$ for $x \ge 1$. The two scalar CDF comparisons follow by integrating $\phi(t) > (39/100)(1 - t^2/2)$. Reflection proves the other endpoint. □

Proof of Theorem 1.3. Delete any deterministic coordinate. A remaining single Bernoulli variable is covered by Lemma 2.4. Otherwise relabel and scale so $h_2 = 1$ and $0 < h_1 = t \le 1$. When $t < 1$, the four configurations have the order $00 < 10 < 01 < 11$. If $x_A = \sum_i (A_i - p_i)h_i$, their CDF values are

A	$F_S^-(x_A)$	$F_S^+(x_A)$
00	0	$q_1 q_2$
10	$q_1 q_2$	q_2
01	q_2	$1 - p_1 p_2$
11	$1 - p_1 p_2$	1.

Lemma 2.4 handles both endpoint atoms. The closed value at 01 is (2.2). At 10, put

$$\alpha = \frac{h_1\sqrt{a_1}}{\sqrt{V}}, \qquad \beta = \frac{h_2\sqrt{a_2}}{\sqrt{V}}; \qquad \alpha^2 + \beta^2 = 1.$$

Its closed deficit becomes (2.3). At $t = 1$ the two middle atoms merge; their left and closed CDF values are q_1q_2 and $1-p_1p_2$. Thus (2.2) is exactly the merged closed deficit. The split value q_2 is not used at this collision. Complementing both Bernoulli variables gives $-S$ with the same variance, third absolute moments, and gap ordering. The identity $F_S^-(y) = 1 - F_{-S}^+(-y)$ supplies every left limit from the closed values just proved. Equation (1.5) completes the proof. A zero diameter or a probability equal to zero or one is handled by deletion before standardization. If no variance remains, the hypotheses of the theorem are not satisfied. □

Corollary 2.5 (Mixtures of the two Esseen orientations)**.** *Fix $c > 0$ and numbers $0 \le \eta_i \le 1$. Suppose the independent variables X_i have laws*

$$\mathcal{L}(X_i) = (1-\eta_i)\mathcal{L}(cZ) + \eta_i\mathcal{L}(-cZ), \qquad Z \sim \mu_{\mathrm{E}}.$$

Then $\Delta \le C_{\mathrm{E}}\ell$.

Proof. Represent each mixture by an independent choice of orientation, followed by an independent draw from that oriented law. Conditional on the orientations, every summand has variance c^2, absolute third moment $c^3 b_{\mathrm{E}}$, where $b_{\mathrm{E}} = (p_{\mathrm{E}}^2+q_{\mathrm{E}}^2)/\sqrt{p_{\mathrm{E}}q_{\mathrm{E}}}$, and common support diameter $c/\sqrt{p_{\mathrm{E}}q_{\mathrm{E}}}$. Theorem 4.1 bounds every conditional CDF by the same $C_{\mathrm{E}}\ell$ relative to the same standard normal CDF. Averaging preserves that bound. For $0 < \eta_i < 1$ the summand has four support points. □

3 Three arbitrary Bernoulli diameters

Write $X_i = h_i(B_i - p_i)$, $q_i = 1 - p_i$, $a_i = p_iq_i$, and $r_i = p_i^2 + q_i^2$. We prove part (i) of Theorem 1.2. The result for fewer than three nondegenerate coordinates follows from Theorem 1.3.

Normalize $B^2 = 1$. The variance-only discrepancy estimate

$$\Delta < \kappa := 271/500 \tag{3.1}$$

follows from Cantelli's inequality and a one-variable normal-tail calculation; its exact scalar certificate is supplied with the companion numerical paper [1]. A negative sharp deficit therefore requires $R < \kappa/C_{\mathrm{E}} < 4/3$.

Suppose a coordinate of variance $w \le 1/25$ is deleted, leaving a Bernoulli pair of variance $v = 1-w$. Its uniform deficit is greater than $1/30$ by Theorem 1.3. Replace that pair by a Gaussian of variance v, and compare the deleted coordinate with a Gaussian of variance w. Taylor's formula to order two gives replacement coefficient

$$L(v) = \frac{\phi(0)(1+4\phi(0))}{6v^{3/2}} < \frac{1}{5} \qquad (v \ge 24/25);$$

the bound uses $\mathbb{E}|Z|^3 = 4\phi(0)$ and $\|\Phi'''\|_\infty = \phi(0)$. If $R_X = \mathbb{E}|X|^3$ and R_T is the sum of third absolute moments of the retained pair, the full deficit is greater than

$$\frac{1}{30} - C_{\mathrm{E}}R_T(v^{-3/2}-1) + (C_{\mathrm{E}} - L(v))R_X > b(w) := \frac{1}{30} - \kappa((1-w)^{-3/2}-1) + \frac{5}{24}w^{3/2}.$$

Here $R_X \geq w^{3/2}$ applies to the single deleted coordinate. On $[0, 1/25]$, $b'(w) \leq -3\kappa/2 + 1/16 < 0$, while

$$b(1/25) > 1/2000, \qquad (25/24)^3 < (1153/1084)^2.$$

Thus a negative deficit requires $w_i > 1/25$ for every coordinate. The Bernoulli moment identities then give

$$R \geq \frac{w_i^{3/2}}{2\sqrt{a_i}}, \qquad R \geq \frac{w_i h_i}{2}, \qquad h_i \geq 2\sqrt{w_i},$$

and hence

$$a_i > 9 \cdot 10^{-6}, \qquad 2/5 < h_i < 200/3. \tag{3.2}$$

In particular it suffices to consider $p_i \in [1/125000, 1 - 1/125000]$.

Order and rescale the raw diameters so that $0 < h_1 \leq h_2 \leq h_3 = 1$. For a configuration $A \in \{0, 1\}^3$, put

$$V = \sum_i a_i h_i^2, \qquad R_h = \sum_i a_i r_i h_i^3, \qquad z_A = (A - p) \cdot h / \sqrt{V}.$$

The following table states the complete finite calculation. Each row asserts

$$C_{\mathrm{E}} R_h / V^{3/2} + \Phi(z_A) - F(p) \geq 0 \tag{3.3}$$

on the probability cube just specified and $s, t \in [0, 1]$.

$$O = (st, s, 1), \qquad T = ((1 - s)/2 + st, (1 + s)/2, 1), \qquad U = (s/2, s/2 + (1 - s)t, 1).$$

Case	h	A	$F(p)$	Accepted leaves
First	O	100	$q_2 q_3$	244953
Second	O	010	$q_3(1 - p_1 p_2)$	373301
Upper pair	O	101	$1 - p_2 p_3$	102405
Last	O	011	$1 - p_1 p_2 p_3$	175197
Triangle lower	T	001	$1 - p_1 p_2 - p_1 p_3 - p_2 p_3 + 2 p_1 p_2 p_3$	263494
Triangle upper	T	110	$q_3 + p_3 q_1 q_2$	62488
Separated lower	U	110	q_3	20070
Separated upper	U	001	$q_3 + p_3 q_1 q_2$	70253

The supplement contains the eight complete recipes, with 1,312,161 accepted leaves and 2,624,314 processed nodes. Every recipe has been replayed from its original cube at 320-bit precision. The replay reconstructs all exact bisections and face projections and recomputes their inequalities; it shares the mathematical formulas with the 192-bit producer. Section 8 specifies the proof rules and the distinction between geometry checks and arithmetic replay.

To see that the table covers every Bernoulli array, the two possible strict orders of the subset sums are

$$000, 100, 010, 001, 110, 101, 011, 111, \qquad 000, 100, 010, 110, 001, 101, 011, 111.$$

The six interior configurations in each order occur in the table. At a collision choose the last configuration in its tied block; that row includes every tied configuration in the closed CDF. This covers equal gaps, $h_1 + h_2 = h_3$, and their intersections. Every other row still satisfies $F_{\mathrm{row}} \leq F_S^+(y)$. Thus a negative row deficit implies a negative deficit for the actual law. The parameter bounds derived under that hypothesis therefore apply in the row's negative-set derivative test. Proposition 2.2 covers the two support endpoints. Reflection supplies strict CDF values, and monotonicity between atoms extends the inequalities to every real threshold.

4 Proof for a common support diameter

Theorem 4.1 (Common diameter)**.** *Part (ii) of Theorem 1.2 holds.*

We minimize the signed deficit over the Bernoulli probabilities. The stationarity equations reduce a hypothetical negative minimum to a binomial law or to a binomial sum with one exceptional Bernoulli coordinate. Schulz's theorem treats the first case. A curvature calculation and the two-summand deficit treat the second.

Let $B_1, \dots, B_n$ be independent Bernoulli variables with parameters $p_1, \dots, p_n$. Put $q_i = 1 - p_i$ and

$$N_p := \sum_{i=1}^{n} B_i, \qquad \mu := \sum_{i=1}^{n} p_i, \qquad V(p) := \sum_{i=1}^{n} p_i q_i, \qquad \mathcal{B}(p) := \sum_{i=1}^{n} p_i q_i (p_i^2 + q_i^2). \tag{4.1}$$

Write $F_j(p) = \mathbb{P}(N_p \le j)$, with $F_j = 0$ below the support and $F_j = 1$ above it. For an integer k and $V(p) > 0$, define the closed and strict deficits

$$\mathcal{H}_+(p;k) := C_{\mathrm{E}} \frac{\mathcal{B}(p)}{V(p)^{3/2}} + \Phi\left(\frac{k-\mu}{\sqrt{V(p)}}\right) - F_k(p), \tag{4.2}$$

$$\mathcal{H}_-(p;k) := C_{\mathrm{E}} \frac{\mathcal{B}(p)}{V(p)^{3/2}} - \Phi\left(\frac{k-\mu}{\sqrt{V(p)}}\right) + F_{k-1}(p). \tag{4.3}$$

By (1.5), the common-diameter bound is equivalent to the nonnegativity of both deficits for every integer k. Complementation gives the exact reflection

$$\mathcal{H}_-(p_1, \dots, p_n; k) = \mathcal{H}_+(q_1, \dots, q_n; n-k). \tag{4.4}$$

Thus it is enough, when convenient, to treat the closed CDF value.

We use the following bound for third differences of a Poisson-binomial mass function.

Lemma 4.2 (A bound for third differences)**.** *Let $c_j = \mathbb{P}(C = j)$ be the mass function of any finite Poisson-binomial law, extended by zero off its support, and let*

$$W = \operatorname{Var} C, \qquad \Delta^3 c_j := c_j - 3c_{j-1} + 3c_{j-2} - c_{j-3}.$$

Then

$$\sup_j |\Delta^3 c_j| < \frac{8C_{\mathrm{E}}}{(W+1)^{3/2}}. \tag{4.5}$$

Proof. Set $A = 8C_{\mathrm{E}}$. The rational estimates $\sqrt{10} > 79/25$ and $\pi < 22/7$ give $A > 16/5$.

Suppose first that $0 \le W \le 1/4$. Write $C = \sum_i B_i$, where the B_i are independent with $B_i \sim \mathrm{Bern}(p_i)$, and put $q_i = 1 - p_i$. Choose the more likely value of each B_i as a base configuration, put $\varepsilon_i = \min(p_i, q_i)$, and let J count deviations from that configuration. If $E = \sum_i \varepsilon_i$ and $P_r = \mathbb{P}(J = r)$, then

$$E \le 2W \le \frac{1}{2}, \qquad P_0 \ge 1 - E \ge \frac{1}{2}.$$

With $R = \sum_i \varepsilon_i/(1-\varepsilon_i)$, one has $W \le E \le R \le 2E \le 1$, $P_1 = P_0 R$, and

$$P_0 \ge e^{-R}, \qquad P_1 \ge Re^{-R} \ge We^{-W}.$$

Let m be the number of ones in the base configuration. At the two central indices $m+1$ and $m+2$, the no-deviation contribution to $\Delta^3 c_j$ has magnitude three. Every one-deviation contribution has

the opposite sign and magnitude at least one; every remaining contribution has magnitude at most three. Since $P_0 \geq 1/2$, the central sign cannot reverse, and hence

$$|\Delta^3 c_j| \leq 3P_0 - P_1 + 3\mathbb{P}(J \geq 2) = 3 - 4P_1 \leq 3 - 4We^{-W} \tag{4.6}$$

at those two indices. At every other index, the base contribution has magnitude at most one, so

$$|\Delta^3 c_j| \leq P_0 + 3(1 - P_0) \leq 1 + 4W. \tag{4.7}$$

The outer bound in (4.7) is at most two, whereas $A(1+W)^{-3/2} > 2$ on this interval. For the central bound (4.6), use $e^{-W} \geq 1 - W$ and

$$(1 + W)^{-3/2} \geq h(W) := 1 - \frac{3}{2}W + \frac{15}{8}W^2 - \frac{35}{16}W^3.$$

For

$$P(W) := Ah(W) - (3 - 4W + 4W^2),$$

one has $h'(W) < 0$ and, because $A > 3$,

$$P'(W) < 3h'(W) + 4 - 8W = -\frac{1}{2} + \frac{13}{4}W - \frac{315}{16}W^2 < 0.$$

At the right endpoint,

$$P(1/4) = A\frac{725}{1024} - \frac{9}{4} > \frac{1}{64}.$$

This proves (4.5) for $W \leq 1/4$.

For $W \geq 1/4$, Fourier inversion and

$$|1 - p + pe^{it}| \leq \exp\{-2p(1 - p)\sin^2(t/2)\}$$

give

$$|\Delta^3 c_j| \leq I(W) := \frac{16}{\pi}\int_0^{\pi/2} \sin^3 x\, e^{-2W\sin^2 x}\, dx. \tag{4.8}$$

Put $J_r(W) = \int_0^{\pi/2} \sin^r x\, e^{-2W\sin^2 x}\, dx$. Integration of the derivative of $\sin^4 x \cos x\, e^{-2W\sin^2 x}$ yields

$$(5 + 4W)J_5 = 4J_3 + 4WJ_7, \qquad \frac{J_5}{J_3} \geq \frac{4}{5 + 4W}.$$

Consequently

$$\frac{d}{dW}\log\{I(W)(1 + W)^{3/2}\} = -2\frac{J_5}{J_3} + \frac{3}{2(1 + W)} \leq -\frac{8}{5 + 4W} + \frac{3}{2(1 + W)} < 0.$$

It remains to check $W = 1/4$. The bound $e^{-u} \leq 1 - u + u^2/2$ gives

$$I(1/4) \leq \frac{16}{\pi}\left(J_3(0) - \frac{1}{2}J_5(0) + \frac{1}{8}J_7(0)\right) = \frac{256}{35\pi}.$$

Finally,

$$\frac{256}{35\pi} < \frac{A}{(5/4)^{3/2}},$$

because, after squaring positive quantities, this is $5760 < 49\pi(19 + 6\sqrt{10})$, which follows from $\pi > 157/50$ and $\sqrt{10} > 79/25$. The normalized majorant in (4.8) is decreasing, completing the proof. □

Proposition 4.3 (Reduction to a binomial sum and one Bernoulli variable)**.** *Fix n, a mean μ, an integer k, and either deficit in (4.2)–(4.3). If that deficit is negative somewhere on*

$$\mathcal{P}_{n,\mu} := \left\{ p \in [0,1]^n : \sum_{i=1}^{n} p_i = \mu \right\},$$

then, after deleting deterministic coordinates and possibly applying the reflection (4.4), a negative minimum on a face has the form

$$N_p \sim \mathrm{Bin}(M,p) + \mathrm{Bern}(x), \qquad M \geq 1, \quad 0 < p, x < 1, \quad p \neq x. \tag{4.9}$$

Proof. Since $\mathcal{B}(p) \geq V(p)/2$, extend the selected deficit by $+\infty$ on $V = 0$. It then has a global minimum away from the zero-variance boundary. Choose a face of least dimension on which the selected deficit is negative somewhere. Among the global minimizers on that face, choose one maximizing $\sum_i p_i^3$. A coordinate at zero can be deleted. A coordinate at one can be deleted after replacing (μ, k) by $(\mu - 1, k - 1)$. Both operations preserve $k - \mu$ and the selected deficit. Each face is identified bijectively with the reduced polytope, so global minimality is preserved; the cubic tie-break is preserved as well, because the deleted cube is the constant zero or one on that face. Minimal dimension therefore makes the chosen minimizer interior.

The next-power-sum choice and the three-coordinate variation adapt Mattner's proof of Theorem 1.1 [8, pp. 5–6]. Here the original absolute cubic sum is also affine along the variation, so the argument applies to the selected deficit. The subsequent exclusion of two repeated probability values is an additional step. Select three coordinates and write e_1, e_2, e_3 for their elementary symmetric polynomials. Their generating polynomial is

$$\prod_{i=1}^{3} \{1 + p_i(w-1)\} = 1 + e_1(w-1) + e_2(w-1)^2 + e_3(w-1)^3. \tag{4.10}$$

Holding e_1, e_2 fixed preserves the mean and the triple variance $V_3 = e_1 - e_1^2 + 2e_2$. At every fixed integer threshold, the CDF of the full sum is affine in e_3, and the contribution of the selected triple to $\mathcal{B}$ is

$$\begin{aligned} \mathcal{B}_3 = {} & e_1 - 3e_1^2 + 4e_1^3 - 2e_1^4 + (6 - 12e_1 + 8e_1^2)e_2 - 4e_2^2 \\ & + 4(3 - 2e_1)e_3. \end{aligned} \tag{4.11}$$

Thus the whole deficit is affine on the feasible e_3 interval. If the three roots are distinct and interior, the current e_3 is interior. A nonzero slope permits a decrease of the deficit; a zero slope permits an increase of e_3 and therefore of

$$p_1^3 + p_2^3 + p_3^3 = e_1^3 - 3e_1e_2 + 3e_3.$$

Both alternatives contradict the choice of minimizer. Hence it has at most two distinct interior probabilities.

Schulz's theorem (1.3) excludes a negative deficit when all probabilities are equal. Apply the reflection (4.4) if necessary, so the selected deficit is closed. Suppose the two values are $0 < a < b < 1$, with multiplicities r and s. It remains to show that $r, s \geq 2$ is impossible. For a selected triple, let d_j be the mass function of the other coordinates and put $\Delta^2 d_j = d_j - 2d_{j-1} + d_{j-2}$. By (4.10), the coefficient of e_3 in the selected CDF is $-\Delta^2 d_k$. Hence the coefficient of e_3 in the deficit is

$$L(e_1; d) = 4\gamma(3 - 2e_1) + \Delta^2 d_k, \qquad \gamma := \frac{C_{\mathrm{E}}}{V^{3/2}}. \tag{4.12}$$

For $f(t) = t^3 - e_1 t^2 + e_2 t$, one has $f''(a) = 2(a-b) < 0$ at (a,a,b) and $f''(b) = 2(b-a) > 0$ at (a,b,b). Thus the values of e_3 at these triples are, respectively, the upper and lower endpoints of the interval for which $t^3 - e_1 t^2 + e_2 t - e_3$ has three roots in $[0,1]$. Global minimality gives

$$L_{aab} \le 0 \le L_{abb}. \tag{4.13}$$

If the other endpoint of this interval first sends a root to zero or one, the same one-sided endpoint conditions hold; no interior endpoint has been omitted.

When $r, s \ge 2$, delete two copies of each parameter and let c_j be the mass function of the remaining core. The deletions associated with the two triples are $c * \mathrm{Bern}(b)$ and $c * \mathrm{Bern}(a)$. Direct substitution in (4.12) gives

$$L_{abb} - L_{aab} = (b-a)\{-8\gamma + \Delta^3 c_k\}. \tag{4.14}$$

The slope-difference identity (4.14) and the signs in (4.13) would force

$$\Delta^3 c_k \ge \frac{8C_{\mathrm{E}}}{V^{3/2}}.$$

If W is the variance of the core, then $V = W + 2a(1-a) + 2b(1-b) \le W + 1$. Lemma 4.2 gives the opposite inequality. Thus one of r, s equals one, which is exactly the family (4.9). □

For a binomial sum and one Bernoulli variable, three adjacent binomial masses bound the second derivative of the deficit. The following inequality will exclude a negative minimum when the binomial sum has at least two summands. The two-summand case is proved in Section 2.

Lemma 4.4 (A curvature inequality on a bounded domain)**.** *Let* $0 < p, x < 1$, $q = 1 - p$, *and*

$$pq \le W \le 3, \qquad 0 \le f \le 1, \qquad A_0 := W - (1-f)q^2 \ge 0.$$

Set

$$B_0 := p^2 + (1-f)(W + 2pq), \qquad V := x(1-x) + W + pq,$$

$$t_x := 1 - 2x, \qquad t_p := 1 - 2p, \qquad y := \frac{f(W+pq)}{1-qf}, \qquad z := \frac{y-x}{\sqrt{V}}.$$

Then

$$\frac{2C_{\mathrm{E}}}{\sqrt{V}}\{4W + 2 + t_x t_p\} + z\phi(z)(2+z^2) - 5V\frac{fB_0}{(2-f)B_0 + A_0} > 0. \tag{4.15}$$

Proof. For $V \le 1/5$, put $S_2 = t_x^2 + t_x t_p + t_p^2$. The inequalities $V \ge \{2 - t_x^2 - t_p^2\}/4$ and $S_2 \ge (t_x^2 + t_p^2)/2$ imply $S_2 \ge 1/3$. Since $4V + S_2 = 4W + 2 + t_x t_p$, the arithmetic–geometric mean inequality bounds the first term in (4.15) from below by

$$C_{\mathrm{E}}\left(8\sqrt{V} + \frac{2S_2}{\sqrt{V}}\right) \ge \frac{8C_{\mathrm{E}}}{\sqrt{3}}.$$

The normal contribution is at least $-4/(e\sqrt{\pi})$, since $u(2+u^2)\phi(u)$ is maximized at $u = \sqrt{2}$. Moreover

$$(2-f)B_0 + A_0 - fB_0 = 2(1-f)B_0 + A_0 \ge 0,$$

so the neighbor fraction is at most one and its entire subtracted contribution is at most $5V \le 1$. The rational bounds

$$C_{\mathrm{E}} > \frac{2}{5}, \qquad \sqrt{3} < \frac{26}{15}, \qquad e > \frac{19}{7}, \qquad \sqrt{\pi} > \frac{7}{4}$$

give the strict margin

$$\frac{8C_{\mathrm{E}}}{\sqrt{3}} - \frac{4}{e\sqrt{\pi}} - 1 > \frac{24}{13} - \frac{16}{19} - 1 = \frac{1}{247}. \tag{4.16}$$

For the complement $V \geq 1/5$, the normal term is bounded below by

$$z\phi(z)(2+z^2) \geq \sqrt{\frac{2}{\pi}}\min\{z, 0\}. \tag{4.17}$$

Indeed the claim is immediate for $z \geq 0$; for $z = -u < 0$, the function $(2+u^2)\phi(u)$ decreases because its derivative is $-u^3\phi(u)$, and its value at zero is $2\phi(0) = \sqrt{2/\pi}$. We verify the remaining domain using 192-bit Arb intervals and the following three parametrizations:

(i) $p = (1+s_p)/2$, $W = pq + s_W(3-pq)$, $f = s_f$;

(ii) $p = s_p/2$, $W = q^2 + s_W(3-q^2)$, $f = s_f$;

(iii) $p = s_p/2$, $W = pq + s_W(q^2-pq)$, $f = 1-(1-s_f)W/q^2$,

where $s_p, s_W, s_f, x \in [0,1]$. The first chart is exactly $p \geq 1/2$. For $p \leq 1/2$, the other two charts split at $W = q^2$; below that interface, $A_0 \geq 0$ is equivalent to $f \geq 1 - W/q^2$, which is precisely the range of the third parametrization. The interfaces overlap. Thus the charts cover the whole stated domain. In the third chart $A_0 = s_f W$; the first two charts give, respectively, $A_0 = qs_p + s_W(3-pq) + fq^2$ and $A_0 = s_W(3-q^2) + fq^2$. On each box the program uses a downward enclosure for the smooth term, the lower bound (4.17) for the normal term, and an upward enclosure for the subtracted mass term in (4.15). A box is accepted only when the resulting outward lower endpoint is positive. Otherwise the widest normalized coordinate is bisected dyadically; an unresolved box at the prescribed depth or box budget aborts the program. Boxes touching both $p = 0$ and $f = 1$ use the weaker limit $y \geq 0$ rather than dividing by an interval containing zero. If interval dependency does not certify the three-neighbor denominator as positive, the program does not divide; it uses the universal bound of one for the neighbor fraction and accepts only if the resulting lower gap is positive. The finite cover and its strict acceptance bounds are recorded in Section 8.

The verifier is `verify/structure/check.py`, run with `--arithmetic` and `--case common-gap`. Its one-exception calculation uses the smaller coefficient 199/500. The coefficient of $2C_{\mathrm{E}}/\sqrt{V}$ in (4.15) is $4W + 2 + t_x t_p \geq 1$, so the certified positive gap remains positive at $C_{\mathrm{E}} > 199/500$. The supporting record is `certificates/structure/common_gap.json`. □

Proposition 4.5 (A binomial sum and one Bernoulli variable)**.** *For every $M \geq 1$, $0 < p, x < 1$, $p \neq x$, and every integer k, both deficits of* $\mathrm{Bin}(M,p) + \mathrm{Bern}(x)$ *are nonnegative.*

Proof. By (4.4), it suffices to treat the closed CDF value. Suppose it has a negative deficit. Proposition 4.3, applied with the same mean and threshold, gives a negative global minimum of the same form with $p \neq x$. Relabel its parameters as M, p, x.

First suppose $M \geq 2$, put $m = M - 1$, and let

$$D \sim \mathrm{Bin}(m,p), \qquad d_j = \mathbb{P}(D = j), \qquad W = mpq.$$

Vary the success probabilities of the exceptional variable and one variable in the repeated group, keeping their sum fixed. Write their probabilities as $\bar{p} + d$ and $\bar{p} - d$, and put $u = d^2$. Their generating polynomial is

$$\{1 + \bar{p}(w-1)\}^2 - u(w-1)^2,$$

so the selected closed CDF along the smoothing curve satisfies the exact affinity

$$\mathcal{F}_k(u) = \mathcal{F}_k(0) + (d_{k-1} - d_k)u. \tag{4.18}$$

At the current point set

$$t_x = 1 - 2x, \qquad t_p = 1 - 2p, \qquad \xi = d_{k-1} - d_k,$$
$$S_2 = t_x^2 + t_x t_p + t_p^2, \qquad z = \frac{k - x - Mp}{\sqrt{V}}, \qquad V = x(1-x) + Mpq.$$

The moving pair satisfies

$$V' = -2, \qquad \mathcal{B}' = -2S_2, \qquad \mathcal{B}'' = -8.$$

If

$$\mathcal{A}(u) = C_{\mathrm{E}} \frac{\mathcal{B}(u)}{V(u)^{3/2}} + \Phi\left(\frac{k-\mu}{\sqrt{V(u)}}\right),$$

then $\mathcal{H}_+ = \mathcal{A} - \mathcal{F}_k$. Since the current unequal pair is an interior point of the u interval, stationarity and (4.18) give $\mathcal{A}' = \xi$, while $\mathcal{H}_+'' = \mathcal{A}''$. Direct differentiation gives

$$\mathcal{A}' = C_{\mathrm{E}} \frac{\mathcal{B}' V + 3\mathcal{B}}{V^{5/2}} + \frac{z\phi(z)}{V},$$
$$\mathcal{A}'' = C_{\mathrm{E}} \frac{\mathcal{B}'' V^2 + 6\mathcal{B}' V + 15\mathcal{B}}{V^{7/2}} + \frac{z\phi(z)(3 - z^2)}{V^2}.$$

Eliminating $\mathcal{B}$ from these two identities with $\mathcal{A}' = \xi$ gives the exact curvature formula

$$V^2 \mathcal{H}_+'' = 5V\xi - C_{\mathrm{E}} \left(8\sqrt{V} + \frac{2S_2}{\sqrt{V}}\right) - z\phi(z)(2 + z^2). \tag{4.19}$$

A local minimum would require the left side to be nonnegative.

Extend d_j by zero. The adjacent difference is

$$\xi = \begin{cases} 0, & k < 0, \\ -d_0, & k = 0, \\ d_{k-1} \dfrac{k - (m+1)p}{kq}, & 1 \le k \le m+1, \\ 0, & k \ge m+2. \end{cases} \tag{4.20}$$

If $\xi \le 0$, the quantity subtracted from $5V\xi$ in (4.19) is strictly positive. Indeed $V \ge x(1-x) + pq$ implies

$$\max\{V, S_2\} \ge \frac{1}{3},$$

by the same elementary calculation used in the proof of Lemma 4.4, and hence

$$C_{\mathrm{E}} \left(8\sqrt{V} + \frac{2S_2}{\sqrt{V}}\right) + z\phi(z)(2 + z^2) \ge \frac{8C_{\mathrm{E}}}{\sqrt{3}} - \frac{4}{e\sqrt{\pi}} > 0.$$

The right side of (4.19) is therefore negative. Thus only the positive range in the third line of (4.20) remains.

Put $y = k - (m+1)p > 0$, $f = y/(kq)$, and $j = k-1$. Then $0 < f \le 1$ and the binomial recurrences give

$$d_{j+1} = (1-f)d_j, \qquad d_{j-1} = r_0 d_j, \qquad r_0 = \frac{(k-1)q}{(m-k+2)p}.$$

The mass of these three atoms is at most one, so

$$\xi = f d_j \le \frac{f}{2 - f + r_0}. \tag{4.21}$$

Eliminating m and k gives

$$r_0 = \frac{A_0}{B_0}, \qquad A_0 = W - (1-f)q^2, \qquad B_0 = p^2 + (1-f)(W + 2pq), \tag{4.22}$$

as well as

$$y = \frac{f(W+pq)}{1-qf}, \qquad C_{\mathrm{E}}\left(8\sqrt{V} + \frac{2S_2}{\sqrt{V}}\right) = \frac{2C_{\mathrm{E}}}{\sqrt{V}}\{4W + 2 + t_x t_p\}. \tag{4.23}$$

Finally, the normal term obeys the lower bound (4.17). Here $W = mpq \ge pq$, and (4.22) gives $A_0 \ge 0$. For $W \le 3$, Lemma 4.4, together with (4.21)–(4.23), makes the right side of (4.19) strictly negative, contradicting local minimality.

For $W \ge 3$, Fourier inversion gives the analytic bound

$$|d_{k-1} - d_k| \le \frac{4}{\pi}\int_0^{\pi/2} \sin u \, e^{-2W \sin^2 u}\, du < \frac{2}{\pi W}. \tag{4.24}$$

The last inequality follows on setting $s = \cos u$ and using $1 - s^2 \ge 1 - s$. Combining (4.24) with the curvature formula (4.19), and using $W \le V \le W + 1/2$, the negative of the right side of (4.19) is at least

$$8C_{\mathrm{E}}\sqrt{V} - \frac{4}{e\sqrt{\pi}} - \frac{10V}{\pi W}.$$

This expression is concave in V, so its minimum is at $V = W$ or $V = W + 1/2$. Both endpoint functions increase with W. At $W = 3$, the rational bounds $C_{\mathrm{E}} > 2/5$, $\sqrt{3} > 5/3$, $\sqrt{7/2} > 11/6$, $\pi > 3$, and $4/(e\sqrt{\pi}) < 1$ give the respective lower bounds 1 and 44/45. This excludes $M \ge 2$.

When $M = 1$, the two-summand deficit in Theorem 1.3 gives the required bound. This excludes the assumed negative closed deficit in the remaining case. Reflection (4.4) supplies the left CDF limit. □

Proof of Theorem 4.1. Delete deterministic coordinates and divide every remaining summand by the common support diameter. Each can then be written $X_i = B_i - p_i$, so the data in (4.1) are exactly the variance and sum of absolute third moments of the sum. If the bound in Theorem 4.1 failed, one of the deficits (4.2)–(4.3) would be negative at an integer threshold. Proposition 4.3 would produce a negative minimum in the family (4.9), contrary to Proposition 4.5. Esseen's i.i.d. Bernoulli sequence has equal support diameters and its ratio tends to C_{E}, so no smaller constant is valid in this class. □

5 Moment optimization and three-point reduction

At a fixed threshold, convolution is affine in each summand law. Fixing its mean and variance therefore gives a moment optimization problem with three affine constraints. Extreme-point theorems for moment sets, including Winkler's characterization [12, Theorem 2.1], provide the broader setting. Strong duality under interior moment constraints is also classical [3, Theorem 2.2]. We give the compactness and separation arguments for the precise coercive objectives needed here.

Theorem 5.1 (Strong quadratic moment duality)**.** *Let $s > 0$ and let $h : \mathbb{R} \to \mathbb{R}$ be upper semicontinuous with*

$$-K_-(1 + |u|^3) \le h(u) \le K_+ - \lambda|u|^3$$

for some finite K_-, K_+ and $\lambda > 0$. Then

$$\max_{P:\ \int dP=1,\ \int u\,dP=0,\ \int u^2\,dP=s,\ \int |u|^3 dP<\infty} \int h\,dP = \min_{\substack{q(u)=A+Bu+Cu^2 \\ q\ge h \ \text{on}\ \mathbb{R}}} (A + Cs). \tag{5.1}$$

Both extrema are attained. An extreme primal optimizer has at most three atoms, and every primal optimizer is concentrated on the contact set of every optimal quadratic majorant.

Proof. Fix any finite-support feasible law P_0. A maximizing sequence may be taken with value at least $\int h\,dP_0 - 1$; together with $h \le K_+ - \lambda|u|^3$, this gives uniformly bounded third moments. The laws are tight, and the uniform bound on third moments makes u^2 uniformly integrable. A weak limit preserves the three moment constraints and has finite third moment by Fatou's lemma. The bounded-above upper semicontinuous version of Portmanteau's theorem shows that the limit attains the maximum. The maximizing set is weakly compact and convex; it is a face of the feasible set because the objective is affine. It has an extreme point, which is also extreme in the feasible set.

If an extreme feasible law P had four disjoint Borel sets $A_1, \dots, A_4$ of positive mass, their vectors

$$\left(P(A_j), \int_{A_j} u\,dP, \int_{A_j} u^2\,dP\right) \in \mathbb{R}^3$$

would be linearly dependent. A nonzero bounded function $g = \sum_j c_j \mathbf{1}_{A_j}$ would then satisfy $\int g\,dP = \int ug\,dP = \int u^2 g\,dP = 0$. For sufficiently small $\epsilon > 0$, the two distinct laws $(1 \pm \epsilon g)P$ remain feasible and have finite third moments, while their average is P. Thus an extreme law has at most three support points.

For strong duality, let $\mathcal{D} \subset \mathbb{R}^4$ consist of

$$\left(\mu(\mathbb{R}), \int u\,d\mu, \int u^2\,d\mu, z\right), \qquad \mu \ge 0, \qquad z \le \int h\,d\mu,$$

where μ has finite third moment. This is a convex cone. It is closed: along a convergent sequence with z bounded below, the inequalities $z \le \int h\,d\mu \le K_+\mu(\mathbb{R}) - \lambda \int |u|^3 d\mu$ give a uniform third-moment bound; weak compactness, uniform integrability of u^2, and Portmanteau for the bounded-above upper semicontinuous function h then retain all four inequalities in the limit.

Let M denote the primal maximum. The point $p = (1, 0, s, M)$ belongs to $\mathcal{D}$, whereas $p + (0, 0, 0, \varepsilon)$ does not for any $\varepsilon > 0$, so p is a boundary point. A nonzero supporting functional (a, b, c, d) exists; since $\mathcal{D}$ is a cone and contains both 0 and $2p$, its supporting level at p is zero.

Thus $am_0 + bm_1 + cm_2 + dz \le 0$ on $\mathcal{D}$ and equality holds at p. The downward vertical ray in $\mathcal{D}$ gives $d \ge 0$. In fact $d > 0$: if $d = 0$, then (a, b, c) would support the moment cone

$$\{0\} \cup \{(m_0, m_1, m_2) : m_0 > 0,\ m_2 \ge m_1^2/m_0\}$$

at $(1, 0, s)$, which is an interior point because $s > 0$, forcing the whole functional to vanish. Normalize $d = 1$. Applying the support inequality to a point mass at u gives $a + bu + cu^2 + h(u) \le 0$, so $q(u) := -a - bu - cu^2$ majorizes h. Equality at p gives $M = -a - cs = A + Cs$. For every feasible P and every feasible q, integration gives $\int h\, dP \le A + Cs$. Thus this attained dual value is minimal. Finally, for any primal and dual optimizers, $0 = (A + Cs) - \int h\, dP = \int (q - h)\, dP$ with $q - h \ge 0$, proving the contact assertion. □

Theorem 5.2 (Three-point reduction)**.** *Every strict failure of $\Delta \le C_{\mathrm{E}}\ell$ can be replaced by a strict failure with the same number of independent centered summands, each supported on at most three points.*

Proof. Fix a strict one-sided failure, its threshold y, its choice $\varepsilon \in \{+1, -1\}$, and the variances of all summands. Put $\lambda = C_{\mathrm{E}}B^{-3} > 0$. When all coordinates except X_i are fixed, let $T_i = \sum_{j \ne i} X_j$ and set

$$h_i(u) = \varepsilon F^{\varepsilon}_{T_i}(y - u) - \lambda |u|^3.$$

The first term is upper semicontinuous for both signs. Indeed the closed CDF composed with $y - u$ is upper semicontinuous, while the strict CDF composed with $y - u$ is lower semicontinuous. The cubic term gives the two bounds required in Theorem 5.1. A coordinate of zero variance is deterministic. Every other coordinate can be replaced by an extreme maximizer with its mean and variance unchanged. This increases or preserves

$$\varepsilon F^{\varepsilon}_{\sum_i X_i}(y) - \varepsilon \Phi(y/B) - \lambda \sum_i \mathbb{E}|X_i|^3.$$

Sequential replacement gives the claimed finite-support failure. □

6 Two arbitrary summands

We prove Theorem 1.1. All variations in this section change the same two marginal laws; their independence is preserved. Normalize the total variance to one. By reflection it suffices to maximize D/R, where $D = \mathbb{P}(X_1 + X_2 \le z) - \Phi(z) > 0$.

Lemma 6.1 (Attainment at a positive ratio)**.** *If $C_2 \ge 49/120$, its value is attained at a finite threshold by two nondegenerate laws, each supported on at most three points.*

Proof. Choose ratios $C_j \to C_2$ bounded below by a fixed $c > 463/1250$. Since

$$R_j \ge v_{1,j}^{3/2} + v_{2,j}^{3/2} \ge 1/\sqrt{2}, \qquad C_j R_j = D_j \le 1,$$

the third moments are uniformly bounded. Each marginal family is tight, and its second moments are uniformly integrable. The thresholds stay bounded: at negative thresholds Chebyshev bounds the closed probability by z_j^{-2}, while at positive thresholds the discrepancy is at most $1 - \Phi(z_j)$. Either unbounded direction contradicts $D_j \ge c/\sqrt{2}$.

Pass to a subsequence on which both marginals converge weakly, $z_j \to z$, and $R_j \to r$. The limits are centered and have total variance one. Their independent sums converge weakly. Portmanteau

for the closed event gives $D \geq C_2 r$, whereas lower semicontinuity gives $R \leq r$. Thus $D/R \geq C_2$, and equality follows from the definition of C_2. In particular $R = r$. The one-variable theorem of Bentkus and Kirsa [2, Theorem 1] gives a constant smaller than $463/1250 < 49/120$, excluding a deterministic marginal.

At fixed threshold and marginal variance, optimize $F^+_{X_{3-i}}(z-u) - C_2|u|^3$ using Theorem 5.1. Replacing one law by a three-atom extreme optimizer cannot decrease $F - \Phi(z) - C_2 R = 0$, and cannot increase it without exceeding C_2. Do this for each marginal. The selected threshold is then a support point: otherwise moving it left to the nearest support point increases D without changing the event. □

At this maximum write $C = C_2$, $s_i = \mathbb{E}X_i^2$, $\beta_i = \mathbb{E}|X_i|^3$, $M_i = \mathbb{E}[X_i|X_i|]$, and

$$A = \frac{z\phi(z) + 3D}{2}, \qquad K = \frac{2A}{3C} = R + \frac{z\phi(z)}{3C}.$$

The companion numerical bound implies $C < 227/500$ [1].

Lemma 6.2 (Full-law contact equations)**.** *For $i = 1, 2$, the quadratic polynomial*

$$q_i(u) = F + As_i - C\beta_i - (\phi(z) + 3CM_i)u - Au^2$$

majorizes $F^+_{X_{3-i}}(z-u) - C|u|^3$ on $\mathbb{R}$, with equality at every support point of X_i. If $f_i = q_i + C|\cdot|^3$, then $f_i' \leq 0$ at all contacts and $f_i' = 0$ at each contact that is not a representation of the selected sum atom.

Proof. Mix the ith law towards δ_u with mass t, center it by subtracting tu, and move the threshold to $z - tu$. At $t = 0$ the derivatives of event probability, variance, and the sum of third absolute moments are respectively

$$F^+_{X_{3-i}}(z-u) - F, \qquad u^2 - s_i, \qquad |u|^3 - \beta_i - 3uM_i.$$

The right derivative of $D(t)V(t)^{3/2}/R(t)$ is nonpositive. Expanding it gives precisely the stated majorant. Its nonnegative gap has integral zero against the original law, proving contact at every atom. To the left of a contact the decreasing deletion CDF retains its contact value, so the derivative of its smooth majorant is nonpositive. Away from a selected representation the deletion CDF is locally constant, giving a local minimum and derivative zero. □

A representation (j, k) of the selected threshold means $x_j + y_k = z$. With ordered supports, the event $x_j + y_k \leq z$ is a downset; encode it by its numbers of included cells in successive rows.

We record the quantitative inputs used to bound its contact parameters. The elementary discrepancy bound $D < 11/20$ and $|z\phi(z)| < 49/200$, together with $R \geq 1/\sqrt{2}$, give

$$1/2 < K < 8/5. \tag{6.1}$$

Indeed $|z\phi(z)|/(3C) < 1/5$, $R > 7/10$, and $R = D/C < 66/49$. The first discrepancy estimate follows from Cantelli: for $z = -s < 0$,

$$D \leq \frac{1}{1+s^2} - \Phi(-s) < \frac{1}{2} + \frac{2}{5}s - \frac{s^2}{1+s^2}.$$

For $s \leq 1/2$ use $s^2/(1+s^2) \geq 4s^2/5$; for $1/2 \leq s \leq 1$ the displayed expression decreases; for $s \geq 1$ Cantelli alone gives $D \leq 1/2$. Positive z also gives $D \leq 1/2$. The density estimate follows from $\sup|z\phi(z)| = 1/\sqrt{2\pi e} < 49/200$.

Each marginal variance satisfies

$$13/200 < s_i < 187/200. \tag{6.2}$$

To prove this, let $v \geq 187/200$ be a dominant variance, b its third absolute moment, and r the other moment. Conditioning, the Bentkus–Kirsa estimate, and the same Gaussian Taylor comparison used in Section 3 give

$$D \leq \frac{463b}{1250v^{3/2}} + L(v)r, \qquad L(v) < 1/5.$$

For $v \geq 15/16$ both coefficients are smaller than $49/120$. For $187/200 \leq v \leq 15/16$, the other coordinate gives $r \geq 1/64$, so $r/(b+r) > 49/4224$. The first coefficient is less than $41/100$, and therefore

$$C < \frac{41}{100} - \frac{21}{100}\frac{49}{4224} = \frac{49}{120} - \frac{13}{16896}.$$

Both cases contradict $C \geq 49/120$.

For a centered three-point law with support $\{-a, c, b\}$, where $a, b > 0$ and $-a < c < b$, put

$$d = \min\{b(a+c), a(b-c)\}.$$

This is the absolute difference between the variances of its two centered two-point components; when $c = 0$, one component is δ_0. Set $d = 0$ for a law with at most two support points.

Lemma 6.3 (Small variance width)**.** *For two laws with at most three support points, let d_i be the nonnegative variance widths just defined. If $d_1 + d_2 \leq B^2/3$, then*

$$C_{\mathrm{E}}\ell - \Delta > 4877/1056500, \qquad \Delta/\ell < (572623/577500)C_{\mathrm{E}} < 49/120.$$

Proof. A centered three-point law is a mixture of δ_0 and a centered pair, or, up to reflection, of two pairs sharing their negative atom. For support $\{-a, b_1, b_2\}$ the pair weights are $m_j(a+b_j)/a$, where m_j is the original mass at b_j; centering makes the weights sum to one. Each conditional third moment is affine in its conditional variance, with nonnegative slope: the slope is the second divided difference of $|x|^3$.

Choose the two mixture labels independently and normalize $B^2 = 1$. Their conditional total variance W has mean one and support width at most $1/3$, so $W \geq 2/3$. Write the conditional sum of third absolute moments as $R + \sum_i c_i(U_i - \mathbb{E}U_i)$, where $c_i \geq 0$. Conditioning on the other label proves $\mathbb{E}[(U_i - \mathbb{E}U_i)W^{-3/2}] \leq 0$. Thus the mixing excess in C_{E} times the normalized sum of third absolute moments is at most $t\{\mathbb{E}W^{-3/2} - 1\}$, where $t = C_{\mathrm{E}}R$. For $N(v) = \varepsilon\Phi(y/\sqrt{v})$, $N''(v) = \varepsilon\phi'''(y/\sqrt{v})/(4v^2) < 7/(50v^2)$; hence its mixing excess is at most $(7/50)\mathbb{E}[-\log W]$.

The fixed scalar certificate proves

$$\mathbb{E}W^{-3/2} - 1 < 113/2000, \qquad \mathbb{E}[-\log W] < 1/70.$$

For either convex function it suffices to check a two-point law on the extreme support points, then enlarge their distance to $d = 1/3$ while keeping the mean one. If the lower displacement is $l \in [0, d]$, its Jensen gap is

$$J_f(l) = \frac{(d-l)f(1-l) + lf(1+d-l)}{d} - f(1).$$

The certificate bounds this explicit function by its midpoint value plus $\frac{1}{2}$ the cell width times a bound for $|J_f'|$ on each of 256 rational cells; all cells satisfy the two displayed inequalities.

Each conditional endpoint array has two Bernoulli coordinates, so its deficit exceeds $3/80$. Averaging gives $C_{\mathrm{E}}R-\Delta > 3/80-(113/2000)t-1/500$. Independently, (3.1) gives $C_{\mathrm{E}}R-\Delta > t-\kappa$. Taking the convex combination which cancels t yields

$$\frac{3/80-1/500-(113/2000)(271/500)}{1+113/2000} = \frac{4877}{1056500}.$$

For positive discrepancy divide $D < C_{\mathrm{E}}R-\delta$ by R and use $D<\kappa$ to obtain $D/R < C_{\mathrm{E}}\kappa/(\kappa+\delta)$. The final rational comparison follows already from $C_{\mathrm{E}} < 2049/5000$. □

For two Bernoulli marginals, Theorem 1.3 gives $D/R < 80C_{\mathrm{E}}/83 < 49/120$, since $D \le 1$. A hypothetical maximum must therefore contain a genuine triple, and its canonical variance widths have sum greater than $1/3$.

Proposition 6.4 (Finite contact exclusion)**.** *Two nondegenerate laws with at most three support points cannot attain a full-law ratio maximum with* $49/120 \le C < 227/500$.

Proof. The contact equations of Lemma 6.2, the bounds (6.1)–(6.2), and Lemma 6.3 apply. The case proofs in Appendix B exhaust their downsets and sets of threshold representations. We spell out the exhaustion here so that the finite calculations have an unambiguous mathematical scope.

For a triple and a pair, the ten downsets are $000, 100, 110, 111, 200, 210, 211, 220, 221, 222$. If the threshold has a unique representation, the two other triple points are smooth minima of $q+C|x|^3$. The selected point cannot be its largest point, since that function increases to the right of its positive minimum. The zero event is impossible. Equal zero minima exclude 100 and 200 by $F < CR$; the largest-point condition excludes 111 and 222. The remaining unique-representation cases are 110, both possible representations of 210, and $211, 220, 221$. Appendix B gives analytic contradictions for $211, 220, 221$ and the middle/low representation of 210; the two compact calculations for 110 and the low/high representation of 210 have respectively 15,081 and 113,322 independent accepted leaves. A collision can occur only at the two maximal cells of 210, 211, or 221. The 211 contradiction is analytic; the 210 and 221 contact calculations have 154,848 and 113,154 accepted leaves.

For two triples, let the unit-variance signed support directions be

$$\eta_i = \frac{\alpha_i\delta_{x_{i,0}} - \delta_{x_{i,1}} + \beta_i\delta_{x_{i,2}}}{u_iv_i}, \qquad \alpha_i = \frac{v_i}{u_i+v_i}, \quad \beta_i = \frac{u_i}{u_i+v_i}.$$

The transfer $(P_1+t\eta_1)\otimes(P_2-t\eta_2)$ preserves total variance, makes the sum of third absolute moments affine, and makes the deficit quadratic with coefficient $g = \iint \mathbf{1}_{x+y\le z}\,d\eta_1 d\eta_2$. Thus $g \ge 0$ at a maximum. For row counts $k=(k_0,k_1,k_2)$, its sign is the sign of $\alpha_1 s(k_0) - s(k_1) + \beta_1 s(k_2)$, where $s(0)=s(3)=0$, $s(1)=\alpha_2$, $s(2)=-\beta_2$. Negative signs exclude $110, 200, 210, 211, 310, 311, 332$. The zero-sign shapes are $000, 111, 222, 300, 330, 333$; the first has zero event probability, and each other shape has a unique maximal cell using a largest triple point. The positive shapes are $100, 220, 221, 320, 321, 322, 331$. The lower-endpoint identity excludes 100. The unique-representation proofs exclude the other six. For collisions, the analytic argument excludes 322 and 331; the three possible representation patterns of 321 are excluded by their complete scalar calculations. The remaining shape 221 and its transpose 320 are covered by two contact charts with 68,963 and 90,233 independently accepted leaves.

Every compact chart includes its closed boundary. Vanishing masses and coalescing support points reduce to the smaller-support cases above. A change of closed downset at a collision is assigned to the larger closed event, already in the classification.

At $C = 49/120$ all analytic contradictions remain strict. The only two lower-floor rejection expressions whose sign convention needs changing are the lower 110 exclusion expression and the

collided triple/pair 221 exclusion expression. All 427 and 34,084 respective leaves pass a strict negative replay at 384 bits. Their largest upper endpoints are below $-4.1742 \cdot 10^{-5}$ and $-2.0860 \cdot 10^{-7}$. For the 321 scalar exclusion expressions, the actual event bounds are strict: the proper downset has probability less than one, the bounded-sum quadratic has an interior support point with positive slack, and Cantelli equality would force the sum to have only two support points. Two genuine triples have at least five. Thus equality at the floor is excluded as well. □

If $C_2 \geq 49/120$, Lemma 6.1 supplies precisely the maximum excluded by Proposition 6.4. This proves the strict supremum bound in Theorem 1.1.

7 Transfer to larger arrays

Normalize the variance of the original array to one. Put

$$K(v) = \frac{\phi(0)}{2} \int_0^1 \frac{s^2\, ds}{[1-(1-v)s^2]^{3/2}}, \qquad 0 < v \leq 1.$$

This is the coefficient in the following Gaussian comparison. Write d_K for the Kolmogorov distance and W_1 for the L^1-Wasserstein distance: the latter is the infimum of $\mathbb{E}|U-V|$ over couplings of the two laws.

Lemma 7.1 (Gaussian comparison)**.** *If T is a sum of independent centered variables with total variance $1-v$ and additive third absolute moment r, and G_v is an independent $N(0,v)$ variable, then $d_K(T+G_v, N(0,1)) \leq K(v)r$.*

Proof. Write $a = 1-v$. For $a > 0$, let T^* have the zero-biased law of T, characterized by $\mathbb{E}[Tf(T)] = a\mathbb{E}f'(T^*)$ [6]. The Goldstein–Tyurin estimate gives $W_1(T, T^*) \leq r/(2a)$ [5, Lemma 1.1][11, Theorem 3 and Corollary 1]. For $h(u) = \Phi((x-u)/\sqrt{v})$, the normal Stein solution has the Ornstein–Uhlenbeck representation

$$f(u) = -\int_0^1 \mathbb{E}h'(su + \sqrt{a(1-s^2)}Z)\, ds, \qquad af'(u) - uf(u) = h(u) - \mathbb{E}h(\sqrt{a}Z).$$

Differentiating twice and convolving the Gaussian derivatives gives

$$\|f''\|_\infty \leq \phi(0) \int_0^1 \frac{s^2\, ds}{(1-as^2)^{3/2}}.$$

The zero-bias identity therefore bounds the discrepancy by $aW_1(T, T^*)\|f''\|_\infty \leq K(v)r$. The case $a = 0$ is immediate. □

Choose two disjoint sets of original coordinates, with independent sums H_1, H_2. One set may be empty. Let $v = \mathrm{Var}(H_1 + H_2) > 0$, let b be the sum of the original cubic moments in these sets, and let $r = R - b$. Define the increase in the sum of third absolute moments under grouping by

$$e = \mathbb{E}|H_1|^3 + \mathbb{E}|H_2|^3 - b.$$

Then

$$\Delta \leq \frac{c_2(b+e)}{v^{3/2}} + K(v)r. \tag{7.1}$$

Indeed, condition on the remainder and use Theorem 1.1 to compare $H_1 + H_2$ with G_v; then apply Lemma 7.1. The quantity $b + e$ is the sum of the third absolute moments of the two grouped sums.

For independent centered X, Y, direct expansion gives

$$\mathbb{E}|X+Y|^3-\mathbb{E}|X|^3-\mathbb{E}|Y|^3 = 2\mathbb{E}\Big[\mathbf{1}_{XY<0}\min(|X|,|Y|)^2\{3\max(|X|,|Y|)-\min(|X|,|Y|)\}\Big] \geq 0. \tag{7.2}$$

The pointwise expansion subtracts $3XY(|X|+|Y|)$; its expectation is zero by independence and centering. Iteration proves $e \geq 0$.

Consequently a sufficient condition for the Esseen bound is

$$\frac{c_2}{v^{3/2}}\left(\frac{b}{R}+\frac{e}{R}\right)+K(v)\left(1-\frac{b}{R}\right) \leq C_{\mathrm{E}}. \tag{7.3}$$

For example,

$$v \geq 11/16, \qquad b \leq r, \qquad e \leq R/500 \quad \Longrightarrow \quad \Delta < C_{\mathrm{E}}R. \tag{7.4}$$

To check this without a numerical search, convexity gives

$$K(v) \leq \frac{\phi(0)}{15}+\frac{\phi(0)}{10v^{3/2}} < \frac{2}{75}+\frac{1}{25v^{3/2}}.$$

Use $v^{3/2} > 57/100$, substitute $b/R \leq 1/2$ and $e/R \leq 1/500$ in (7.3), and compare with the smaller constant $c_2 < C_{\mathrm{E}}$; the cross-multiplied rational slack is $1/6000$. For two original coordinates, $e = 0$.

Proof of Corollary 1.4. For an arbitrary pair, (7.1) with $e = 0$ proves the result as soon as $c_2/v^{3/2} \leq C_{\mathrm{E}}$ and $K(v) \leq C_{\mathrm{E}}$. The elementary bounds $C_{\mathrm{E}} > 40973/100000$ and

$$(40973/100000)^2(439/440)^3-(49/120)^2 = \frac{4093395655159}{76665600000000000000} > 0$$

prove the first comparison for $v \geq 439/440$; the displayed chord bound proves the second.

For a Bernoulli pair, the uniform deficit gives instead

$$\Delta < C_{\mathrm{E}}b/v^{3/2}-3/80+K(v)r.$$

If the original array violated the Esseen bound, (3.1) would give $C_{\mathrm{E}}R < \kappa$. Since $K(v) < C_{\mathrm{E}}$ on the variance range in question,

$$\Delta - C_{\mathrm{E}}R < \kappa(v^{-3/2}-1)-3/80.$$

This is nonpositive when $v^{3/2} \geq 1084/1159$. The exact comparison $(22/23)^3 > (1084/1159)^2$ proves the claim. □

7.1 Signed convolution estimates

The same comparison can retain the signed error of the selected sum. For $H = X_1 + X_2$ of variance v and independent remainder T, define

$$f_H(x) = F_H(x)-\Phi(x/\sqrt{v}), \qquad \mathcal{N}_T(f_H) = \sup_x |\mathbb{E}f_H(x-T)|.$$

The exact convolution identity and Lemma 7.1 give

$$\Delta(H+T) \leq \mathcal{N}_T(f_H)+K(v)r. \tag{7.5}$$

Replacing $\mathcal{N}_T(f_H)$ by $\|f_H\|_\infty$ can lose the gain needed for sharpness. For three independent symmetric signs of size $1/\sqrt{3}$, the two-sign partial sum has $\|f_H\|_\infty = 1/4$, whereas

$$\mathcal{N}_T(f_H) = \frac{1}{4} - \frac{1}{2}\Phi(-\sqrt{2}).$$

Both statements follow by checking the sum atoms and their left limits. Together with the chord bound for $K(2/3)$, Mills' lower bound gives

$$\frac{1}{4} - \frac{1}{2}\Phi(-\sqrt{2}) + \frac{K(2/3)}{3\sqrt{3}} < \frac{C_{\mathrm{E}}}{\sqrt{3}}.$$

Thus (7.5) proves the sharp estimate in this example using the actual convolution. The corresponding unsmoothed two-summand norm already exceeds the target.

There is also an explicit Gaussian version which uses the sign of the integrated error. If $f = F_U - F_V$ for equal-mean laws, $\|f\|_\infty \le d$, and $\int |f| \le w$, then $\int f = 0$ and each of f_+, f_- has integral at most $w/2$. Among functions bounded by d with that integral, a centered interval of height d maximizes the integral against a centered Gaussian density. Hence, for $a > 0$,

$$\|f * \phi_a\|_\infty \le d\{2\Phi(w/(4d\sqrt{a})) - 1\}.$$

The right side increases with both d and w. For a two-summand head, Theorem 1.1 gives $d \le c_2 b/v^{3/2}$, while the mean normal-approximation bound of Goldstein and Tyurin [5, Theorem 1.1][11, Theorem 4, equation (10)] gives $w \le b/v$. Therefore

$$d_K(H + G_a, G_{v+a}) \le \frac{c_2 b}{v^{3/2}} \left\{ 2\Phi\left(\frac{\sqrt{v}}{4c_2\sqrt{a}} \right) - 1 \right\}. \tag{7.6}$$

The Gaussian is an explicit hypothesis of this last estimate.

8 Computational supplement

The finite calculations assert inequalities on complete parameter domains. The supplied record distinguishes an arithmetic evaluation from a check of its stored output. The default command

```
uv run --frozen python verify/finite/current/check.py
```

checks source and archive hashes, reconstructs every original contact partition, verifies the three Bernoulli-pair forests and all eight Bernoulli-triple recipes, and checks the complete scalar Cartesian partitions. It performs no contact arithmetic and does not search. Adding `--arithmetic` recomputes every necessary interval bound on those frozen cells. The optional `--case` selects one of the mathematical domains listed in the record. A failed enclosure is an error; the verifier never refines a cell to make it pass.

For the Bernoulli-pair margin, the three roots are the two unit cubes in (2.2)–(2.3), with $\theta = \pi t/2$ in the second, and the endpoint square $[0, 99/100]^2$. For an interval $p \in [l, u]$, its variance cap is $\max_{p\in[l,u]} p(1-p)$, computed exactly. The functions $a/(1-2a)^2$ and $(1-2a)/\sqrt{a}$ are respectively increasing and decreasing on $[0, 1/4]$. These observations bound the Hölder third-moment expression, the endpoint odds, and the two terms involving variance shares. Monotonicity of Φ supplies the probability bounds. The supplied partition has 63,792 leaves covering the three stated roots, evaluated at 320 bits by direct interval substitution in separately arranged squared-variance-share formulas.

For the Bernoulli triple, a recipe consists of exact midpoint bisections, face projections, and terminal inequalities. A terminal inequality is a direct interval lower bound, an unconditional whole-box mean-value bound, or one of the proved analytic complements. A face projection uses a derivative sign on the possible negative set: a negative point would continue to the indicated face without increasing the deficit. The bounds in (3.2), and their raw consequences $h_{\min} > 3/500$, $\sqrt{V} > 3/200$, and $\sqrt{V} > 3h_{\min}/8$, are valid for this negative-set derivative test. The mean-value bound instead uses the unmodified full-box derivatives. The 320-bit replay recomputes the formulas of the frozen 192-bit producer with a separate traversal; it does not constitute a separately derived arithmetic implementation.

For the contact calculations, each leaf excludes a necessary system. The lower 110, 210 with a low/high representation, and collided triple/pair 221 implementations use direct support moments and independently arranged height and Gaussian equations. The collided lower-gap triple/pair and two-triple implementations repeatedly intersect intervals for actual candidate values. Each intersection is an equality or a proved inequality in Appendix B; an empty intersection excludes the entire box. These value intersections are never used to narrow a full-box derivative estimate. The two-triple signed divided-difference inequality has the normalization displayed in the appendix.

The contact counts in Proposition 6.4 refer to the independent accepted leaves. Exact path reconstruction checks the full original domains, including overlapping closed faces. The lower-floor arithmetic is reevaluated at 384 bits with strict negative rejection of a lower-floor exclusion expression, accounting for equality $C = 49/120$. The scalar 321 calculations use native Arb series to bound derivatives on the complete rectangles; their accepted counts are 4,096 three-representation cells, 4,096 adjacent-representation cells (four represented by eight fixed subcells each), and 3,840 endpoint-representation cells.

The common-diameter curvature calculation uses the three charts in Lemma 4.4. At coefficient 199/500, its complete 192-bit evaluation processed 550,563 boxes and accepted 275,283 boxes, with no remaining domain. Monotonicity in the coefficient supplies the stated inequality at C_{E}. This record supplies counts and a deterministic subdivision program; the arithmetic check reconstructs its partition. The endpoint-odds proof uses exact Bernstein coefficients in $\mathbb{Q}(\sqrt{5})$, checked by a separate exact polynomial conversion. Its source programs and fixed controls are supplied beside the current certificates. The algebraic calculation uses SymPy [9].

The public archive contains exact leaf paths and boxes, projection recipes, and scalar witnesses. Its manifest identifies the supplied arithmetic source and the corresponding frozen source generation. The default record check does not by itself prove an interval inequality; the proof uses the stated formulas, analytic domain reductions, and interval evaluations.

A Support endpoint estimates

Use the Bernoulli representation from Section 2. In this appendix $V = B^2$ and $\mathcal{B}_h = B^3\ell$.

Lemma A.1 (Normal tails at the support endpoints)**.** *Let $n \geq 1$, and let $B_1, \ldots, B_n$ be independent, with $B_i \sim \mathrm{Bern}(p_i)$, $p_i \in (0,1)$, and $h_i > 0$. Put $q_i = 1 - p_i$ and define*

$$V = \sum_i p_i q_i h_i^2, \qquad \mathcal{B}_h = \sum_i p_i q_i (p_i^2 + q_i^2) h_i^3,$$

and set

$$u = \sum_i q_i h_i, \qquad m = \sum_i p_i h_i.$$

Then

$$C_{\mathrm{E}} \frac{\mathcal{B}_h}{V^{3/2}} > \max\left\{1 - \Phi\left(\frac{u}{\sqrt{V}}\right), \Phi\left(-\frac{m}{\sqrt{V}}\right)\right\}. \tag{A.1}$$

Consequently, the closed deficit at the upper support and the strict deficit at the lower support are both strictly positive.

Proof. It is enough to prove the first inequality in (A.1); complementing every B_i exchanges p_i and q_i, preserves V and $\mathcal{B}_h$, and gives the second. Normalize $V = 1$ and write

$$x_i = q_i h_i, \qquad v_i = p_i q_i h_i^2, \qquad b_i = p_i q_i (p_i^2 + q_i^2) h_i^3, \qquad x = \sum_i x_i.$$

For $c = 2(\sqrt{2} - 1)$,

$$b_i = \frac{p_i^2 + q_i^2}{p_i} \frac{v_i^2}{x_i}, \qquad \frac{p_i^2 + q_i^2}{p_i} - c = \frac{(\sqrt{2} p_i - 1)^2}{p_i} \geq 0.$$

Since $\sum_i v_i = 1$, Cauchy–Schwarz gives

$$\mathcal{B}_h \geq c \sum_i \frac{v_i^2}{x_i} \geq \frac{c}{x}. \tag{A.2}$$

We claim that

$$x\{1 - \Phi(x)\} < C_{\mathrm{E}} c \qquad (x > 0). \tag{A.3}$$

If $0 < x \leq 2/3$, then the left side is at most $1/3$. The elementary bounds

$$\sqrt{10} > \frac{79}{25}, \qquad \sqrt{2} > \frac{169}{120}, \qquad \pi < \frac{22}{7},$$

give $C_{\mathrm{E}} > 49/120$ and $c > 49/60$. For the first comparison,

$$20(\sqrt{10} + 3) > \frac{616}{5} > 49\sqrt{\frac{44}{7}} > 49\sqrt{2\pi};$$

the middle inequality follows on squaring, with difference $2156/25$. The bound on π follows, for example, from

$$\frac{22}{7} - \pi = \int_0^1 \frac{t^4 (1-t)^4}{1 + t^2}\, dt > 0.$$

Hence $C_{\mathrm{E}} c > 2401/7200 > 1/3$. For $x \geq 2/3$, Mills' inequality and monotonicity of ϕ give

$$x\{1 - \Phi(x)\} < \phi(x) \leq \phi(2/3) = \phi(0) e^{-2/9}.$$

Now $e^{-2/9} < 9/11 < c$, where the last inequality follows from $\sqrt{2} > 31/22$, and $C_{\mathrm{E}}/\phi(0) = (\sqrt{10} + 3)/6 > 1$. This proves (A.3). Combining it with (A.2) proves the upper-support inequality. At that support the closed CDF equals one. At the lower support the strict CDF equals zero, and exact complementation exchanges the two CDF limits. The two deficit statements follow. □

Proposition A.2 (An inequality for Bernoulli odds)**.** *Let $o_1, \dots, o_n \geq 0$, not all zero, and put*

$$s = \sum_i o_i, \qquad \Pi = \prod_i (1+o_i)^{-1}, \qquad g(o) = \frac{o(1+o)^2}{(1+o^2)^2}, \qquad \mathcal{A} = \sum_i g(o_i). \tag{A.4}$$

Then

$$\Pi^2 \mathcal{A} \leq \frac{1}{3} \qquad (0 < s \leq 1), \tag{A.5}$$

$$\Pi^2 \mathcal{A} \leq \frac{s}{(1+s/2)^2(1+s^2/4)^2} \qquad \left(1 \leq s \leq \frac{16}{9}\right), \tag{A.6}$$

and

$$\Pi^2 \mathcal{A} \leq \frac{944784}{6076225} \qquad \left(s \geq \frac{16}{9}\right). \tag{A.7}$$

For every fixed nonzero direction $v \in [0,\infty)^n$, set $\mathcal{F}_v(t) = \Pi(tv)^2 \mathcal{A}(tv)$. Then

$$\frac{d \log \mathcal{F}_v(t)}{d \log t} < -\frac{3}{100} \qquad \left(t \sum_i v_i \geq \frac{16}{9}\right).$$

In particular, $\mathcal{F}_v$ is strictly decreasing on this range. Equality in (A.6) holds exactly when two odds are positive and equal to $s/2$, up to zero coordinates and permutation.

Proof. We first prove the intermediate-range envelope. After relabeling, isolate $x = o_1$ and $y = o_2$, and write

$$a = x + y, \qquad k = xy, \qquad 0 < k \leq \frac{a^2}{4}.$$

For the untouched coordinates, set

$$H = \sum_{j=3}^{n} g(o_j), \qquad \Pi_0 = \prod_{j=3}^{n} (1+o_j)^{-1}.$$

With

$$z_a(k) = \frac{1}{(1+a+k)^2}, \qquad R_a(k) = z_a(k)\{g(x) + g(y)\},$$

the full functional retains the untouched coordinates exactly:

$$\Pi^2 \mathcal{A} = \Pi_0^2 \{R_a(k) + H z_a(k)\}. \tag{A.8}$$

Put $D = 1 + a^2 - 2k + k^2$. Direct symmetric expansion gives

$$R_a(k) = \frac{N(a,k)}{D^2(1+a+k)^2},$$

where

$$\begin{aligned} N(a,k) = a + 2a^2 + a^3 + (a^3 - a - 4)k \\ + (2a^2 - a + 8)k^2 + (a-4)k^3. \end{aligned}$$

The pair endpoints are a merge and an equal split. Their values are

$$\mathcal{C}_1(a) = \frac{a}{(1+a^2)^2}, \qquad \mathcal{C}_2(a) = \frac{a}{(1+a/2)^2(1+a^2/4)^2},$$

and

$$\mathcal{C}_2(a) - \mathcal{C}_1(a) = -\frac{a^2(a^2 - 6a + 4)(a^3 + 10a^2 + 4a + 16)}{(a+2)^2(a^2+1)^2(a^2+4)^2}.$$

Thus the endpoints exchange order at $a_* = 3 - \sqrt{5}$.

Two exact Bernstein certificates control the whole pair curve. For $0 \le a \le a_*$, define

$$\mathcal{S}_{\mathrm{raw}}(a,k) := \frac{aD^2(1+a+k)^2 - (1+a^2)^2 N(a,k)}{k}, \qquad \mathcal{S}(a,k) := \frac{\mathcal{S}_{\mathrm{raw}}(a,k)}{(1+a)^2}.$$

Both quotients are polynomials, and

$$R_a(0) - R_a(k) = \frac{k\mathcal{S}_{\mathrm{raw}}(a,k)}{(1+a^2)^2 D^2 (1+a+k)^2}.$$

Under

$$a = a_* r, \qquad k = \frac{a^2 t}{4}, \qquad (r,t) \in [0,1]^2,$$

the uncancelled mapped quotient has monomial bidegree $(13,5)$. Cancelling the positive common factor $(1+a)^2$ gives a reduced quotient of bidegree $(11,5)$. Their degree-$(60,20)$ Bernstein tables differ but have the same sign pattern: exactly 1280 entries are positive, and the sole zero is the corner coefficient $(60,20)$. Hence

$$R_a(k) \le R_a(0),$$

strictly for $k > 0$ except at $(a,k) = (a_*, a_*^2/4)$. Since z_a decreases, the same merge inequality holds after the term $Hz_a(k)$ in (A.8) is added.

For $a_* \le a \le 16/9$, define $\theta \in [0,1]$ by

$$z_a(k) = \theta z_a(0) + (1-\theta) z_a(a^2/4).$$

Define $\mathcal{L}(a,k)$ by the exact identity

$$\begin{aligned} &\theta R_a(0) + (1-\theta) R_a(a^2/4) - R_a(k) \\ &\quad = \frac{k(a^2 - 4k)\mathcal{L}(a,k)}{(1+a^2)^2(4+a^2)^2(1+a+k)^2(8+8a+a^2)D^2}. \end{aligned}$$

Under

$$a = a_* + \left(\frac{16}{9} - a_*\right) r, \qquad k = \frac{a^2 t}{4},$$

the remaining quotient has bidegree $(13,4)$ and all 70 tensor Bernstein coefficients are positive. Therefore

$$R_a(k) < \theta R_a(0) + (1-\theta) R_a(a^2/4) \qquad \left(0 < k < \frac{a^2}{4}\right). \tag{A.9}$$

The untouched term $Hz_a(k)$ is affine in the same coordinate, so (A.9) holds for the complete pair profile, uniformly in $H \ge 0$.

Fix $s \in [1, 16/9]$ and maximize $\Pi^2\mathcal{A}$ on the closed odds simplex. No positive pair in a maximizer can have sum at most a_*. For $a < a_*$, merging the pair strictly increases the objective. At $a = a_*$, this is also true for an equal pair when $H > 0$; if $H = 0$, then $s = a_* < 1$. Every remaining pair lies in the strict chord region, so it must be equal. Thus all positive odds in any maximizer are equal. If their number is $d \ge 2$, then

$$d < \frac{2s}{a_*} \le \frac{8}{9}(3 + \sqrt{5}) < 5.$$

The case $d = 1$ is already in the resulting list, so in all cases $d \in \{1, 2, 3, 4\}$. The equal-block value is

$$\mathcal{C}_d(s) = \frac{s}{(1 + (s/d)^2)^2 (1 + s/d)^{2(d-1)}}.$$

After positive factors are removed, the three differences $\mathcal{C}_2 - \mathcal{C}_d$, $d = 1, 3, 4$, have the respective factors

$$-(s^2 - 6s + 4), \qquad 8s^3 - 33s^2 - 18s + 108, \qquad s^4 + 12s^3 - 64s^2 + 256.$$

They are strictly positive on the intermediate range. The first polynomial inside the minus sign decreases there and starts at -1. The second decreases there and has right-endpoint value $12140/729$. The third is at least $256 - 64(16/9)^2 > 0$. This proves (A.6) and its equality statement.

For the small range, expand term by term:

$$\Pi^2 \mathcal{A} = \sum_i \frac{o_i}{(1 + o_i^2)^2 \prod_{j \neq i} (1 + o_j)^2}.$$

If $r_i = s - o_i$, then the product over $j \neq i$ is at least $1 + r_i$. The elementary inequality

$$(1 + x^2)^2 (1 + y)^2 \geq 3(x + y) \qquad (x > 0,\ y \geq 0) \tag{A.10}$$

therefore bounds every summand with $o_i > 0$ by $o_i/(3s)$; a zero-odds summand vanishes. To verify (A.10), put $A = (1 + x^2)^2$. The difference is

$$Ay^2 + (2A - 3)y + (A - 3x).$$

Its constant term is positive because $\min_{x>0} A/x = 16/(3\sqrt{3}) > 3$. If $2A - 3 < 0$, then $x < 1/2$, and four times the leading coefficient times the quadratic minimum is $12A(1 - x) - 9 > 0$, since $A(1 - x) \geq 25/32$. This proves (A.5).

It remains to prove the large-range envelope. Discard the zero coordinates of a fixed nonnegative direction. Along the resulting positive ray $o_i = tv_i$, let $\mathcal{F}(t) = \Pi(tv)^2 \mathcal{A}(tv)$ and put

$$e(o) = \frac{o g'(o)}{g(o)}.$$

Logarithmic differentiation gives

$$\frac{d}{d \log t} \log \mathcal{F}(t) = \frac{\sum_i g(o_i) e(o_i)}{\mathcal{A}} - 2 \sum_i \frac{o_i}{1 + o_i}.$$

The exact identities

$$\frac{5}{4} - e(o) = \frac{9o^3 + 17o^2 - 7o + 1}{4(1 + o)(1 + o^2)} > 0,$$

$$17o^2 - 7o + 1 = 17 \left(o - \frac{7}{34} \right)^2 + \frac{19}{68},$$

and

$$\frac{x}{1 + x} + \frac{y}{1 + y} - \frac{x + y}{1 + x + y} = \frac{xy(x + y + 2)}{(1 + x)(1 + y)(1 + x + y)} > 0$$

imply

$$\frac{d}{d \log t} \log \mathcal{F}(t) < \frac{5}{4} - \frac{2s}{1 + s} \leq -\frac{3}{100} \qquad \left(s \geq \frac{16}{9} \right).$$

Scale back to total odds $16/9$ and apply (A.6). This gives (A.7).

The two tables are generated by `verify/finite/support_odds/exact_certificate.py`; the independent basis conversion is in the corresponding `independent_certificate.py`. For a polynomial $P(x) = \sum_\alpha a_\alpha x^\alpha$ of multidegree d, the Bernstein coefficients are

$$b_\beta = \sum_{\alpha \le \beta} a_\alpha \prod_j \frac{\binom{\beta_j}{\alpha_j}}{\binom{d_j}{\alpha_j}}. \tag{A.11}$$

The programs check every coefficient sign. Their source hashes and the reported sign counts are in `certificates/finite/results.json`. □

Lemma A.3 (Endpoint atom comparison)**.** *Under the hypotheses and notation of Lemma A.1, the closed deficit at the lower support and the strict deficit at the upper support are strictly positive.*

Proof. Normalize $V = 1$, put $o_i = p_i/q_i$, and set

$$w_i = p_i q_i h_i^2, \qquad \sum_i w_i = 1.$$

The distance to the lower support, its atom, and the cubic sum are

$$x_0 = \sum_i \sqrt{o_i w_i}, \qquad \Pi = \prod_i (1 + o_i)^{-1},$$

$$\mathcal{B}_h = \sum_i \frac{1 + o_i^2}{\sqrt{o_i}(1 + o_i)} w_i^{3/2}.$$

Use s and $\mathcal{A}$ from (A.4). With

$$a_i = \frac{o_i}{(1 + o_i)^2}, \qquad r_i = \frac{1 + o_i^2}{(1 + o_i)^2}, \qquad z_i = \sqrt{a_i} h_i,$$

we have $\sum_i z_i^2 = 1$ and $\mathcal{B}_h = \sum_i (r_i/\sqrt{a_i}) z_i^3$. Cauchy–Schwarz, twice, gives

$$1 \le \mathcal{B}_h \sum_i \frac{\sqrt{a_i}}{r_i} z_i \le \mathcal{B}_h \sqrt{\mathcal{A}}, \qquad x_0^2 \le s.$$

Let $Q(x) = 1 - \Phi(x)$. Since Q decreases, it is enough to prove

$$\frac{C_{\mathrm{E}}}{\sqrt{\mathcal{A}}} + Q(\sqrt{s}) > \Pi. \tag{A.12}$$

Put

$$f(s) := (1 + s) Q(\sqrt{s}), \qquad f'(s) < 0 \quad (s > 0). \tag{A.13}$$

To prove the derivative sign, take $x = \sqrt{s}$, $M(x) = Q(x)/\phi(x)$, and $u(x) = (1 + x^2)/(2x)$, the required derivative sign is $M(x) < u(x)$. For $x \ge 1$ this follows from the Mills bound $M(x) < 1/x$. On $(0, 1)$, put $d = u - M$. We have $d(0+) = \infty$ and $d(1) > 0$. Since $M' = xM - 1$, at any interior critical point of d,

$$M(x) = \frac{3x^2 - 1}{2x^3}, \qquad d(x) = \frac{(1 - x^2)^2}{2x^3} > 0.$$

Thus d cannot have a nonpositive minimum, proving (A.13).

First suppose $0 < s \le 1$. By (A.5),

$$\frac{C_{\mathrm{E}}}{\Pi\sqrt{\mathcal{A}}} > \frac{7}{10},$$

using $C_{\mathrm{E}} > 49/120$ and $\sqrt{3} > 12/7$. Also $\Pi \le (1+s)^{-1}$. The alternating exponential bound on $[0,1]$ and $\phi(0) < 2/5$ give

$$Q(1) > \frac{1}{2} - \frac{2}{5}\left(1 - \frac{1}{6} + \frac{1}{40}\right) = \frac{47}{300}.$$

By (A.13),

$$\frac{Q(\sqrt{s})}{\Pi} \ge (1+s)Q(\sqrt{s}) \ge 2Q(1) > \frac{47}{150} > \frac{3}{10}.$$

The two strict bounds prove (A.12) in the small range.

For $1 \le s \le 16/9$, write $u = s/2$. The intermediate-range envelope gives

$$\frac{C_{\mathrm{E}}}{\Pi\sqrt{\mathcal{A}}} \ge C_{\mathrm{E}}\frac{(1+u)(1+u^2)}{\sqrt{2u}} > \frac{49}{64}. \tag{A.14}$$

The factor after C_{E} increases for $u \ge 1/2$; its logarithmic derivative has the sign of $5u^3+3u^2+u-1 > 0$. At $s = 16/9$, the alternating bound

$$e^{-t^2/2} \le 1 - \frac{t^2}{2} + \frac{t^4}{8} - \frac{t^6}{48} + \frac{t^8}{384} \qquad \left(0 \le t \le \frac{4}{3}\right)$$

and $\phi(0) < 2/5$ give

$$Q\left(\frac{4}{3}\right) > \frac{16727087}{186004350}, \qquad \frac{25}{9}Q\left(\frac{4}{3}\right) - \frac{15}{64} > \frac{33055039}{2142770112} > 0.$$

Consequently,

$$\frac{Q(\sqrt{s})}{\Pi} \ge (1+s)Q(\sqrt{s}) > \frac{15}{64}. \tag{A.15}$$

The bounds in (A.14) and (A.15) again sum to more than one.

Finally, if $s \ge 16/9$, Proposition A.2 and the exact margin

$$\frac{2401}{14400} - \frac{944784}{6076225} = \frac{7873013}{699981120} > 0$$

give $\Pi^2\mathcal{A} < C_{\mathrm{E}}^2$. The cubic term alone proves (A.12). Hence, in every range,

$$C_{\mathrm{E}}\mathcal{B}_h + Q(x_0) - \Pi \ge \frac{C_{\mathrm{E}}}{\sqrt{\mathcal{A}}} + Q(\sqrt{s}) - \Pi > 0.$$

This is the closed deficit at the lower support. Complementing every B_i preserves V and $\mathcal{B}_h$ and carries it to the strict deficit at the upper support. □

Proof of Proposition 2.2. The two preceding endpoint lemmas supply the two signs at each endpoint. □

B Contact calculations

Throughout this appendix, suppose that two nondegenerate independent laws attain the maximum ratio $C = C_2 \geq 49/120$. The companion upper bound gives $C < 227/500$. Each law has at most three support points, and at least one is a genuine triple: three distinct points with positive masses. Initially normalize the total variance to one. Write F for the selected closed event probability, $D = F - \Phi(z)$, and R for the sum of the two third absolute moments. Strict Lyapunov convexity gives $R > 1/\sqrt{2}$. A representation of the threshold is a pair of support points whose sum is z. The row-count notation for the event is defined in Proposition 6.4.

That proposition disposes of the zero-event, largest-point, and negative-mixed-derivative cases. The remaining arguments are indexed below. Transposition interchanges the two marginal laws. The certificate identifiers refer to domains in `certificates/finite/current-results.json`; the partitions and formulas are checked as described in Section 8.

Marginals and event	Representations	Argument or certificate
Triple/pair 211	Unique	Section B.5
Triple/pair 220	Unique middle/high	Section B.9
Triple/pair 221	Unique middle/high	Section B.10
Triple/pair 210	Unique middle/low	Section B.11
Triple/pair 210	Unique low/high	Section B.12; `low210`
Triple/pair 110	Unique	Section B.13; `lower110`
Triple/pair 211	Two	Section B.14
Triple/pair 221	Two, upper-gap collision	Section B.15; `collision221`
Triple/pair 210	Two, lower-gap collision	Sections B.16–B.18; `collision210`
Two triples 220	Unique	Sections B.6–B.8
Two triples 221, 320	Unique	Section B.19
Two triples 321	Unique middle/middle	Section B.20
Two triples 322, 331	Two	Section B.21
Two triples 321	Three	Sections B.23–B.27; `three_owner321`
Two triples 321	Two adjacent	Sections B.28–B.29; `adjacent_owner321` and its completion
Two triples 321	Two endpoints	Sections B.28, B.30; `endpoint_owner321`
Two triples 221, 320	Two	Sections B.31–B.37; `tt221_negative`, `tt221_positive`

Changes of scale The discrepancy D and ratio C are unchanged by a common positive rescaling. In any raw scale, let V be the total variance, R_0 the sum of the centered third absolute moments, and o the centered threshold. Then

$$z = \frac{o}{\sqrt{V}}, \qquad R_{\text{normalized}} = \frac{R_0}{V^{3/2}}, \qquad C = \frac{DV^{3/2}}{R_0}.$$

Two choices of raw scale are used:

Scale	Definition and conversion
Contact scale	Divide the variance-one supports by $K = R_{\text{normalized}} + z\phi(z)/(3C)$. Then $V = K^{-2}$, $\tau = CK^3 = C/V^{3/2}$, and the contact slope is $\sigma = \phi(z)/(CK^2)$. After division by τ, the cubic majorant has quadratic coefficient $-3/2$.
Unit-span scale	Translate the supports to start at zero and divide by the chosen outer support span. Center these raw laws when computing V, R_0, o. Setting a support span to one does not set the contact parameter to one.

Each family states its scale. In particular the three-representation calculations use the unit-span scale, while the collided 221 systems use the contact scale.

B.1 Variance derivatives at a triple

The total variance is one in this calculation.

Let $x_1 < x_2 < x_3$ be the support of a genuine triple. Its second divided difference, normalized by $[x_1, x_2, x_3]x^2 = 1$, satisfies

$$[x_1, x_2, x_3]|x|^3 \le \frac{15D + \phi'''(z)}{12C}. \tag{B.1}$$

Indeed, take a nonzero signed measure η on these points with zero mass and first moment. Put $v = \int x^2\, d\eta$ and $r = \int |x|^3\, d\eta$. The Vandermonde determinant gives $v \neq 0$, and $r/v = [x_1, x_2, x_3]|x|^3$. Both signs of t are allowed in $P + t\eta$. The total variance becomes $V(t) = 1 + tv$, the sum of third absolute moments becomes $R + tr$, and the event probability becomes $g(t) = F + tg'$. Maximality gives

$$J(t) = V(t)^{3/2}\{g(t) - \Phi(z/\sqrt{V(t)})\} - C(R + tr) \le 0, \qquad J(0) = 0.$$

With $A = (3D + z\phi(z))/2$, differentiation gives

$$g' + Av - Cr = 0, \qquad J''(0) = v^2 \left\{ 3C\frac{r}{v} - \frac{15D + \phi'''(z)}{4} \right\} \le 0.$$

This proves (B.1).

For any three real knots of diameter h,

$$[x_1, x_2, x_3]|x|^3 \ge h/2. \tag{B.2}$$

On one side of zero the divided difference is the sum of the absolute values. Otherwise scale h to one and reflect to write the knots as $-a, c, 1 - a$, where $0 < a < 1$ and $0 \le c \le 1 - a$. The divided difference q obeys

$$q = 1 + c - 2a + \frac{2a^3}{a + c}, \qquad (a + c)(q - 1/2) = 2a(a - 1/2)^2 + c(c + 1/2 - a).$$

The right side is nonnegative for $a \le 1/2$. For $a > 1/2$, put $d = a - 1/2$ and rewrite it as $(c - d/2)^2 + (2a - 1/4)d^2 \ge 0$. Equality requires the symmetric knots $(-h/2, 0, h/2)$.

B.2 A joint curvature and diameter bound

For every centered variance-one law,

$$15\{F(z) - \Phi(z)\} + \phi'''(z) < 79/10. \tag{B.3}$$

For $z \ge 0$ use $F(z) \le 1$. On $[0, \sqrt{3}]$ the derivative of $15(1 - \Phi(z)) + \phi'''(z)$ is $(z^4 - 6z^2 - 12)\phi(z) < 0$; its value at zero is $15/2$. Beyond $\sqrt{3}$ the third derivative is nonpositive, giving the same upper bound.

For $z = -x < 0$, Cantelli gives $D \le d(x) = (1+x^2)^{-1} - (1 - \Phi(x))$. The ratio $2x/\{(1+x^2)^2\phi(x)\}$ increases, since its logarithmic derivative is $(x^2 - 1)^2/\{x(1 + x^2)\}$. At $x = 1/2$ the ratio exceeds one; moreover $1 - \Phi(1/2) > 3/10$. Hence d decreases on $[1/2, \infty)$ and $d(1/2) < 1/2$. On $[1/2, \sqrt{3}]$, $\phi'''(-x) \le 0$, so the quantity in (B.3) is less than $15/2$. For $x \ge \sqrt{3}$, $d(x) < 1/4$ and Fourier inversion gives $\|\phi'''\|_\infty \le 2/\pi < 2/3$, yielding $53/12$.

It remains to consider $0 \le x \le 1/2$. The bounds $1-\Phi(x) \ge 1/2-\phi(0)x$, $\phi(x) \ge \phi(0)(1-x^2/2)$, and $\phi(0) < 2/5$ give the upper bound

$$U(x) = \frac{15}{1+x^2} - \frac{15}{2} + \frac{24}{5}x + x^3 - \frac{x^5}{5}.$$

Here

$$10(1+x^2)\{79/10 - U(x)\} = 4 - 48x + 154x^2 - 58x^3 - 8x^5 + 2x^7 =: p(x).$$

On each interval $[i/64, (i+1)/64]$, $0 \le i < 32$, expand $p((i+t)/64)$ in the degree-seven Bernstein basis. Direct rational conversion gives a positive coefficient minimum 11964025693/3848290697216. For completeness this assertion means the following finite rational identity and comparison: if $p(x) = \sum_{j=0}^{7} a_j x^j$, compute

$$c_{i,k} = \sum_{j=k}^{7} a_j \binom{j}{k} \frac{i^{j-k}}{64^j}, \qquad b_{i,l} = \sum_{k=0}^{l} c_{i,k} \frac{\binom{l}{k}}{\binom{7}{k}} \quad (0 \le i < 32,\ 0 \le l \le 7).$$

The minimum of these 256 fractions is the stated number. The nonnegative Bernstein basis functions sum to one, proving positivity.

Combining (B.1)–(B.3) with $C \ge 49/120$ shows that every normalized triple diameter is less than 158/49. In raw coordinates of total variance V and triple diameter A_i this is

$$24964V > 2401A_i^2. \tag{B.4}$$

If the two raw spans equal one, it follows that

$$V > 2401/24964, \qquad R \ge V^{3/2}/\sqrt{2} > \frac{49^3}{158^3\sqrt{2}} > 1/48.$$

The last strict comparison follows on squaring from $49^6\,48^2 > 2\,158^6$.

B.3 Threshold and density bounds

For every centered variance-one law with positive closed-CDF discrepancy

$$D = F(z) - \Phi(z) \ge d_0 := 49/(120\sqrt{2}),$$

one necessarily has

$$\boxed{-5/4 < z < 3/5, \qquad \phi(z) > 9/50.} \tag{B.3.1}$$

This scalar lemma does not assume stationarity or a finite support. Its application to a violating fixed-C_2 maximum uses only the already proved $R_{\text{normalized}} \ge 1/\sqrt{2}$ and $C \ge 49/120$. It does not supply a uniform threshold bound for arbitrary numbers of summands, whose normalized additive third absolute moment sum can be arbitrarily small.

The negative threshold For $z = -s < 0$, Cantelli's inequality, including the closed event, gives

$$D \le g(s) := \frac{1}{1+s^2} - \Phi(-s).$$

The ratio

$$H(s) = \frac{2s}{(1+s^2)^2\phi(s)}$$

satisfies

$$\frac{d}{ds}\log H(s) = \frac{(s^2-1)^2}{s(1+s^2)} \geq 0.$$

Also $2s/(1+s^2)^2 = 640/1681 > 1/4$ at $s = 5/4$, while $\phi(5/4) < \phi(1) < 1/4$. The last inequality follows from $2\pi e > 16$, using $\pi > 3$ and $e > 8/3$. Thus $H(s) > 1$ and $g'(s) < 0$ for every $s \geq 5/4$.

Taylor's formula with the signed remainder gives, for $u \geq 0$, $\exp(-u) \leq 1 - u + u^2/2 - u^3/6 + u^4/24$. At $s = 5/4$, let

$$I = s - s^3/6 + s^5/40 - s^7/336 + s^9/3456.$$

Integrating the polynomial and using $\phi(0) < 2/5$ gives

$$\Phi(-s) > 1/2 - (2/5)I,$$
$$g(5/4) < 16/41 - 1/2 + (2/5)I = \frac{37147949759}{130006646784} =: B_-.$$

Both B_- and d_0 are positive, and the exact comparison is

$$d_0^2 - B_-^2 = \frac{727351123347544172903}{4225432052004934385664 00} > 0.$$

Consequently $D \geq d_0$ excludes every $z \leq -5/4$.

The positive threshold The classical $\pi < 22/7$ implies $\phi(0) > 79/200$, because

$$7/44 - (79/200)^2 = 1349/440000 > 0.$$

For $x \geq 0$, $\exp(-x^2/2) \geq 1 - x^2/2$. Hence at $s = 3/5$,

$$1 - \Phi(s) < 1/2 - (79/200)(s - s^3/6) = 13861/50000 =: B_+.$$

The exact difference $d_0^2 - B_+^2 = 146635361/22500000000$ is positive. Since $F(z) \leq 1$, all $z \geq 3/5$ have $D \leq 1 - \Phi(3/5) < d_0$.

A density bound and the raw contact slope By (B.3.1), $|z| < 5/4$, so $\phi(z) > \phi(5/4)$. Put $x = 25/32$. The degree-five Taylor polynomial is a lower bound for $\exp(-x)$ for all $x \geq 0$, giving

$$\phi(5/4) > \frac{79}{200}\sum_{j=0}^{5}\frac{(-25/32)^j}{j!} = \frac{29108943797}{161061273600} > 9/50.$$

The final exact slack is $117914549/161061273600 > 0$.

By the contact-scale definition, $\sigma = \phi(z)/(CK^2)$. Since $C < 227/500$ and $K < 8/5$,

$$\sigma > \frac{9/50}{(227/500)(8/5)^2} > \frac{9}{200}.$$

Thus

$$\boxed{\sigma > 9/200.} \tag{B.3.2}$$

B.4 The raw contact scale

Here supports are divided by K, and V, R, o denote their raw total variance, sum of third absolute moments, and centered threshold.

The collided triple/pair 210 and two-triple 221 contact charts satisfy

$$C = V^{3/2}/G, \quad D = R/G, \quad z = o/\sqrt{V}, \quad \sigma\sqrt{V} = G\phi(z), \quad 3(V - R) = \sigma o.$$

Consequently

$$\boxed{C/\sqrt{V} = D + z\phi(z)/3.} \tag{B.4.1}$$

The threshold result Section B.3 gives $-5/4 < z < 3/5$. On $0 \le z < 3/5$, $z\phi(z)$ increases, and $\phi(3/5) < 1/3$ as verified below. For negative z, the product is negative. Thus in all these contact charts $z\phi(z) < 1/5$. Combining (B.4.1) with $D < 11/20$ and $C \ge 49/120$ proves

$$\frac{1}{\sqrt{V}} < \frac{\frac{11}{20} + \frac{1}{15}}{49/120} = \frac{74}{49}, \qquad \boxed{V > 2401/5476.} \tag{B.4.2}$$

If the actual representation satisfies $o \le 0$, then $z \le 0$ and the stronger bound is

$$\frac{1}{\sqrt{V}} < \frac{11/20}{49/120} = \frac{66}{49}, \qquad \boxed{V > 2401/4356.} \tag{B.4.3}$$

The entire negative 210 chart has $o = c - ph < 0$, so (B.4.3) holds on that whole candidate class. In another chart it can be used when the enclosing interval proves $o \le 0$ for every surviving candidate. It cannot be applied to a box that still permits both signs without an additional valid case split.

These scale formulas are specific to the chart normalization fixing the quadratic coefficient of the cubic contact potential. They are not asserted for the unit-span three-representation 321 chart, which uses a different raw scale.

For the elementary density endpoint, the classical Archimedean lower bound $\pi > 223/71$ gives $\phi(0) < 399/1000$, because

$$(399/1000)^2 - 71/446 = 1823/223000000 > 0.$$

The degree-four Taylor upper bound for $\exp(-x)$ at $x = 9/50$ now gives

$$\phi(3/5) < \frac{399}{1000}\sum_{j=0}^{4}\frac{(-9/50)^j}{j!} = \frac{16663671213}{50000000000} < 1/3.$$

B.5 One triple: contact identities and threshold shapes

In this subsection the support coordinates have total variance one; factors of K are retained explicitly.

Derivative identities for the Bernoulli law Write the Bernoulli support as $-u, v$, with $u, v > 0$, and set

$$h_B = u + v, \qquad K_B = \frac{u^2 + v^2}{u + v}.$$

Its variance is uv, its third absolute moment is uvK_B, and its signed second moment is $M_B = uv(v-u)/(u+v)$. Its dual derivative is

$$f'_B(x) = 3C\{x|x| - Kx - M_B\} - \phi(z). \tag{B.5.1}$$

The nonselected point is a smooth minimum. Applying (B.5.1) there gives

Bernoulli representation	exact identity	consequence	
$-u$	$\phi(z) = 3Cv(K_B - K)$	$K_B > K,$	$2v > K$
v	$\phi(z) = 3Cu(K - K_B)$	$K_B < K,$	$2u > K.$

(B.5.2)

The second derivative gives the last inequalities. Equality would be a stationary inflection rather than a local minimum. Also

$$K_B \geq \frac{h_B}{2}, \qquad K_B < h_B. \tag{B.5.3}$$

Exhaustive grid and three elementary eliminations List the number of included Bernoulli support points in each successive triple row. The ten possible downsets are

$$000, 100, 110, 111, 200, 210, 211, 220, 221, 222.$$

A unique selected representation must be a maximal included cell. A genuine triple cannot have its largest support point as the selected representation, by the two smooth-minimum argument in the proof of Proposition 6.4. This excludes 111 and 222; 000 has negative signed discrepancy.

Shapes 100 and 200 have a selected lowest triple point, while both nonselected triple points have deletion-CDF value zero. They are thus two equal zero minima. The cubic identity gives them as $-K, K$, with

$$f_T(x) = C|x|^3 - \frac{3}{2}CKx^2 + \frac{1}{2}CK^3.$$

The selected triple point is $-a$, with $a > K$, and let its mass be l. All event probability comes from that triple point, so

$$F = lf_T(-a) = Cl(a-K)^2(a+K/2) < Cla^3 \leq CR.$$

This contradicts $F - \Phi(z) = CR > 0$. Shape 200 need not be the full sum's lower endpoint, so this slightly extends the endpoint argument.

For shape 211, the only possible representation is triple-low/Bernoulli-high. The two nonselected triple points have the same deletion level, hence are the symmetric smooth minima $-K, K$. Write the selected triple point as $-a$, with $a > K$. The included triple-high/Bernoulli-low cell is strictly below the representation, giving

$$K - u < -a + v, \qquad h_B > a + K > 2K.$$

But (B.5.2)–(B.5.3), for a high Bernoulli representation, require $h_B/2 \leq K_B < K$. This is a contradiction. Thus 211 is also excluded.

Exactly five representation-specific configurations remain: 110 with the middle/low representation, the two representations of 210, and 220 or 221 with the middle/high representation.

Exact data for the five remaining configurations When the selected representation uses the middle triple point, write the triple support as $-a, c, b$. Its two nonselected minima satisfy

$$\frac{K}{2} < a < K < b, \qquad a^2 + b^2 = K(a+b). \tag{B.5.4}$$

Let its masses be l, m, r, put $q_B = v/(u+v)$, and let $u_T = a+c$, $v_T = b-c$. The actual deletion levels and strict geometry are:

Shape and representation	Triple levels	Bernoulli levels	Gap condition
110, middle/low	$(q_B, q_B, 0)$	$(l+m, 0)$	$h_B > u_T$
210, middle/low	$(1, q_B, 0)$	$(l+m, l)$	$h_B < u_T$
220, middle/high	$(1, 1, 0)$	$(l+m, l+m)$	$h_B < v_T$
221, middle/high	$(1, 1, q_B)$	$(1, l+m)$	$h_B > v_T$

In the middle/low 210 case, (B.5.2)–(B.5.3) give

$$a + c > h_B > K_B > K, \qquad c > K - a > 0. \tag{B.5.5}$$

For the low/high representation of 210, instead write the triple support as $-t, -a, b$, where $t > a$ and its two nonselected minima $-a, b$ satisfy (B.5.4). Its deletion levels are $(1, q_B, 0)$, the Bernoulli deletion levels are $(l+m, l)$, and its exact strict geometry is

$$t - a < h_B < t + b, \qquad K_B < K. \tag{B.5.6}$$

Every table entry follows directly from the indicated threshold cell; none replaces the actual convolution probabilities by free parameters.

B.6 Two triples: symmetry of a unique-representation rectangle

Common shape coordinates Use the full-law cubic dual from Lemma 6.2, and normalize the total variance to one. Put

$$K = \frac{2A}{3C} = R + \frac{z\phi(z)}{3C} > 0.$$

For coordinate i, write its support as $K(-a_i, c_i, b_i)$, where $-a_i, b_i$ are the two nonselected smooth minima and c_i is the selected middle point. These are dimensionless support coordinates. The minimum equations and the strictly decreasing contact levels give

$$\frac{1}{2} < a_i < 1 < b_i, \qquad a_i^2 + b_i^2 = a_i + b_i, \qquad b_i = \frac{1 + \sqrt{1 + 4a_i - 4a_i^2}}{2}. \tag{B.6.1}$$

Set

$$h_i = a_i + b_i, \quad u_i = a_i + c_i, \quad v_i = b_i - c_i,$$

$$D(a) = \frac{1}{2}(b-a)(a^2 + 4ab + b^2), \qquad H(a,c) = 3\{a(1-a) + c - c|c|\}. \tag{B.6.2}$$

Thus $CK^3 D(a_i)$ is the difference of the two minimum contact levels, and $CK^2 H(a_i, c_i)$ is the negative slope at the representation. Let m_i be the middle mass and q_i the combined mass of the first two support points. The unique threshold representation and centering imply

$$\sigma = \frac{\phi(z)}{CK^2} > 0, \qquad m_i = \frac{\sigma}{H(a_i, c_i)}, \qquad q_i = \frac{b_i}{h_i} + \frac{\sigma u_i}{h_i H(a_i, c_i)}. \tag{B.6.3}$$

For shape 220, the dual equals q_j at the first and middle support points of coordinate i, and equals zero at its last support point. Consequently

$$f_i(c_i K) = f_i(-a_i K), \qquad q_j = CK^3 D(a_i). \tag{B.6.4}$$

In particular,

$$q_1 D(a_1) = q_2 D(a_2). \tag{B.6.5}$$

The contact equations force identical laws For each a in $(1/2, 1)$, equation $f(c) = f(-a)$ selects one middle point on the descending part of the cubic between its local maximum and its positive minimum. Its normalized location is a strictly increasing function of a.

For $1/2 < a \le 3/4$, factorization on the negative half-line gives

$$c = 2a - 3/2, \qquad H = (a + c)^2. \tag{B.6.6}$$

For $3/4 \le a < 1$, the middle point is nonnegative and satisfies $c < a$. Indeed, the linear coefficient of the dual is negative, so $f(Ka) - f(-Ka) = -6CK^3 a^2(1-a) < 0$, whereas f decreases strictly along this positive part of the descending branch. Set $k = c/a$, so $0 \le k < 1$. The equal-height equation is exactly

$$a = \frac{3(1+k)^2}{2(k^3 + 3k + 2)}, \qquad c = ka. \tag{B.6.7}$$

Its derivative is

$$a'(k) = \frac{3(1+k)(1-k)(k^2+4k+1)}{2(k^3+3k+2)^2} > 0 \quad (0 \le k < 1). \tag{B.6.8}$$

We claim that, for every fixed $\sigma > 0$, the expression in (B.6.3) obeys

$$a \longmapsto q(a)D(a) = D(a)\frac{b}{a+b} + \sigma\frac{D(a)(a+c)}{(a+b)H(a,c)} \quad \text{strictly decreasing on } (1/2, 1). \tag{B.6.9}$$

First, direct differentiation using (B.6.1) gives $D'(a) = -3(2a-1)(a+b) < 0$. Also b decreases with a, so $b/(a+b)$ strictly decreases. Hence the first term in (B.6.9) strictly decreases. For the second term, put

$$T(a) = \frac{D(a)(a+c)}{(a+b)H(a,c)}.$$

On the nonpositive branch, (B.6.6) gives $T = D/[(a+b)(a+c)]$. Here D decreases, while $a+b$ and $a+c$ increase, so T decreases.

On the positive branch, the cubic factors at its positive minimum:

$$D(a) = (b-c)^2(c+2b-3/2), \qquad H(a,c) = 3(b-c)(b+c-1).$$

Thus the exact function to check is

$$T(k) = \frac{(b-c)(a+c)(c+2b-3/2)}{3(a+b)(b+c-1)}, \tag{B.6.10}$$

with (B.6.1), (B.6.7). The denominator stays positive through $k = 1$, so (B.6.10) and its derivative extend continuously to the closed interval $[0, 1]$. The fixed certificate `rectangle`, checked by the rules in Section 8, proves

$$\boxed{T'(k) < -1/4 \quad \text{for every } 0 \le k \le 1.} \tag{B.6.11}$$

The two formulas for T agree at $a = 3/4$, so (B.6.9) follows. Equation (B.6.5) now gives $a_1 = a_2$. Equations (B.6.1), (B.6.6)–(B.6.8), and (B.6.3) then give identical supports, middle masses, and outer masses for the two coordinates. In particular $c_1 = c_2 = c$.

B.7 Two triples: nonpositive rectangle representations

Here $-a, c, b$ are the support points in the variance-one scale, with K retained explicitly.

The equal-height contacts Write the discrepancy as $D = CR$, the selected threshold as z, and set

$$K = \frac{2A}{3C} = R + \frac{z\phi(z)}{3C}.$$

For each coordinate, write its two nonselected smooth minima as $-a, b$ and its selected middle atom as c, of mass m. The cubic dual has derivative

$$f'(u) = 3Cu|u| - 3CKu - L.$$

The two minimum equations imply

$$a^2 + b^2 = K(a+b), \qquad L = 3Ca(K-a). \tag{B.7.1}$$

The decreasing deletion CDF has strictly larger value at $-a$ than at b in shape 220. The minimum-height difference is

$$f(-a) - f(b) = \frac{C}{2}(b-a)(a^2+4ab+b^2) > 0.$$

Thus $b > a$. The strict local-minimum conditions and (B.7.1) consequently give

$$K/2 < a < K < b, \qquad b = \frac{K + \sqrt{K^2 + 4Ka - 4a^2}}{2}. \tag{B.7.2}$$

In this rectangle, the deletion CDF has the same value at $-a$ and c, so $f(c) = f(-a)$. Suppose $c \le 0$. On the negative half-line, direct factorization gives

$$f(c) - f(-a) = C(c+a)^2(2a - 3K/2 - c).$$

Since the support points are distinct, this yields

$$c = 2a - 3K/2, \qquad K/2 < a \le 3K/4, \qquad -f'(c) = C(a+c)^2. \tag{B.7.3}$$

The unique-representation support derivative is $mf'(c) = -\phi(z)$. Therefore

$$m = \frac{\phi(z)}{C(a+c)^2}. \tag{B.7.4}$$

The use of (B.7.4) is exactly where uniqueness of the selected representation is needed. The proof does not extend (B.7.4) to a collision.

A variance loss forced by the selected mass For a centered law on $-a, c, b$ with middle mass m, its variance is

$$v = ab - m(a+c)(b-c) = ab - \frac{\phi(z)}{C}\frac{b-c}{a+c}. \tag{B.7.5}$$

This follows by solving the two remaining masses, or by averaging the identity $x^2+(a-b)x-ab = 0$ at the two outer support points.

On $K/2 < a \le 3K/4$, the function ab in (B.7.2) increases with a. Its maximum is

$$\frac{3(2+\sqrt{7})}{16}K^2 < \frac{7}{8}K^2.$$

Indeed, after division by K, the derivative of $xb(x)$ is positive for $1/2 \le x \le 3/4$, while $\sqrt{7} < 8/3$ proves the displayed bound. Also $b > 9K/8$, since the minimum of b in this range is $(2+\sqrt{7})K/4$ and $\sqrt{7} > 5/2$. Equation (B.7.3) gives $a + c \le 3K/4$, so

$$\frac{b-c}{a+c} > \frac{3}{2}.$$

Consequently every such coordinate satisfies

$$\boxed{v < \frac{7}{8}K^2 - \frac{3}{2}\frac{\phi(z)}{C}.} \tag{B.7.6}$$

If both selected middle points are nonpositive, their sum is $z = -t$ with $0 \le t < K$, by (B.7.3). Adding (B.7.6) over the two coordinates gives

$$1 < \frac{7}{4}K^2 - 3\frac{\phi(t)}{C}. \tag{B.7.7}$$

The right side is always less than one The proved global upper bound gives $C < 1/2$, and $C \ge 49/120 > 2/5$. The public variance-only discrepancy bound, with $\kappa = 271/500$, gives $CR < \kappa$. Hence

$$K < \frac{\kappa - t\phi(t)/3}{C}. \tag{B.7.8}$$

First, (B.7.8) and $t < K$ imply $t < \kappa/(2/5) < 7/5$ and

$$t\{2/5 + \phi(t)/3\} < \kappa.$$

The left side increases on $[0, 7/5]$: its derivative is $2/5+(1-t^2)\phi(t)/3 > 2/5-(24/25)(2/5)/3 > 0$. At $t = 6/5$ it exceeds

$$\frac{6}{5}\left(\frac{2}{5} + \frac{19}{300}\right) = \frac{139}{250} > \frac{271}{500},$$

using $\phi(6/5) > 19/100$. Thus $t < 6/5$.

Put $B(t) = \kappa - t\phi(t)/3$. Since $\sup_{t\geq 0} t\phi(t) = \phi(1) < 1/4$, we have $B(t) > 9/20$. For $2/5 \leq C \leq 1/2$, the derivative in C of

$$\frac{7B(t)^2}{4C^2} - \frac{3\phi(t)}{C}$$

has the sign of $3C\phi(t) - (7/2)B(t)^2$, which is negative because $\phi(t) < 2/5$, $C \leq 1/2$, and $B(t) > 9/20$. Equations (B.7.7)–(B.7.8) therefore imply $1 < U(t)$, where

$$U(t) = \frac{7B(t)^2}{4(2/5)^2} - \frac{3\phi(t)}{2/5}, \qquad 0 \leq t \leq 6/5.$$

This scalar function has no interior maximum. In fact, with $c_0 = 2/5$,

$$\frac{U'(t)}{\phi(t)} = \frac{7B(t)(t^2-1)}{6c_0^2} + \frac{3t}{c_0},$$

whose derivative is

$$\frac{7}{6c_0^2}\left\{\frac{(t^2-1)^2\phi(t)}{3} + 2tB(t)\right\} + \frac{3}{c_0} > 0.$$

Thus U' changes sign at most once, from negative to positive. At the two endpoints, the elementary bounds $\phi(0) > 39/100$ and $\phi(6/5) > 19/100$ give

$$U(0) < \frac{46087}{160000}, \qquad U(6/5) < \frac{152023}{160000} < 1.$$

For the second comparison, the expression defining U is decreasing in $\phi(t)$ at fixed t, so replacing $\phi(6/5)$ by $19/100$ gives an upper bound. This contradicts $1 < U(t)$ and completes the exclusion.

The numerical-looking Gaussian comparisons above require no floating point arithmetic. For example $\pi < 22/7$ gives $\phi(0) > 199/500$. At $x = 18/25$, the Taylor bound

$$e^x \leq 1 + x + x^2/2 + x^3/6 + \frac{x^4}{24(1-x/5)} < 25/12$$

gives $\phi(6/5) > (199/500)(12/25) > 19/100$. Also $\pi e > 8$ gives $\phi(1) < 1/4$, and $\pi > 25/8$ gives $\phi(0) < 2/5$.

B.8 Two triples: positive rectangle representations

Return to the contact-scale coordinates $(-a, c, b)$ of Section B.6.

Positive representations Suppose $c > 0$ in the identical normalized laws obtained above. Omit the coordinate index. The rectangle condition is

$$b - a > 2c, \qquad \text{equivalently } v = b - c > u = a + c. \tag{B.8.12}$$

The inequality is strict because a collision between a corner sum and the selected middle-middle sum would give a second representation of the threshold.

Before division by K, each coordinate has variance $1/2$. Writing its middle mass as m, centering and the exact variance formula give

$$\frac{1}{2} = K^2(ab - muv), \qquad q = \frac{b + mu}{h}.$$

Eliminating m yields

$$q = 1 + \frac{ac}{hv} - \frac{1}{2hvK^2}. \tag{B.8.13}$$

By (B.6.1), $h \le 2$, $ab \le 1$, and $b < (1+\sqrt{2})/2 < 5/4$. Also $v < b$, while (B.8.12) gives $hv > 2u^2$. Therefore

$$hv < 5/2, \qquad \frac{ac}{hv} < \frac{ac}{2(a+c)^2} \le \frac{1}{8},$$

and (B.8.13) implies

$$q < Q(K) := \frac{9}{8} - \frac{1}{5K^2}, \qquad K \ge 1/\sqrt{2}. \tag{B.8.14}$$

The selected threshold is $z = 2Kc > 0$, and the event probability is q^2. Since $CR = CK - z\phi(z)/3$, its ratio identity would require

$$0 = q^2 - \Phi(z) - CK + z\phi(z)/3 < q^2 - \frac{1}{2} - \frac{2}{5}K. \tag{B.8.15}$$

Here $C \ge 49/120 > 2/5$, and $\Phi(z) \ge 1/2 + z\phi(z)$ for $z \ge 0$. For $K \ge 5/4$, the right side of (B.8.15) is at most $1/2 - (2/5)K \le 0$, since $q \le 1$, giving a contradiction.

For $1/\sqrt{2} \le K \le 5/4$, define

$$G(K) = Q(K)^2 - 1/2 - (2/5)K.$$

Its second derivative is

$$G''(K) = \frac{4}{5K^4}\left(\frac{1}{K^2} - \frac{27}{8}\right) < 0.$$

Also $G'(5/4) = 654/78125 > 0$. Thus G increases throughout this interval and

$$G(K) \le G(5/4) = \left(\frac{997}{1000}\right)^2 - 1 = -\frac{5991}{10^6} < 0.$$

Equations (B.8.14)–(B.8.15) again give a contradiction. This excludes positive representations of the 220 event; the only numerical input is (B.6.11).

B.9 One triple: the rectangle

Normalized contacts and exact variance Normalize total variance to one, and use the dimensionless supports

$$T: \quad K(-a, c, b), \qquad B: \quad K(-d, e), \qquad K = R + \frac{z\phi(z)}{3C}.$$

The representation is the middle triple point and the high Bernoulli point, so $z = K(c + e)$. The nonselected triple points are smooth minima with deletion levels $1, 0$; the representation has level 1. Consequently

$$\frac{1}{2} < a < 1 < b, \quad a^2 + b^2 = a + b, \quad f_T(Kc) = f_T(-Ka), \quad -\frac{1}{2} < c < 1. \tag{B.9.1}$$

Both Bernoulli deletion levels are the same, namely $q = \mathbb{P}(T \le Kc)$. Its low point is a smooth minimum, and its high point has negative derivative. The selected Bernoulli point identity gives

$$d > \frac{1}{2}, \quad \frac{d^2 + e^2}{d + e} < 1.$$

Set $k = e/d > 0$. Factoring the equal-height equation first, without yet restricting k, gives

$$d = d(k) = \frac{3(1+k)^2}{2(k^3 + 3k + 2)}, \qquad e = kd(k), \qquad k > 0. \tag{B.9.2}$$

Substitution into the strict Bernoulli representation inequality then gives

$$1 - \frac{d^2 + e^2}{d + e} = \frac{(1-k)(k^2 + 4k + 1)}{2(k^3 + 3k + 2)} > 0.$$

Hence $k < 1$. Differentiating (B.9.2) on $(0, 1)$ shows $d'(k) > 0$, while $d(0) = 3/4$ and $d(1) = 1$. Consequently $3/4 < d < 1$ and $0 < e < d$.

This is the equal-height equation factored in Section B.6.

Let m, r be the triple's middle and high masses. Centering gives

$$r = \frac{a - m(a+c)}{a+b} > 0, \qquad \mathrm{Var}(T) = K^2\{ac + r(a+b)(b-c)\}.$$

Writing $h = a + b$, exact total variance is therefore

$$\boxed{1 = K^2\{ac + de + rh(b-c)\}.} \tag{B.9.3}$$

The full event is simply $T \le Kc$, so $F = 1 - r$.

The selected standardized threshold is positive If $c > 0$ this is immediate. Suppose $c \le 0$. The triple equal-height factorization yields

$$\frac{1}{2} < a \le \frac{3}{4}, \qquad c = 2a - \frac{3}{2}, \qquad H_T = -f_T'(Kc)/(CK^2) = (a+c)^2.$$

Put $\sigma = \phi(z)/(CK^2)$. The unique-representation derivative identity is $mH_T = \sigma$. Since the high mass in (B.9.3) is positive,

$$\sigma < a(a+c) \le \frac{9}{16}. \tag{B.9.4}$$

For the Bernoulli law, its smooth low-point derivative instead gives

$$\sigma = 3d\left(1 - \frac{d^2 + e^2}{d + e}\right) = \frac{9(1+k)^2(1-k)(k^2 + 4k + 1)}{4(k^3 + 3k + 2)^2}. \tag{B.9.5}$$

We claim that (B.9.4)–(B.9.5) imply $e > 1/2$. The functions $d(k)$ and $e(k) = kd(k)$ strictly increase on $(0, 1)$, by differentiating (B.9.2), and

$$e(11/20) = \frac{10571}{20354} > \frac{1}{2}.$$

If $e \le 1/2$, then $0 < k < 11/20$. The difference between the numerator in (B.9.5) and the value $9/16$ has the sign of

$$kB(k), \qquad B(k) = 8 + 7k - 20k^2 - 26k^3 - 4k^4 - k^5.$$

The polynomial B is strictly concave on $[0, 11/20]$, because $B'' = -40 - 156k - 48k^2 - 20k^3 < 0$, and its two endpoint values are

$$B(0) = 8, \qquad B(11/20) = \frac{3385269}{3200000} > 0.$$

Thus $\sigma > 9/16$, contradicting (B.9.4). Hence $e > 1/2$. Since $c > -1/2$, we have proved in both sign cases that

$$\boxed{z = K(c + e) > 0.} \tag{B.9.6}$$

Elementary uniform bounds For every centered coordinate, its third absolute moment is at least its variance to the power $3/2$. With two coordinates and total variance one, this gives $R \ge 1/\sqrt{2} > 7/10$. Also $C \ge 49/120$, so $C^2 > 1/6$ and $CR > 343/1200$. Since $F \le 1$,

$$\Phi(z) < \frac{857}{1200}.$$

The elementary bounds $\phi(0) > 39/100$ and $\exp(-x) \ge 1 - x$ give

$$\Phi(3/5) > \frac{1}{2} + \frac{39}{100}\left\{\frac{3}{5} - \frac{(3/5)^3}{6}\right\} = \frac{17999}{25000} > \frac{857}{1200}.$$

Consequently $0 < z < 3/5$. We will also use

$$\phi(z) > \phi(3/5) > \frac{33}{100}. \tag{B.9.7}$$

For a rational verification of the last bound, $\pi < 22/7$ implies $\phi(0) > 199/500$; the cubic alternating exponential lower bound gives

$$e^{-9/50} > 1 - \frac{9}{50} + \frac{(9/50)^2}{2} - \frac{(9/50)^3}{6} = \frac{208807}{250000}.$$

Their product is $41552593/125000000 > 33/100$.

Set $N(z) = \Phi(z) - z\phi(z)/3$. The ratio identity is

$$1 - r = N(z) + CK,$$

and the decreasing positive Gaussian density gives $N(z) \ge 1/2 + (2/3)z\phi(z)$. Thus

$$K < \frac{1}{2C} < \frac{5}{4}, \qquad r < \frac{1}{2} - CK - \frac{2}{3}z\phi(z). \tag{B.9.8}$$

Variance contradiction We need only the following support estimates:

$$ac + de < c + e + \frac{1}{4}, \qquad h(b-c) < 3. \tag{B.9.9}$$

For the first, $d < 1$ and $e > 0$ give $de < e$. If $c > 0$, also $ac < c$; if $c \le 0$, then $ac = c + (1-a)|c| < c + 1/4$, by (B.9.1).

For the second, when $c > 0$ we have $a > 3/4$, and

$$h(b-c) < hb \le \frac{17 + 7\sqrt{7}}{16} < \frac{9}{4} < 3.$$

Here $(hb)' = h(3-4a)/(2b-1) \le 0$, so the bound is its value at $a = 3/4$. When $c \le 0$, substitute $c = 2a - 3/2$. Direct differentiation gives

$$\{h(b-c)\}' = \frac{2(a^2 - b^2 - 6ab + 3b)}{2b-1} < 0 \qquad (1/2 < a \le 3/4),$$

because $a^2 < b^2$ and $6ab \ge 3b$. Its upper endpoint at $a = 1/2$ is $(1 + \sqrt{2}/2)^2 = 3/2 + \sqrt{2} < 3$. This proves (B.9.9).

Insert (B.9.8)–(B.9.9) into the exact variance (B.9.3):

$$\begin{aligned} 1 &< Kz + \frac{K^2}{4} + 3K^2 r \\ &< \frac{K^2}{4} + 3K^2(1/2 - CK) + zK\{1 - 2K\phi(z)\}. \end{aligned} \tag{B.9.10}$$

Each term has an elementary upper bound. Equations (B.9.7)–(B.9.8) give

$$\frac{K^2}{4} < \frac{25}{64}, \qquad 3K^2(1/2 - CK) \le \frac{1}{18C^2} < \frac{1}{3}.$$

For the last term in (B.9.10), if its brace is nonpositive the term is already nonpositive. Otherwise, using $z < 3/5$ and $\phi(z) > 33/100$,

$$zK\{1 - 2K\phi(z)\} < \frac{3}{5} \max_{x \ge 0} \left(x - \frac{33}{50} x^2 \right) = \frac{5}{22}.$$

Therefore (B.9.10) yields the impossible strict inequality

$$\boxed{1 < \frac{25}{64} + \frac{1}{3} + \frac{5}{22} = \frac{2009}{2112} < 1.}$$

This completes the exclusion of all unique-representation 220 configurations.

B.10 One triple: the upper shape

We exclude a maximizing pair with a unique middle/high representation and event 221. Use the contact scale:

$$T: \quad K(-a, c, b), \qquad B: \quad K(-d, e), \qquad K = R + \frac{z\phi(z)}{3C}.$$

The selected representation is middle/high, with $z = K(c + e)$. Let r be the triple's high mass, let $p = d/(d + e)$ be the Bernoulli high mass, and put $\varepsilon = rp$. The actual event probability is

$$F = 1 - rp = 1 - \varepsilon. \tag{B.10.1}$$

Contact signs force a positive threshold The triple deletion levels are $(1, 1, 1-p)$, so the two nonselected smooth minima and the selected middle point satisfy

$$\frac{1}{2} < a < 1 < b, \quad a^2 + b^2 = a + b, \quad f_T(Kc) = f_T(-Ka), \quad -\frac{1}{2} < c < 1. \tag{B.10.2}$$

The Bernoulli low point is a smooth minimum, and its high representation has negative derivative. The representation identity gives

$$d > \frac{1}{2}, \qquad \frac{d^2 + e^2}{d + e} < 1. \tag{B.10.3}$$

We also need $d < 1$, which does not follow from (B.10.3) alone. The actual Bernoulli deletion levels are $1, 1-r$. Set $t = e/d > 0$ and $P(t) = t^3 + 3t + 2$. Subtracting the two contact heights, using the smooth derivative equation at $-Kd$, gives exactly

$$-r = f_B(Ke) - f_B(-Kd) = CK^3 d^2 P(t) \left\{ d - \frac{3(1+t)^2}{2P(t)} \right\} < 0. \tag{B.10.4}$$

Since

$$2P(t) - 3(1+t)^2 = (t-1)^2(2t+1) \geq 0,$$

equation (B.10.4) gives $d < 1$. The strict grid geometry for 221 is $d + e > b - c$. Consequently

$$c + e > b - d > 0, \qquad z > 0. \tag{B.10.5}$$

Moreover (B.10.3) and $d > 1/2$ imply

$$\frac{1 + t^2}{1 + t} < 2, \qquad 0 < t < 1 + \sqrt{2} < \frac{17}{7}. \tag{B.10.6}$$

Universal scalar bounds and exact variance The elementary Gaussian comparisons in Section B.9, apply here. They only use $F \leq 1$, $R \geq 1/\sqrt{2}$, $C \geq 49/120$, and $z > 0$, all valid here. Thus

$$0 < z < \frac{3}{5}, \qquad \phi(z) > \frac{33}{100}, \qquad C^2 > \frac{1}{6}. \tag{B.10.7}$$

Writing $N(z) = \Phi(z) - z\phi(z)/3$, the ratio identity (B.10.1) is

$$1 - \varepsilon = N(z) + CK.$$

Since $N(z) \geq 1/2 + (2/3)z\phi(z)$,

$$\varepsilon < \frac{1}{2} - CK - \frac{2}{3} z\phi(z). \tag{B.10.8}$$

Centering gives the exact variance formula

$$1 = K^2\{ac + de + r(a+b)(b-c)\}. \tag{B.10.9}$$

The equal-height triple contact in (B.10.2) gives the support estimate $(a+b)(b-c) < 3$, proved separately on the two sign branches in Section B.9. Also

$$ac + de < c + e + \delta, \qquad \delta = \frac{1}{4} - (1-d)e. \tag{B.10.10}$$

Indeed, $(a-1)c < 1/4$: it is negative for positive c, and is less than $1/4$ when $-1/2 < c \le 0$ and $1/2 < a < 1$. Let

$$L = 3(1+t), \qquad p = \frac{1}{1+t}.$$

Then (B.10.1), (B.10.9)–(B.10.10) imply

$$1 < Kz + \delta K^2 + LK^2\varepsilon. \tag{B.10.11}$$

A sharper intercept when the high Bernoulli mass is small For $0 < t \le 1$, simply use $\delta < 1/4$. For $1 \le t < 17/7$, the slope condition (B.10.3) gives

$$\frac{1}{2} < d < d_* := \frac{1+t}{1+t^2} \le 1.$$

The function $d(1-d)$ decreases on $(1/2, 1)$. Hence

$$(1-d)e = td(1-d) > td_*(1-d_*) = \frac{t^2(t^2-1)}{(1+t^2)^2} \ge \frac{5}{12}(t-1). \tag{B.10.12}$$

The last inequality is a direct polynomial comparison. After dividing by $t-1$ when $t > 1$, its numerator is

$$P_0(t) = -5t^4 + 12t^3 + 2t^2 - 5.$$

Its derivative $4t(-5t^2 + 9t + 1)$ changes sign at most once on the positive axis, from positive to negative. Thus its minimum on $[1, 17/7]$ is at an endpoint, and

$$P_0(1) = 4, \qquad P_0(17/7) = \frac{11404}{2401} > 0.$$

The assertion also holds at $t = 1$ directly. We have proved

$$\delta < \bar{\delta}(t) := \begin{cases} 1/4, & 0 < t \le 1, \\ 2/3 - 5t/12, & 1 \le t < 17/7. \end{cases} \tag{B.10.13}$$

Maximization of a cubic variance bound Insert (B.10.8), (B.10.13) into (B.10.11), and put $A = \bar{\delta}(t) + L/2 > 0$. We obtain

$$1 < AK^2 - LCK^3 + zK\left\{1 - \frac{2}{3}LK\phi(z)\right\}. \tag{B.10.14}$$

The first two terms have the exact maximum

$$\max_{K \ge 0}(AK^2 - LCK^3) = \frac{4A^3}{27L^2C^2} < \frac{8A^3}{9L^2}, \tag{B.10.15}$$

using (B.10.7). For the remaining term, a nonpositive brace gives a nonpositive contribution. When the brace is positive, (B.10.7) gives

$$zK\left\{1-\frac{2}{3}LK\phi(z)\right\}<\frac{3}{5}\max_{x\geq 0}\left(x-\frac{11}{50}Lx^2\right)=\frac{15}{22L}. \tag{B.10.16}$$

It remains to check the two explicit upper functions in (B.10.15)–(B.10.16). Write $x=1+t$. For $0<t\leq 1$, we have $1<x\leq 2$, $L=3x$, and $A=(6x+1)/4$. The bound equals

$$H_0(x)=\frac{(6x+1)^3}{648x^2}+\frac{5}{22x}=\frac{x}{3}+\frac{1}{6}+\frac{101}{396x}+\frac{1}{648x^2}.$$

This function is strictly convex. Its maximum on $[1,2]$ is at an endpoint, with the larger value at $x=2$:

$$H_0(x)\leq H_0(2)=\frac{27407}{28512}<1. \tag{B.10.17}$$

For $1\leq t<17/7$, we have $2\leq x<24/7$ and $A=(13/12)(x+1)$. The bound is

$$H_1(x)=\frac{2197}{17496}\frac{(x+1)^3}{x^2}+\frac{5}{22x}.$$

Put $c_0=2197/17496$. Then

$$H_1(x)=c_0x+3c_0+\frac{3c_0+5/22}{x}+\frac{c_0}{x^2},$$

again strictly convex. Its two endpoint values are

$$H_1(2)=\frac{27407}{28512}<1,\qquad H_1(24/7)=\frac{771397337}{775982592}<1. \tag{B.10.18}$$

The second endpoint is larger, and its positive deficit is $4585255/775982592$. Convexity bounds the whole interval by those endpoints. Equations (B.10.14)–(B.10.18) force $1<1$, a contradiction. This excludes every unique-representation 221 configuration .

B.11 One triple: the middle representation of the triangular event

Consider shape 210 with the middle-triple/low-Bernoulli representation.

Normalize total variance to one and write the dimensionless supports as

$$T:\quad K(-a,c,b),\qquad B:\quad K(-d,e),\qquad K=R+\frac{z\phi(z)}{3C}>0.$$

Let the triple masses be l,m,r, all positive. The selected threshold is $z=K(c-d)$. The actual deletion levels are

$$T:\quad (1,q_B,0),\qquad B:\quad (l+m,l),\qquad q_B=\frac{e}{d+e}. \tag{B.11.1}$$

The strict representation geometry is

$$d+e<a+c. \tag{B.11.2}$$

The positive Bernoulli minimum must be at least K The nonselected Bernoulli point Ke is a smooth minimum, so $e > 1/2$. Suppose $e < 1$. Its cubic dual then also has a smooth negative minimum at $-K\alpha$, where

$$\alpha = \frac{1 + \sqrt{1 + 4e - 4e^2}}{2} > 1, \qquad \alpha^2 + e^2 = \alpha + e.$$

The difference of the minimum heights is exactly

$$f_B(-K\alpha) - f_B(Ke) = \frac{CK^3}{2}(e - \alpha)(\alpha^2 + 4\alpha e + e^2) < 0. \tag{B.11.3}$$

But $f_B(Ke) = l$ by (B.11.1). The deletion threshold at the new negative minimum is

$$z + K\alpha = K(c - d + \alpha) > K(-a + e + \alpha) > -Ka,$$

using (B.11.2). Its deletion CDF is therefore at least l. Equation (B.11.3) contradicts the global cubic majorant at this point. Hence

$$\boxed{e \geq 1.} \tag{B.11.4}$$

The nonselected triple minima have levels $1, 0$, so

$$\frac{1}{2} < a < 1 < b, \qquad a^2 + b^2 = a + b. \tag{B.11.5}$$

Equations (B.11.2), (B.11.4)–(B.11.5) give

$$c - d > e - a > 0, \qquad z > 0,$$

and in particular $c > 0$.

Contact height bounds the negative triple minimum Define the dimensionless minimum-height gap

$$D(a) = \frac{1}{2}(b - a)(a^2 + 4ab + b^2), \qquad b = \frac{1 + \sqrt{1 + 4a - 4a^2}}{2}.$$

The levels $1, 0$ in (B.11.1) give the exact identity

$$CK^3 D(a) = 1. \tag{B.11.6}$$

The scalar comparisons in Section B.9 give

$$0 < z < \frac{3}{5}, \qquad \phi(z) > \frac{33}{100}, \qquad C^2 > \frac{1}{6}, \tag{B.11.7}$$

because $R \geq 1/\sqrt{2}$, $F \leq 1$, and $C \geq 49/120$. Also $F = N(z) + CK$, where $N(z) = \Phi(z) - z\phi(z)/3 > 1/2$; hence $CK < 1/2$. Together with (B.11.6),

$$1 = CK^3 D(a) < \frac{D(a)}{8C^2}, \qquad D(a) > 8C^2 > \frac{4}{3}. \tag{B.11.8}$$

Direct differentiation gives

$$D'(a) = -3(2a - 1)(a + b) < 0.$$

At $a = 2/3$, its value is

$$D(2/3) = \frac{71 + 17\sqrt{17}}{108} < \frac{191}{144} < \frac{4}{3},$$

using $\sqrt{17} < 17/4$. Therefore (B.11.8) forces

$$\boxed{a < \frac{2}{3}.} \tag{B.11.9}$$

The selected triple gap is bounded away from one

$$D(a) < D(1/2) = \frac{3}{4} + \frac{\sqrt{2}}{2} < \frac{3}{2}.$$

Equation (B.11.6) consequently gives

$$K^3 > \frac{1000}{681} > \left(\frac{25}{22}\right)^3. \tag{B.11.10}$$

The second strict difference is the positive rational 7375/7251288. Put $\sigma = \phi(z)/(CK^2)$. From (B.11.6)–(B.11.8), (B.11.10),

$$\sigma = \phi(z)D(a)K > \frac{33}{100}\frac{4}{3}\frac{25}{22} = \frac{1}{2}. \tag{B.11.11}$$

The Bernoulli low-representation identity is

$$\sigma = 3e\left\{\frac{d^2 + e^2}{d + e} - 1\right\} > 0.$$

Let $u = a + c$. Since the Bernoulli third-moment slope is strictly smaller than its support diameter, and (B.11.2) gives $d + e < u$,

$$\sigma < 3u(u - 1).$$

Combining this with (B.11.11), and noting $3(8/7)(1/7) = 24/49 < 1/2$, we obtain

$$\boxed{u > \frac{8}{7}.} \tag{B.11.12}$$

The actual variance Hessian gives the opposite bound Let $\gamma = [-Ka, Kc, Kb]|x|^3$ be the third-moment slope in the centered triple's variance direction. Because $c > 0$, its normalized value is

$$g(a, u) := \frac{\gamma}{K} = b + u - 2a + \frac{2a^3}{(a + b)u}. \tag{B.11.13}$$

On $1/2 < a \le 2/3$, $u \ge 8/7$, this function increases with u and decreases with a. For the first derivative,

$$g_u = 1 - \frac{2a^3}{(a + b)u^2} > 0,$$

since $a+b>2a$ and $u>a$. For the second, write $h=a+b$. Here $b'<0$, $h'>0$, and

$$g_a = b' - 2 + \frac{2}{u}\left(\frac{3a^2}{h} - \frac{a^3 h'}{h^2}\right) < -2 + \frac{6a^2}{uh} < -2 + \frac{14}{9} < 0,$$

using $a \le 2/3$, $h > 3/2$, $u \ge 8/7$. Equations (B.11.9), (B.11.12) therefore imply

$$\frac{\gamma}{K} > g(2/3, 8/7) = \frac{499 + 35\sqrt{17}}{504} > \frac{71}{56} > \frac{5}{4}. \tag{B.11.14}$$

The strict lower bound uses only $\sqrt{17} > 4$.

On the other hand, both signs of the centered triple variance direction are admissible, because all three masses are positive. The actual second-order condition derived in Section B.1 is

$$3C\gamma - \frac{15CR + \phi'''(z)}{4} \le 0.$$

Use $R = K - z\phi(z)/(3C)$ and $\phi'''(z) = (3z - z^3)\phi(z)$ to rewrite it as

$$12C\gamma \le 15CK - z(z^2+2)\phi(z) < 15CK,$$

because $z > 0$. Thus $\gamma/K < 5/4$, contradicting (B.11.14). This excludes the middle/low 210 representation .

B.12 One triple: the low representation of the triangular event

For the low-triple/high-Bernoulli representation of 210, divide support coordinates by $K = R + z\phi(z)/(3C) > 0$.

Exact contact reduction Write

$$T: \quad -(a+w), -a, b, \qquad B: \quad -d, e, \qquad w > 0,$$

with triple masses ℓ, m, r. The triple's two nonselected minima have levels $q, 0$, while its selected lowest point has level 1. The Bernoulli masses at $-d, e$ are q, p, respectively, and its deletion levels are $\ell + m, \ell$. The representation geometry is

$$w < d + e < a + b + w. \tag{B.12.1}$$

The smooth triple minima give

$$\frac{1}{2} < a < 1 < b, \quad a^2 + b^2 = a + b, \quad b = \frac{1 + \sqrt{1 + 4a - 4a^2}}{2}.$$

Put

$$h = a + b, \quad D = \frac{1}{2}(b-a)(a^2 + 4ab + b^2), \quad P = w^2(3a - 3/2 + w), \quad \tau = CK^3.$$

The minimum-height gap is $q = \tau D$. Factoring the negative cubic at its smooth minimum gives the other level gap as $p = \tau P$. Therefore

$$\boxed{\tau = \frac{1}{D+P}, \qquad p = \frac{P}{D+P}, \qquad q = \frac{D}{D+P}.} \tag{B.12.2}$$

The Bernoulli negative smooth minimum has $d > 1/2$. Its high-representation derivative gives $k_B = (d^2 + e^2)/(d + e) < 1$. The strict level difference $f_B(e) - f_B(-d) = -m < 0$ also gives $d < 1$, by the exact factorization in Section B.10.

Thus, after choosing a, w, d, every quantity is determined. Write

$$x = d/p = d + e, \quad e = qx, \quad r_B = p^2 + q^2, \quad k_B = r_B x,$$

$$\ell = \frac{d(1 - k_B)}{w(2a - 1 + w)}, \quad m = \frac{b - \ell(h + w)}{h}, \quad r = 1 - \ell - m. \tag{B.12.3}$$

The first formula follows by equating the two unique-representation slope identities:

$$\sigma = 3\ell w(2a - 1 + w) = 3d(1 - k_B), \qquad \sigma = \frac{\phi(z)}{CK^2}.$$

The other two formulas in (B.12.3) are centering and total mass.

It remains to impose the exact Bernoulli height difference. Put

$$M = q^3 + 3p^2 q + 2p^3.$$

Using its smooth derivative at $-d$, the difference is

$$\boxed{m = \tau x^2(3/2 - Mx).} \tag{B.12.4}$$

For fixed a, w, this is a cubic equation in x. In particular the CDF levels and coordinate probabilities have not been replaced by independent optimization parameters.

Variance and two necessary ratio inequalities The dimensionless variances are

$$V_T = ab + \ell w(h + w), \qquad V_B = pqx^2, \qquad V_0 = V_T + V_B.$$

Total variance one gives $K = V_0^{-1/2}$, hence

$$C = \tau V_0^{3/2}, \qquad \boxed{\tau^2 V_0^3 < (227/500)^2.} \tag{B.12.5}$$

The dimensionless Bernoulli third absolute moment is $R_B = pqx^3(p^2 + q^2)$. The triple dual has constant term τB_0, where $B_0 = b^2(2b - 3/2)$, since it vanishes at its positive smooth minimum b. Averaging that dual over the centered triple and using $F - CR = \Phi(z) > 0$ gives the further necessary inequality

$$\boxed{B_0 - \frac{3}{2}V_T - R_B > 0.} \tag{B.12.6}$$

Proof of the compact closed domain Since $V_T > ab$ and $R_B > 0$, equation (B.12.6) implies

$$0 < B_0 - \frac{3}{2}ab = \frac{b}{2}(b^2 - 3a^2).$$

If $a \geq 3/4$, then $b < 5/4$ gives $b^2 < 25/16 < 27/16 \leq 3a^2$, a contradiction. Thus $a < 3/4$. The strictly decreasing height function satisfies

$$1 < D(3/4) < D(a) < D(1/2) < 3/2.$$

Here $D(3/4) = (17 + 7\sqrt{7})/32 > 1$, and $D(1/2) = 3/4 + \sqrt{2}/2 < 3/2$.

The inequalities $d > 1/2$ and $k_B < 1$ give $e/d < 1 + \sqrt{2} < 3$, hence $q < 3/4$. Equation (B.12.2) implies $P > D/3 > 1/3$. If $w \le 1/2$, then $P = w^2(3a - 3/2 + w) \le (1/4)(3/4 + 1/2) = 5/16$, a contradiction. Thus $w > 1/2$.

Conversely, (B.12.1) and $d = px < 1$ give $pw < 1$, or $(w-1)P < D$. If $w \ge 3/2$, then $P > w^3$ and $(w-1)P \ge (1/2)(27/8) = 27/16 > 3/2 > D$, again a contradiction. Thus $w < 3/2$.

Every candidate therefore lies in the interior of the single closed box

$$\boxed{(a, w, d) \in [1/2, 3/4] \times [1/2, 3/2] \times [1/2, 1].} \tag{B.12.7}$$

Keeping all its faces makes the computational cover larger than needed. The fixed certificate `low210` proves that no point of (B.12.7) satisfies all the necessary contact, probability, variance, and ratio conditions above. Thus this representation cannot occur at a maximizing pair. The enclosure and partition rules are given in Section 8.

B.13 One triple: the lower event

For shape 110, the unique selected representation is the triple's middle point and the Bernoulli's low point.

Two equal-height charts Divide all support points by $K = R + z\phi(z)/(3C) > 0$, and write

$$T: \quad -a, c, b, \qquad B: \quad -d, e.$$

The triple's deletion levels are $(q, q, 0)$, where $q = e/(d+e)$. Thus its nonselected minima and selected point satisfy

$$\frac{1}{2} < a < 1 < b, \quad a^2 + b^2 = a + b, \quad f_T(c) = f_T(-a).$$

Both possible representation signs have exact closed parameter charts $0 \le k \le 1$:

chart	a	c
negative	$1/2 + k/4$	$2a - 3/2$
positive	$3(1+k)^2/[2(k^3 + 3k + 2)]$	$ka.$

(B.13.1)

The two charts share the $c = 0$ boundary and cover the entire genuine equal-height representation range. Their other faces include limiting contacts that need not be genuine triples. Keep all faces in the certificate.

Put

$$b = \frac{1 + \sqrt{1 + 4a - 4a^2}}{2}, \quad h = a + b, \quad u = a + c, \quad v = b - c,$$

$$D = \frac{1}{2}(b - a)(a^2 + 4ab + b^2), \quad H = 3\{a(1-a) + c - c|c|\}.$$

On the negative chart the stable exact formula is $H = u^2$. On the positive chart it is $H = 3[a(1-a) + c(1-c)]$. In either case $0 < H \le 3/2$ at a genuine contact.

The middle mass replaces the singular derivative quotient Let the triple middle mass be the independent parameter m, with $0 < m < 1$. Its other masses and its representation derivative parameter are

$$\ell = \frac{b - mv}{h}, \qquad r = \frac{a - mu}{h}, \qquad \sigma = mH = \frac{\phi(z)}{CK^2} > 0. \tag{B.13.2}$$

The Bernoulli's nonselected positive point is a smooth minimum at height zero. Its entire cubic dual majorizes a nonnegative CDF. If $1/2 < e < 1$, the second smooth minimum is at the negative point

$$-\alpha = -\frac{1 + \sqrt{1 + 4e - 4e^2}}{2}, \qquad \alpha > 1,$$

and its height is strictly below the positive minimum's height zero. This is impossible for the global majorant. Hence $e \geq 1$.

The low Bernoulli representation identity is

$$\sigma = 3e \left\{ \frac{d^2 + e^2}{d + e} - 1 \right\}. \tag{B.13.3}$$

Since $\sigma < 3/2$ and $e \geq 1$, its third-moment slope is less than $3/2$, so its diameter is less than three. Consequently $d < 2$ and

$$0 < t := d/e < 2.$$

Set $f = (t^2+1)/(t+1)$. Solving the positive quadratic root in (B.13.3) determines the Bernoulli support and probabilities without dividing by H:

$$\boxed{e = \frac{1 + \sqrt{1 + (4/3)f\sigma}}{2f}, \quad d = te, \quad q = \frac{1}{1+t}, \quad k_B = ef.} \tag{B.13.4}$$

Every actual candidate therefore lies in one of the two closed cubes

$$\boxed{(k, m, t) \in [0, 1] \times [0, 1] \times [0, 2].} \tag{B.13.5}$$

The use of m as a parameter avoids artificial singularities where the representation derivative H vanishes on a chart face.

Exact height, ratio, and normal-density equations The strict 110 representation geometry is $d + e > u$. The minimum-level difference gives

$$\tau := CK^3 = q/D.$$

For the Bernoulli dual, whose positive smooth minimum has height zero, the low-representation contact value is the dimensionless cubic

$$L = e^3(t^3 + 3t + 2) - \frac{3}{2}e^2(1 + t)^2.$$

Its deletion-CDF value is $\ell + m = 1 - r$. Thus the exact remaining height equation, cleared of its positive denominator, is

$$\boxed{(1 - r)D - qL = 0.} \tag{B.13.6}$$

The dimensionless variances and Bernoulli third absolute moment are

$$V_T = ab - muv, \quad V_B = de, \quad V_0 = V_T + V_B, \quad R_B = de\, k_B.$$

Total variance one gives $K = V_0^{-1/2}$, $z = (c-d)/\sqrt{V_0}$, and

$$C = \frac{qV_0^{3/2}}{D}.$$

The established upper and lower ratio bounds require

$$\boxed{(49/120)^2 D^2 \le q^2 V_0^3 < (227/500)^2 D^2.} \tag{B.13.7}$$

As in the low/high 210 reduction, averaging the triple cubic and using $F - CR = \Phi(z) > 0$ supplies the strict necessary inequality

$$\boxed{b^2(2b - 3/2) - \frac{3}{2}V_T - R_B > 0.} \tag{B.13.8}$$

Finally the definition of σ in (B.13.2) is essential: pure algebraic contacts alone can have ratios below 227/500. Its exact squared form is

$$\boxed{\sigma^2 q^2 V_0 - \frac{D^2}{2\pi} \exp\left(-\frac{(c-d)^2}{V_0}\right) = 0.} \tag{B.13.9}$$

Squaring is harmless as a necessary condition. It only enlarges the candidate set, while all actual quantities are positive. Equation (B.13.9) is evaluated only on boxes with a strictly positive variance enclosure. Other necessary constraints remain available on wider boxes. The fixed certificate `lower110` proves that neither closed chart in (B.13.5) contains a point satisfying all these necessary conditions, including the ratio floor $C = 49/120$. Section 8 gives the enclosure and partition rules. This excludes the unique 110 representation.

B.14 One triple: collision across the full span

The collided 211 event has its two maximal cells at the selected threshold.

Normalize total variance to one. Write R for the sum of third absolute moments, z for the threshold, $\phi_0 = \phi(0)$, and

$$K = R + \frac{z\phi(z)}{3C} > 0. \tag{B.14.1}$$

The global cubic dual is the one in Lemma 6.2. It majorizes the decreasing deletion CDF, touches it at every support point, and has nonpositive derivative at every contact. At a nonselected support point its contact is a smooth local minimum. The two threshold representations use the actual marginal masses.

Collision geometry and the sole smooth triple contact The selected fiber of the collided 211 event is exactly $(triple - low, Bernoulli - high)$ and $(triple - high, Bernoulli - low)$. The Bernoulli diameter therefore equals the triple's total diameter. In units of K, write the supports as

$$T/K: \quad -t, -a, b, \qquad B/K: \quad -d, e, \qquad t > a, \quad b, d, e > 0.$$

Set

$$h = t + b = d + e, \qquad u = t - a > 0, \qquad p = \frac{d}{h}, \quad q = \frac{e}{h}, \qquad \tau = CK^3. \tag{B.14.2}$$

The Bernoulli probabilities at its low and high atoms are respectively q, p. The triple deletion levels are exactly $(1, q, q)$.

The middle triple atom is nonselected and is therefore a smooth minimum. It must be negative. If it were a positive minimum, the cubic dual would have positive derivative at the still larger triple contact; zero cannot be a local minimum because its quadratic coefficient is $-3CK/2 < 0$. The negative minimum can thus be written $-Ka$, with $a > 1/2$.

After dividing its dual values by τ and subtracting a constant, the triple dual is

$$g(x) = |x|^3 - \frac{3}{2}x^2 - 3a(1-a)x. \tag{B.14.3}$$

The equality of the two contact levels at $-a, b$ gives $g(b) = g(-a)$. Put $k = b/a > 0$ and $P_0 = k^3 + 3k + 2$. Direct factorization gives

$$a = \frac{3(1+k)^2}{2P_0}. \tag{B.14.4}$$

Moreover,

$$\frac{g'(b)}{3} = a\{a(1+k^2) - (1+k)\} = \frac{a(1+k)(k-1)(k^2+4k+1)}{2P_0}.$$

The contact derivative is nonpositive, so $0 < k \le 1$. The identities

$$2P_0 - 3(1+k)^2 = (k-1)^2(2k+1),$$

$$a - \frac{3}{4} = \frac{3k(1+2k-k^2)}{4P_0} > 0$$

therefore prove

$$\boxed{\frac{3}{4} < a \le 1, \qquad 0 < b \le 1.} \tag{B.14.5}$$

The boundary $a = b = 1$ is retained. A selected contact may have zero derivative at a collision, so it cannot be discarded as a smooth-representation contradiction.

The remaining triple contact level difference is $1 - q = p$. Factoring the negative cubic at its minimum gives

$$\boxed{p = \tau u^2(t + 2a - 3/2).} \tag{B.14.6}$$

Every factor is positive, including $t + 2a - 3/2 > t$ by (B.14.5).

Combine the first and second variance derivatives Let η be the centered signed three-point direction whose variance derivative is one, as in Proposition 6.4, The support and threshold stay fixed when the triple law is changed to $P_T + s\eta$; both signs of sufficiently small s are admissible. Since the support stays fixed, the collided closed event presents no differentiability issue.

Let $\gamma = \int |x|^3 \eta(dx)$ and let L be the corresponding CDF derivative. The levels $(1, q, q)$ and gaps $Ku, K(a+b)$ give

$$\boxed{L = \frac{p}{K^2 uh} > 0.} \tag{B.14.7}$$

Along this direction the signed excess at price C is exactly

$$J(s) = F + sL - \Phi\left(\frac{z}{\sqrt{1+s}}\right) - C(R + s\gamma)(1+s)^{-3/2}.$$

At the ratio maximum, $J'(0) = 0$ and $J''(0) \le 0$. Differentiation gives

$$C\gamma = L + \frac{3}{2}CR + \frac{1}{2}z\phi(z),$$

$$3C\gamma - \frac{15CR + \phi'''(z)}{4} \le 0.$$

Using $\phi'''(z) = (3z - z^3)\phi(z)$ and eliminating γ yields the exact necessary condition

$$\boxed{12L + 3CR + (z^3 + 3z)\phi(z) \le 0.} \tag{B.14.8}$$

In particular $z = -s < 0$, since L, C, R are positive. Equation (B.14.1) now reads $CR = CK + s\phi(s)/3$, so (B.14.8) becomes

$$12L + 3CK \le s(s^2+2)\phi(s) \le 2s\phi_0. \tag{B.14.9}$$

The last inequality is elementary: the derivative of $(s^2+2)\phi(s)$ is $-s^3\phi(s)$, so that function decreases on $[0, \infty)$ from $2\phi_0$.

Geometry makes the positive CDF derivative too large The threshold is the high-triple/low-Bernoulli sum, hence

$$s = K(d - b) < Kt. \tag{B.14.10}$$

Here $d < h = t + b$ because the Bernoulli law is nondegenerate. Since $L > 0$, (B.14.9) gives $3CK < 2s\phi_0$. Combining this with (B.14.10) and $C \ge 49/120 > \phi_0$ gives

$$\boxed{t > \frac{s}{K} > \frac{3C}{2\phi_0} > \frac{3}{2}.} \tag{B.14.11}$$

Substitute the exact probability identity (B.14.6) into (B.14.7). Equations (B.14.5), (B.14.10), and (B.14.11) imply

$$\begin{aligned} L &= CK\,\frac{(t-a)(t+2a-3/2)}{t+b} \\ &> CK\,\frac{t(t-1)}{t+1} \\ &> Cs\,\frac{t-1}{t+1} \\ &> \frac{Cs}{5}. \end{aligned} \tag{B.14.12}$$

The first strict comparison uses $a \le 1$, $b \le 1$, and $2a - 3/2 > 0$; the final one uses $t > 3/2$. Inserting (B.14.12) into (B.14.9) yields

$$\frac{12Cs}{5} < 2s\phi_0, \qquad \text{hence} \qquad C < \frac{5}{6}\phi_0.$$

The identity (B.14.8) is also a general necessary condition for any genuine triple at a full-law ratio maximum, with its actual unit-variance CDF derivative L. In particular $L \ge 0$ forces a negative selected threshold. Its successful use here depends on the exact collided 211 geometry and the actual Bernoulli probability in (B.14.6)–(B.14.7).

B.15 One triple: collision across the upper gap

Normalize variance to one and write

$$K = R + z\phi(z)/(3C) > 0, \qquad \tau = CK^3, \qquad \sigma = \phi(z)/(CK^2) > 0.$$

All following support variables are divided by K.

The exact compact support and probability charts The selected fiber is $(triple - middle, Bernoulli - high)$ and $(triple - high, Bernoulli - low)$. Write

$$T:\ -a, c, b, \qquad B:\ -d, e, \qquad a, b, d, e > 0, \quad -a < c < b.$$

The collision forces

$$h = b - c = d + e, \qquad p = d/h, \quad q = e/h, \tag{B.15.1}$$

where q, p are the low and high Bernoulli probabilities. The triple deletion levels are $(1, 1, q)$; its low point is the sole nonselected contact and is a smooth negative minimum. Thus $a > 1/2$ and, up to a constant, the divided triple dual is

$$g(x) = |x|^3 - \frac{3}{2}x^2 - 3a(1-a)x. \tag{B.15.2}$$

The middle contact satisfies $g(c) = g(-a)$ and $g'(c) \le 0$. The two equal-height charts in Section B.13 apply:

	a	c
negative, $0 \le k \le 1$	$1/2 + k/4$	$(k-1)/2$
positive, $0 \le k \le 1$	$3(1+k)^2/[2(k^3+3k+2)]$	$ka.$

(B.15.3)

At a genuine configuration $a < 1$: the common endpoint $a = c = 1$ would put the still larger contact b to the right of the positive minimum. The highest triple contact has $g'(b) \le 0$, so

$$0 < b \le B(a) := \frac{1+\sqrt{1+4a(1-a)}}{2}.$$

Use $b = wB(a)$, $0 \le w \le 1$, retaining the unnecessary $b \le c$ region and rejecting it by the strict gap condition $h > 0$.

Let the triple middle mass m be an independent parameter. Its low and high masses, and the useful contact derivatives, are

$$A = a + b, \quad u = a + c, \qquad \ell = \frac{b - mh}{A}, \qquad r = \frac{a - mu}{A}, \tag{B.15.4}$$

$$H_c = -g'(c) = 3\{a(1-a) + c - c|c|\}, \qquad H_b = -g'(b) = 3\{a(1-a) + b - b^2\} \ge 0.$$

On the negative chart use the exact identity $H_c = u^2$. Averaging the dual derivative gives

$$\boxed{\sigma = mH_c + rH_b.} \tag{B.15.5}$$

Thus every candidate lies in one of the two full closed four-cubes

$$\boxed{(k, w, p, m) \in [0,1]^4.} \tag{B.15.6}$$

The denominator A is bounded below by $1/2$ throughout these cubes, including extraneous boundary points. No small-gap or probability division is needed in this parameterization.

Actual contact and collision-preserving support equations Put

$$U = g(-a) - g(b) = a^2(3/2 - 2a) - b^3 + \frac{3}{2}b^2 + 3a(1-a)b.$$

The triple contact levels give

$$\boxed{p = \tau U, \qquad U > 0.} \tag{B.15.7}$$

For the Bernoulli law, set $k_B = h(p^2 + q^2)$. Its divided derivative at the high support point is

$$f'_B(Ke)/(CK^2) = 3e(k_B - 1) - \sigma.$$

$$mf'_T(Kc) = pf'_B(Ke).$$

Substitute (B.15.5) and the displayed derivative. The exact necessary collision equation is

$$\boxed{qmH_c - prH_b - 3pqh(1 - k_B) = 0.} \tag{B.15.8}$$

This support motion preserves the collision and uses the actual marginal masses.

The Bernoulli deletion levels are $(1, \ell + m) = (1, 1 - r)$. Its signed second moment is $M_B = de(e-d)/h$, and its divided derivative everywhere is $3[x|x| - x - M_B] - \sigma$. Integrating between $-d, e$ gives its exact value difference

$$f_B(Ke) - f_B(-Kd) = \tau\{J - \sigma h\} = -r,$$

$$J = (e-d)\{(e-d)^2 - 3h/2\}.$$

Using (B.15.7) clears the denominator:

$$\boxed{p(J-\sigma h) + rU = 0.} \tag{B.15.9}$$

Variance, normal comparisons, and exact bounds Use the following dimensionless moments:

$$V_T = ab - muh, \qquad V_B = de, \qquad V_0 = V_T + V_B,$$

$$R_T = \ell a^3 + m|c|^3 + rb^3, \qquad R_B = de\,k_B, \qquad R_0 = R_T + R_B.$$

Normalization and (B.15.7) give

$$K = V_0^{-1/2}, \qquad z = \frac{o}{\sqrt{V_0}}, \quad o = c + e, \qquad C = \frac{pV_0^{3/2}}{U}. \tag{B.15.10}$$

The definition of K and (B.15.5) imply the further necessary polynomial identity

$$\boxed{3(V_0 - R_0) - \sigma o = 0.} \tag{B.15.11}$$

The bounds $1/2 < K < 8/5$ in (6.1) also give the strict necessary bound

$$\boxed{25/64 < V_0 < 4.} \tag{B.15.11a}$$

This holds before any contact division and removes zero-variance limits.

The strict ratio interval gives

$$(49/120)^2U^2 \le p^2V_0^3 < (227/500)^2U^2. \tag{B.15.12}$$

These denominator-cleared tests are available even when a box's variance enclosure crosses zero. On boxes with a strictly positive variance enclosure, also impose the unsquared density and actual CDF equations

$$\boxed{\sigma p\sqrt{V_0} - U\phi(z) = 0,} \tag{B.15.13}$$

$$\boxed{(1-pr)U - pR_0 - U\Phi(z) = 0.} \tag{B.15.14}$$

The true event probability is $F = 1 - pr$. A simple necessary CDF positivity test without transcendental evaluation follows by averaging the triple dual, which equals one at $-a$:

$$\boxed{U - p\{a^2(3/2 - 2a) + (3/2)V_T + R_B\} > 0.} \tag{B.15.15}$$

At the maximizing pair, each marginal variance lies strictly between 13/200 and 187/200 of the total variance, by (6.2). Hence

$$\boxed{V_T > \frac{13}{200}V_0, \qquad V_B > \frac{13}{200}V_0.} \tag{B.15.16}$$

Finally, the triple's actual unit-variance CDF derivative is $L = -p/(K^2hA)$. The first/second variance identity (B.14.8) of Section B.14 applies without changing the closed event. Substitute (B.15.10), (B.15.13), and multiply by the positive quantity $UhAV_0/p$. The necessary second-order condition becomes the polynomial inequality

$$\boxed{-12V_0^2U + 3R_0hAV_0 + \sigma hAo(o^2 + 3V_0) \le 0.} \tag{B.15.17}$$

The fixed certificate `collision221` proves that the two closed support charts above contain no point satisfying their contact equations, probability conditions, and necessary inequalities. Its strict-floor checks include $C = 49/120$, as described in Section 8. Thus a collision across the upper gap cannot occur at a maximizing pair.

B.16 One triple: collision across the lower gap

Use the contact scale: the support coordinates below are divided by K.

$$K = R + z\phi(z)/(3C), \quad \tau = CK^3, \quad \sigma = \phi(z)/(CK^2).$$

The bounds in (6.1) give

$$\boxed{1/2 < K < 8/5, \qquad 1/20 < \tau < 2.} \tag{B.16.1}$$

The lower bound for τ uses $C > 2/5$; the upper bound uses $C < 227/500$ and $(227/500)(8/5)^3 < 2$.

Compact support charts from the zero minimum Divide all support points by K and write

$$T: \ -t, c, b, \qquad B: \ -d, e, \quad t, b, d, e > 0, \quad -t < c < b.$$

The selected fiber is $(triple - low, Bernoulli - high)$ and $(triple - middle, Bernoulli - low)$, so

$$h = t + c = d + e, \qquad p = d/h, \quad q = e/h. \tag{B.16.2}$$

The low/high Bernoulli probabilities are q, p; the triple deletion levels are $(1, q, 0)$. Its high contact is nonselected and is a smooth positive minimum of height zero. The whole dual is nonnegative, so $b \ge 1$. Indeed a positive minimum with $1/2 < b < 1$ has a strictly lower negative local minimum, contradicting the nonnegative global majorant.

The triple dual divided by τ is exactly

$$g_b(x) = |x|^3 - \frac{3}{2}x^2 - 3b(b-1)x + B_0, \qquad B_0 = b^2(2b - 3/2). \tag{B.16.3}$$

It vanishes at b. Put

$$G = g_b(-t) = t^3 - \frac{3}{2}t^2 + 3b(b-1)t + B_0, \qquad G_c = g_b(c).$$

The contact values give

$$\boxed{\tau = 1/G, \quad q = G_c/G, \quad p = 1 - q, \quad 1/2 < G < 20.} \tag{B.16.4}$$

The height bounds in (B.16.4) follow from (B.16.1). They also bound the supports. If $b \ge 5/2$, then $g_b(-t)$ increases for $t \ge 0$, since its derivative is $3[(t - 1/2)^2 + b(b-1) - 1/4] > 0$. Hence $G \ge B_0 \ge 175/8 > 20$, impossible. Since $b \ge 1$,

$$G \geq t^3 - \frac{3}{2}t^2 + \frac{1}{2}.$$

For $t \geq 7/2$, the right side is increasing and is at least 25, also impossible. Thus

$$1 \leq b < 5/2, \qquad 0 < t < 7/2. \tag{B.16.5}$$

Use independent parameters $(s_b, s_t, w, m) \in [0,1]^4$ and maps

$$b = 1 + \frac{3}{2}s_b, \qquad t = \frac{7}{2}s_t,$$

$$c = -t(1-w) \quad \text{on the negative chart}, \qquad c = wb \quad \text{on the positive chart}. \tag{B.16.6}$$

Here m is the triple middle mass. Both closed cubes are retained, including all extra boundary points and the shared $c = 0$ face.

Let $A = t + b$, $v = b - c$. Centering determines

$$\ell = \frac{b - mv}{A}, \qquad r = \frac{t - mh}{A}. \tag{B.16.7}$$

The denominator $A \geq 1$ throughout the whole parameter domain.

Exact collision and Bernoulli contact equations Define the nonnegative selected-contact slopes

$$H_\ell = 3\{t^2 - t + b(b-1)\}, \qquad H_m = 3\{b(b-1) + c - c|c|\}.$$

Averaging the triple dual derivative gives

$$\boxed{\sigma = \ell H_\ell + m H_m > 0.} \tag{B.16.8}$$

Set $d = ph$, $e = qh$, and $k_B = h(p^2 + q^2)$. Move the raw low triple point and the raw high Bernoulli point by opposite amounts, holding all other raw support points and all probabilities fixed. Both selected raw sums and the closed event stay fixed. Recenter the same two marginals and move the threshold by their total mean shift. The two-sided ratio derivative gives

$$\ell f_T'(-Kt) = p f_B'(Ke).$$

Using the Bernoulli derivative $f_B'(Ke)/(CK^2) = 3e(k_B - 1) - \sigma$ and (B.16.8), this is

$$\boxed{q\ell H_\ell - pmH_m - 3pqh(1 - k_B) = 0.} \tag{B.16.9}$$

The Bernoulli deletion levels are $(\ell + m, \ell)$. Integrating its divided derivative, as in the collided 221 reduction, gives

$$\tau\{J - \sigma h\} = -m, \qquad J = (e-d)\{(e-d)^2 - 3h/2\}.$$

Since $\tau = 1/G$, its denominator-cleared contact equation is

$$\boxed{J - \sigma h + mG = 0.} \tag{B.16.10}$$

Normalization, actual CDF, and second-order constraints Write the dimensionless variance and sum of third absolute moments as

$$V_T = tb - mhv, \quad V_B = de, \quad V_0 = V_T + V_B,$$

$$R_T = \ell t^3 + m|c|^3 + rb^3, \quad R_B = de\,k_B, \quad R_0 = R_T + R_B.$$

Then

$$K = V_0^{-1/2}, \qquad C = V_0^{3/2}/G, \qquad o = c - d, \qquad z = o/\sqrt{V_0}.$$

The definition of K gives the necessary polynomial identity

$$\boxed{3(V_0 - R_0) - \sigma o = 0.} \tag{B.16.11}$$

The compact bounds and ratio interval imply

$$25/64 < V_0 < 4, \qquad (49/120)^2 G^2 \le V_0^3 < (227/500)^2 G^2. \tag{B.16.12}$$

At the maximizing pair, each marginal variance lies strictly between 13/200 and 187/200 of the total variance, by (6.2). Hence

$$V_T > \frac{13}{200} V_0, \qquad V_B > \frac{13}{200} V_0. \tag{B.16.13}$$

The actual event probability is $F = \ell + mq$. The density and CDF equations are

$$\boxed{\sigma\sqrt{V_0} - G\phi(z) = 0,} \tag{B.16.14}$$

$$\boxed{G\{\ell + mq - \Phi(z)\} - R_0 = 0.} \tag{B.16.15}$$

Averaging the zero-minimum triple dual (B.16.3) also supplies the necessary positivity test

$$\boxed{B_0 - \frac{3}{2} V_T - R_B > 0.} \tag{B.16.16}$$

For the triple's centered unit-variance direction, the actual CDF derivative at the fixed closed event is

$$L = \frac{(pA - h)V_0}{Ahv}.$$

Combine its first and second variance derivatives and use (B.16.14). After multiplication by the positive factor $GAhvV_0$, the necessary second-order condition is

$$\boxed{12(pA - h)GV_0^2 + 3R_0 AhvV_0 + \sigma Ahv\, o(o^2 + 3V_0) \le 0.} \tag{B.16.17}$$

B.17 Lower-gap collision: elimination of the middle mass

Continue with the supports divided by K in Section B.16.

A positive affine coefficient Retain the previous notation

$$A = t + b, \quad h = t + c, \quad v = b - c, \quad \ell = (b - mv)/A, \quad r = (t - mh)/A,$$

with $H_\ell, H_m \geq 0$, $\sigma = \ell H_\ell + mH_m > 0$, and Bernoulli probabilities $q, p > 0$. The collision support equation is

$$q\ell H_\ell - pmH_m - 3pqh\{1 - h(p^2 + q^2)\} = 0.$$

Substitution of ℓ makes it affine in m, and gives exactly

$$\boxed{m = \frac{qbH_\ell - 3Apqh\{1 - h(p^2 + q^2)\}}{qvH_\ell + ApH_m}.} \tag{B.17.1}$$

The denominator $L = qvH_\ell + ApH_m$ is strictly positive: every coefficient is positive, both slopes are nonnegative, and $\sigma > 0$ excludes their simultaneous vanishing. Thus there is no additional zero-denominator collision case to retain.

Let $N = G_c$, $P = G - G_c$, $W = P^2 + N^2$, so $q = N/G$, $p = P/G$. Define

$$\begin{aligned} \mathcal{D} &= G^3(NvH_\ell + APH_m) = G^4 L, \\ M &= NbH_\ell G^3 - 3APNhG^2 + 3APNh^2 W. \end{aligned} \tag{B.17.2}$$

Then $m = M/\mathcal{D}$. The four natural mass conditions become polynomial:

$$M > 0, \quad \mathcal{D} - M > 0, \quad b\mathcal{D} - vM > 0, \quad t\mathcal{D} - hM > 0. \tag{B.17.3}$$

A uniform positive denominator bound The canonical mixture-width theorem applies to this triple and its Bernoulli partner without an optimality assumption. Consequently every violating candidate in this system must satisfy

$$\min\{bh, tv\} > V/3 > 25/192. \tag{B.17.4}$$

The Bernoulli variance bound gives

$$pqh^2 > \frac{13}{200}V > \frac{13}{512}.$$

As $h < A = t + b < 6$, this yields $p, q > pq > 13/18432$. Moreover $A \geq 1$.

By (B.3.2), $\sigma > 9/200 > 1/80$. Since $\ell, m < 1$, one has $H_\ell + H_m \geq \sigma$, and hence

$$L > \frac{13}{18432}\frac{25}{672}\frac{1}{80}.$$

Multiplication by $G^4 > 1/16$ proves the explicit bound

$$\boxed{\mathcal{D} > \frac{65}{3170893824} > \frac{1}{50000000}.} \tag{B.17.5}$$

The final comparison has exact numerator slack $65\,50000000 - 3170893824 = 79106176$. Thus the eliminated mass is a smooth rational function on an open neighborhood of every candidate in the compact support charts. In interval arithmetic, (B.17.5) is only a necessary test; division is permitted only if the direct enclosure of $\mathcal{D}$ is positive on the whole current box.

The two three-parameter support charts Delete the independent mass variable from the two charts. For $(s_b, s_t, w) \in [0,1]^3$, set

$$b = 1 + 3s_b/2, \qquad t = 7s_t/2,$$

and take $c = -t(1-w)$ on the negative chart or $c = wb$ on the positive chart. Compute m by (B.17.1) or (B.17.2) and retain all genuine-support, probability, height, slope, ratio, variance, width and global-dual constraints. Every original candidate has a preimage in one of these two closed three-cubes; the shared $c = 0$ face remains covered. Conversely, any candidate satisfying the retained tests and four remaining equations satisfies the eliminated support equation identically. This statement only identifies necessary systems; it does not assert that every solution is a maximizing law.

Four explicit equations without an independent mass variable The following formulas permit denominator-free evaluation before any normal function. Define

$$L_* = b\mathcal{D} - vM, \quad H_* = t\mathcal{D} - hM, \quad S_* = L_* H_\ell + AMH_m, \quad O = cN - tP,$$
$$V_* = (tb\mathcal{D} - Mhv)G^2 + \mathcal{D}PNh^2,$$
$$R_* = (L_* t^3 + AM|c|^3 + H_* b^3)G^4 + A\mathcal{D}PNh^3 W, \quad F_* = GL_* + AMN.$$

Then the actual quantities are

$$V = \frac{V_*}{\mathcal{D}G^2}, \quad R = \frac{R_*}{A\mathcal{D}G^4}, \quad \sigma = \frac{S_*}{A\mathcal{D}}, \quad F = \frac{F_*}{A\mathcal{D}G}, \quad z = O\sqrt{\mathcal{D}/V_*}. \tag{B.17.6}$$

The retained Bernoulli height, scale, squared density and CDF equations are, respectively,

$$A\mathcal{D}h(N-P)\{h^2(N-P)^2 - 3hG^2/2\} - S_* hG^3 + AMG^4 = 0,$$
$$3(AG^2 V_* - R_*) - S_* OG^3 = 0,$$
$$S_*^2 V_* - A^2\mathcal{D}^3 G^4 \phi(z)^2 = 0,$$
$$G^4\{F_* - A\mathcal{D}G\Phi(z)\} - R_* = 0. \tag{B.17.7}$$

Each clearing factor is positive at every candidate. In particular the two equations preceding the normal functions are exact polynomial equations in the three support parameters on either sign chart.

The ratio test is $(49/120)^2\mathcal{D}^3G^8 \le V_*^3 < (227/500)^2\mathcal{D}^3G^8$. The added width tests are $3bh\mathcal{D}G^2 > V_*$ and $3tv\mathcal{D}G^2 > V_*$.

B.18 Lower-gap collision: direct necessary equations

Use the contact scale of Section B.16.

Independent law coordinates Write the centered triple supports $(-t, c, b)$, set $A = t + b$, $h = t + c$, $v = b - c$, and use its variance $W = V_T$ as the mass coordinate. Exact centering gives

$$\ell = \frac{W + bc}{Ah}, \qquad m = \frac{tb - W}{hv}, \qquad r = \frac{W - tc}{Av}. \tag{B.18.1}$$

These formulas are intersected with normalization, the direct second moment, and the usual middle-mass expression $W = tb - mhv$. For the Bernoulli, low/high probabilities are q, p, supports are $(-d, e)$, and $d = ph$, $e = qh$. Its third absolute moment is evaluated as

$$V_B = de, \qquad R_B = V_B \left(h - \frac{2V_B}{h} \right), \qquad h/2 \le h - 2V_B/h \le h. \tag{B.18.2}$$

$$R_* = \frac{tb(t^2 + b^2)}{A}, \qquad M_* = \frac{tb(b - t)}{A},$$

the exact moments are

$$R_T = R_* + \gamma(W - tb), \qquad \mathbb{E}[T|T|] = M_* + \eta(W - tb). \tag{B.18.3}$$

The direct three-term sum of third absolute moments is also intersected. The bounds $A/2 \le \gamma \le 2A$ and $-1 \le \eta \le 1$ are the established divided- difference bounds. All adjacent and outer gap divisions are performed separately with their positive candidate bounds; a product enclosure that straddles zero is never inverted.

Dual contacts and support pairings The triple majorant is evaluated using the factorization

$$g_b(x) = (x - b)^2(x + 2b - 3/2) + 2(-x)_+^3. \tag{B.18.4}$$

Thus $G = g_b(-t)$, $q = g_b(c)/G$, and $p = 1 - q$. A differently arranged expanded value of G is also intersected. The high minimum gives

$$\sigma = 3\{b(b - 1) - \mathbb{E}[T|T|]\} = \ell H_t + m H_m,$$

where $H_t = 3(t^2 - t + b(b - 1))$ and $H_m = 3(b(b - 1) + c - c|c|)$ are nonnegative. Put

$$Q = 3pqh\{1 - h(p^2 + q^2)\}.$$

The two collision-preserving derivative pairings are evaluated as

$$\ell H_t = p\sigma + Q, \qquad m H_m = q\sigma - Q. \tag{B.18.5}$$

The Bernoulli height equation is

$$mG = \sigma h - (e - d)\{(e - d)^2 - 3h/2\}. \tag{B.18.6}$$

Normalization and inverse identities The independently stored values satisfy the contact equations

$$C = V^{3/2}/G, \quad D = R/G = F - \Phi(z), \quad \sigma\sqrt{V} = G\phi(z), \quad 3(V-R) = \sigma o, \quad o = c-d = e-t = z\sqrt{V}. \tag{B.18.7}$$

They are evaluated both forward and backward, including $V = (CG)^{2/3}$, $\sqrt{V} = G\phi/\sigma$, and the exact inverse density

$$z^2 = -\log(2\pi\phi(z)^2). \tag{B.18.8}$$

Equation (B.18.8) has a known negative square-root branch on the negative chart. On the positive chart a branch is used only when the current candidate interval has proved its sign; otherwise its symmetric hull is retained. A nonnegative interval for z^2 is intersected before taking a square root. The CDF is evaluated through the error function.

The following bounds hold at every candidate: $49/120 \le C \le 227/500$, $49/(120\sqrt{2}) \le D \le 11/20$, $-5/4 \le z \le 3/5$, $\phi \ge 9/50$, $V \ge 2401/5476$ (or $2401/4356$ on the negative chart), and $V \le 4$. It also uses the established coordinate variance fraction bounds $13V/200 \le V_i \le 187V/200$ and Hölder third-moment bounds. References are Section B.3, Section B.4, and Section B.17. Strict bounds are relaxed to closed bounds for enclosure.

The canonical mixture-width inequalities $3bh > V$, $3tv > V$ are also used backward to obtain necessary bounds on each positive factor. The elementary Bernoulli bound $h \ge 2\sqrt{V_B}$ is included. Initial gap and probability floors are exactly the proved floors: $h \ge \sqrt{13/128}$, $v \ge 25/672$, $p, q \ge 13/18432$, and $t \ge 13/1280$. The fixed certificate `collision210` proves that no point of either closed support chart satisfies the necessary conditions in Sections B.16–B.18. The enclosure and partition rules are given in Section 8. This excludes the lower-gap collision.

B.19 Two triples: asymmetric events with a unique representation

Normalize total variance to one and use the dimensionless coordinates of Section B.6. Thus the two coordinate supports are $K(-a_i, c_i, b_i)$, with

$$\frac{1}{2} < a_i < 1 < b_i, \qquad a_i^2 + b_i^2 = a_i + b_i, \qquad h_i = a_i + b_i,$$

and the two nonselected minima have height difference

$$CK^3 D(a_i), \qquad D(a) = \frac{1}{2}(b-a)(a^2 + 4ab + b^2).$$

The decreasing deletion CDF gives these same strict inequalities for a_i, b_i in 221 and 320, because the two nonselected contact levels are distinct. Along the curve $a^2 + b^2 = a + b$,

$$D'(a) = -3(2a-1)(a+b) < 0, \qquad h'(a) = \frac{2(b-a)}{2b-1} > 0. \tag{B.19.1}$$

For shape 221, the first coordinate's three deletion-CDF values are q_2, q_2, l_2, where l_2 is the lowest mass, m_2 the middle mass, and $q_2 = l_2 + m_2$. The second coordinate's three values are $1, q_1, 0$. Therefore

$$CK^3 D(a_1) = m_2 < 1 = CK^3 D(a_2).$$

Equation (B.19.1) forces $a_1 > a_2$ and hence $h_1 > h_2$.

On the other hand, the selected representation is middle-middle. The high-low corner is included, and the low-high corner is excluded. The unique threshold representation makes both relevant comparisons strict:

$$b_1 - a_2 < c_1 + c_2 < -a_1 + b_2.$$

With $u_i = a_i + c_i$, these are exactly

$$h_1 < u_1 + u_2 < h_2,$$

contradicting $h_1 > h_2$. Interchanging the two coordinates transposes 221 to 320, so that shape is excluded as well.

B.20 Two triples: triangular event with a unique representation

Contact geometry and notation Normalize total variance to one. Write the discrepancy as $D = CR$, the selected threshold as z, and put

$$K = \frac{2A}{3C} = R + \frac{z\phi(z)}{3C}.$$

The full-law cubic contacts from Proposition 6.4 give supports $K(-a_i, c_i, b_i)$, with the selected point in the middle of each coordinate. The two nonselected points are smooth minima. In shape 321 their deletion-CDF values are 1 and the other coordinate's lowest mass, so the minimum-height difference is positive. Consequently

$$\frac{1}{2} < a_i < 1 < b_i, \qquad a_i^2 + b_i^2 = a_i + b_i =: h_i \le 2. \tag{B.20.1}$$

Define $u_i = a_i + c_i$ and $v_i = b_i - c_i$. The two cross corners are strictly below the selected middle-middle sum. Thus

$$v_1 < u_2, \qquad v_2 < u_1, \qquad c_1 + c_2 > \max\{b_1 - a_2, b_2 - a_1\} > 0. \tag{B.20.2}$$

In particular $z = K(c_1 + c_2) > 0$ and $K > 0$. Let l_i, m_i, r_i denote the lowest, middle, and highest masses, and put $p_i = m_i + r_i$. The actual selected event probability and its complement are

$$F = 1 - p_1 p_2 + m_1 m_2, \qquad P_{\text{out}} = r_1 p_2 + r_2 p_1 - r_1 r_2. \tag{B.20.3}$$

A divided-difference inequality on the actual triangle For a pair a, b satisfying (B.20.1), let

$$d_a(c) = [-a, c, b]|x|^3.$$

The divided-difference convention is $[x_0, x_1, x_2]x^2 = 1$. Direct interpolation gives

$$d_a(c) = \begin{cases} b + c - a + \dfrac{2a^3}{(a+b)(a+c)}, & c \ge 0, \\ a - b - c + \dfrac{2b^3}{(a+b)(b-c)}, & c \le 0. \end{cases} \tag{B.20.4}$$

This is a strictly convex function of c on $(-a, b)$, with

$$d_a(0) = 1, \qquad d_a'(0) = \frac{b-a}{a+b} > 0. \tag{B.20.5}$$

We prove that (B.20.2) implies

$$\boxed{d_{a_1}(c_1) + d_{a_2}(c_2) > 2.} \tag{B.20.6}$$

Relabel so $a_1 \le a_2$. Then $b_1 \ge b_2$, $h_1 \le h_2$, and the slopes $k_i = (b_i - a_i)/h_i$ satisfy $k_1 \ge k_2$. Put $z_0 = b_2 - a_1 > 0$. We need one derivative comparison:

$$d_{a_2}'(z_0) \ge k_1. \tag{B.20.7}$$

By (B.20.4), this is equivalent to

$$a_2^3 h_1 \le a_1 h_2 (h_2 - a_1)^2.$$

Use $h_1 \le 2$ and $h_2 \ge a_2 + 1$. With $a = a_1$, $b = a_2$, it is sufficient that

$$a(b+1)(b+1-a)^2 \ge 2b^3, \qquad 1/2 \le a \le b \le 1. \tag{B.20.8}$$

For fixed b, the function $a(b+1-a)^2$ has at most one interior critical point, a maximum. Its minimum on $[1/2, b]$ is at an endpoint. At $a = b$, (B.20.8) follows from $b + 1 \ge 2b^2$. At $a = 1/2$, it follows from

$$(b+1)(b+1/2)^2 - 4b^3 = \frac{1}{2} + (1-b)(3b^2 + b - 1/4) > 0.$$

This proves (B.20.7).

If both middle points are nonnegative, (B.20.6) follows immediately from (B.20.5), since their sum is positive. If $c_1 < 0$, then (B.20.2) gives $c_2 > z_0 - c_1$. Along s from c_1 to zero, strict convexity and (B.20.7) give

$$\frac{d}{ds}\{d_{a_1}(s) + d_{a_2}(z_0 - s)\} \le k_1 - d_{a_2}'(z_0) \le 0.$$

The sum is therefore at least $1 + d_{a_2}(z_0) > 2$. The replacement argument stays within the middle-knot intervals because $z_0 - c_1 < c_2 < b_2$. If $c_2 < 0$, the same argument uses $d_{a_1}'(z_0) \ge k_1 \ge k_2$. This proves (B.20.6), including mixed signs.

The actual probability Hessian forces K>1 Use the centered unit-variance signed probability directions on the two three-point supports displayed in Proposition 6.4. Their third-moment derivatives are

$$\gamma_i = K d_{a_i}(c_i).$$

For their two perturbation parameters s, t, the total variance is $1 + s + t$, the sum of third absolute moments is $R + \gamma_1 s + \gamma_2 t$, and the selected CDF is $F + f_1 s + f_2 t + gst$. Both signs of each sufficiently small parameter are admissible. At the maximizing ratio, differentiate

$$J(s,t) = (1+s+t)^{3/2}\{F(s,t) - \Phi(z/\sqrt{1+s+t})\} - C(R + \gamma_1 s + \gamma_2 t).$$

It has a local maximum zero. First stationarity and direct second differentiation give its Hessian entries

$$J_{ii} = 3C\gamma_i - B_* = -b_i^*, \qquad J_{12} = g - \frac{b_1^* + b_2^*}{2},$$

$$B_* = (15CR + \phi'''(z))/4, \qquad b_i^* = B_* - 3C\gamma_i \geq 0. \tag{B.20.9}$$

This is the two-coordinate extension of Section B.1. Negative semidefiniteness implies

$$g \leq \frac{b_1^* + b_2^*}{2} + \sqrt{b_1^* b_2^*} \leq b_1^* + b_2^*. \tag{B.20.10}$$

The actual 321 grid gives, exactly,

$$g = \frac{1 - v_1 v_2/(h_1 h_2)}{K^4 u_1 v_1 u_2 v_2} > \frac{3}{4K^4}. \tag{B.20.11}$$

Indeed (B.20.2) gives $h_1 h_2 > (v_1 + v_2)^2 \geq 4 v_1 v_2$, and $u_i v_i \leq h_i^2/4 \leq 1$.

Since $CR = CK - z\phi(z)/3$ and $\phi'''(z) = (3z - z^3)\phi(z)$, equations (B.20.6), (B.20.9)–(B.20.11) yield

$$\frac{3}{4K^4} < \frac{3}{2} CK - \frac{1}{2} z(z^2 + 2)\phi(z). \tag{B.20.12}$$

Put $N(z) = \Phi(z) - z\phi(z)/3$. It is strictly increasing and exceeds $1/2$ for positive z. The ratio identity is $CK = F - N(z)$, so $CK < 1/2$. Equation (B.20.12) now implies

$$\boxed{K > 1, \qquad K < 1/(2C).} \tag{B.20.13}$$

The Hessian also forces z<1/4 Let

$$L(z) = 1 - N(z) = Q(z) + z\phi(z)/3, \qquad A_*(z) = \frac{3}{2} Q(z) - \frac{1}{2} z(z^2 + 1)\phi(z).$$

We have $CK < L(z)$. Equation (B.20.12), and $C \geq 49/120$, imply

$$\frac{3}{4} < K^4 A_*(z) < \left(\frac{L(z)}{49/120} \right)^4 A_*(z). \tag{B.20.14}$$

The middle expression must be positive, so (B.20.14) is only used when $A_*(z) > 0$. First, $z < 1$: the moment inequality $R \geq 1/\sqrt{2}$ gives $CR > 1/4$, whereas $Q(1) < \phi(1) < 1/4$. A threshold $z \geq 1$ would thus contradict $CR = F - \Phi(z) \leq Q(z)$.

On $[0, 1]$, both L and A_* strictly decrease, since

$$L'(z) = -(2 + z^2)\phi(z)/3, \qquad A_*'(z) = -(2 + z^2 - z^4/2)\phi(z).$$

At $z = 1/4$, use $Q(z) \leq 1/2 - z\phi(z)$ and $\phi(1/4) > 3/8$ to get

$$L(1/4) < 7/16, \qquad A_*(1/4) < 573/1024.$$

If $z \geq 1/4$, (B.20.14) would consequently give

$$\frac{3}{4} < \left(\frac{15}{14} \right)^4 \frac{573}{1024} < \frac{3}{4},$$

a contradiction. The exact slack in the last comparison is $495363/39337984$. We have proved

$$\boxed{0 < z < 1/4.} \tag{B.20.15}$$

The Gaussian comparisons are elementary: $\phi(0) > 39/100$ and $\exp(-1/32) > 31/32$ give $\phi(1/4) > 3/8$; also $\pi e > 8$ gives $\phi(1) < 1/4$.

The necessary outside probability is too large By (B.20.2), (B.20.13), and (B.20.15),

$$b_j - a_i < c_1 + c_2 = z/K < 1/4.$$

Since $b_j > 1$, this forces $a_i > 3/4$ for each coordinate. Its selected middle point also has $c_i > 0$. Indeed its contact value is strictly less than the left-minimum value one, whereas for $c \le 0$ the exact negative branch difference is

$$f(Kc) - f(-Ka) = CK^3(c+a)^2(2a - 3/2 - c) > 0 \quad (a > 3/4).$$

Centering and the variance identity give

$$p_i = \frac{a_i + m_i v_i}{h_i} > \frac{a_i}{h_i} > \frac{39}{100}, \qquad \mathrm{Var}(X_i)/K^2 = a_i c_i + r_i h_i v_i. \tag{B.20.16}$$

For the first estimate, $a_i > 3/4$ and $b_i < (2+\sqrt{7})/4 < 7/6$ give $a_i/h_i > 9/23 > 39/100$. Moreover

$$h_i v_i < h_i b_i \le \frac{17 + 7\sqrt{7}}{16} < \frac{9}{4}. \tag{B.20.17}$$

To check the middle bound, differentiating along (B.20.1) gives $(hb)' = h(3-4a)/(2b-1) \le 0$ for $a \ge 3/4$.

Let $r = r_1 + r_2$. From (B.20.13), (B.20.16)–(B.20.17), $a_i < 1$, and $c_i > 0$,

$$\frac{9}{4} r > \frac{1}{K^2} - \sum_i a_i c_i > \left(\frac{49}{60}\right)^2 - z.$$

Hence

$$r > r_0 - 4z/9 =: r_*(z), \qquad r_0 = \frac{2401}{8100} < 3/10. \tag{B.20.18}$$

Also $0 < r < 1$, because $r_i < a_i/h_i < 1/2$. Equation (B.20.3), (B.20.16), and $r_1 r_2 \le r^2/4$ now imply

$$P_{\mathrm{out}} > f(r) := \frac{39}{100} r - r^2/4 \ge f(r_*(z)). \tag{B.20.19}$$

For the last inequality, f is concave, so its minimum on $[r_*(z), 1]$ is at an endpoint. The second endpoint has $f(1) = 7/50$, whereas $f(r_*(z)) < (39/100)(3/10) < 7/50$.

On the other hand, $\phi(z) > 3/8$ gives $N(z) \ge 1/2 + (2/3) z\phi(z) > 1/2 + z/4$. The ratio identity and (B.20.13) give

$$P_{\mathrm{out}} = 1 - N(z) - CK < \frac{11}{120} - \frac{z}{4}. \tag{B.20.20}$$

Equations (B.20.19)–(B.20.20) are incompatible. At zero their required gap is exactly

$$f(r_0) - \frac{11}{120} = \frac{103447}{52488000} > 0.$$

For $0 \le z \le 1/4$, the derivative of $f(r_0 - 4z/9) - (11/120 - z/4)$ is at least $1/4 - (4/9)(39/100) = 23/300 > 0$. The gap therefore stays positive. This is the final contradiction and proves the 321 exclusion.

B.21 Two triples: upper collisions

Normalize the total variance to one; factors of K in the supports are displayed explicitly below.

$$K = R + z\phi(z)/(3C) > 0, \qquad \phi_0 = \phi(0).$$

For shape 322, the maximal included cells are exactly $(0, 2)$ and $(2, 1)$. A genuine collision at the selected threshold therefore has precisely these two representations. Let the second coordinate's high atom have mass $r_2 > 0$, and put $q = 1 - r_2$.

The first triple forces a substantially negative threshold The first coordinate's deletion levels are $(1, q, q)$. If its successive support gaps are $u, v > 0$, its actual centered unit-variance CDF derivative is

$$L = \frac{r_2}{u(u+v)} > 0.$$

The first and second variance derivatives are valid at collisions because the support and closed threshold are fixed. Their elimination, derived directly in Section B.14, gives

$$12L + 3CR + (z^3 + 3z)\phi(z) \le 0.$$

Consequently $z = -s < 0$, and substitution of $CR = CK + s\phi(s)/3$ yields

$$12L + 3CK \le s(s^2 + 2)\phi(s) \le 2s\phi_0. \tag{B.21.1}$$

The last inequality follows by differentiating $(s^2 + 2)\phi(s)$.

The second triple's equal-level contact prevents that threshold The second coordinate's deletion levels are $(1, 1, \ell_1)$, where ℓ_1 is the positive low mass of the first triple. Its low support point is nonselected, hence a smooth negative minimum of the global cubic dual. Write its first two support points as $-Ka, Kc$.

The contact at Kc has the same value as the negative minimum, and its derivative is nonpositive. Therefore the elementary equal-level cubic calculation gives

$$c \ge -1/2. \tag{B.21.2}$$

For clarity, if $c \le 0$, factor the negative cubic at $-a$: its other equal-height contact is $c = 2a - 3/2$ with $a > 1/2$, so $c > -1/2$. If $c > 0$, inequality (B.21.2) is immediate. No unique-representation assumption is used for this contact; the nonpositive derivative follows from the global majorant at any deletion-CDF jump.

The selected representation $(2, 1)$ uses the first coordinate's highest support point, which is strictly positive by centering and nondegeneracy, and the second coordinate's middle point Kc. Thus

$$z > -K/2, \qquad 0 < s < K/2. \tag{B.21.3}$$

Combining (B.21.1) and (B.21.3) gives

$$3CK < 2s\phi_0 < K\phi_0, \qquad C < \phi_0/3,$$

contrary to $C \geq 49/120 > \phi_0$. This excludes 322. Exchanging the two coordinates transposes its grid to 331 and proves that exclusion too.

B.22 Unit-span divided differences

For $s, \mu \in [0,1]$, put $T = s(1-s)$ and

$$\rho(s,\mu) = (1-s)\mu^3 + s(1-\mu)^3 - |s-\mu|^3.$$

This is exactly the third-moment slope in the three-representation 321 mass-elimination chart. One has the unconditional bounds

$$\boxed{T/2 \leq \rho(s,\mu) \leq T\max\{1+s, 2-s\} \leq 2T.} \tag{B.22.1}$$

At $0 < s < 1$, ρ/T is the second divided difference of $|x|^3$ on $(-\mu, s-\mu, 1-\mu)$. The elementary three-point divided-difference bound in Section B.1 is half the support diameter, giving the lower bound $\rho/T \geq 1/2$. The endpoint cases $s = 0, 1$ satisfy $\rho = T = 0$ directly.

For the upper bound, direct differentiation in μ gives

$$\partial_\mu^2 \rho = 6\{(1-s)\mu + s(1-\mu) - |s-\mu|\} = \begin{cases} 12\mu(1-s), & \mu \leq s, \\ 12s(1-\mu), & \mu \geq s. \end{cases}$$

Thus ρ is convex in μ on $[0,1]$, including its common derivative at $\mu = s$, and its maximum is at an endpoint. The endpoint values are exactly $\rho(s,0) = T(1+s)$ and $\rho(s,1) = T(2-s)$, proving (B.22.1).

In the complementary-gap two-coordinate chart, both third-moment slopes have this same T. Their half difference $a = (\rho_1 - \rho_2)/2$ therefore satisfies

$$\boxed{|a| \leq 3T/4.} \tag{B.22.2}$$

B.23 Three representations: contact and variance equations

Unit-span coordinates Translations and a common positive scale put the raw supports and selected raw threshold into the form

$$X_{\mathrm{raw}} = (0, s, 1), \qquad Y_{\mathrm{raw}} = (0, 1-s, 1), \qquad y_{\mathrm{raw}} = 1, \qquad 0 < s < 1. \tag{B.23.1}$$

Let $p = (p_0, p_1, p_2)$ and $q = (q_0, q_1, q_2)$ be the positive probability vectors of the two laws, with means μ, ν. Put $m = p_1$ and $n = q_1$. Write x, y for their centered support vectors, v_i, r_i for their variances and third absolute moments, and $M_1 = \sum_j p_j x_j |x_j|$, $M_2 = \sum_j q_j y_j |y_j|$. Set

$$V = v_1 + v_2, \qquad R = r_1 + r_2, \qquad o = 1 - \mu - \nu, \qquad z = o/\sqrt{V}. \tag{B.23.2}$$

Event and contact equations The event probability is the actual six-cell sum

$$F = 1 - p_1 q_2 - p_2(q_1 + q_2), \qquad D = F - \Phi(z) > 0, \qquad C = DV^{3/2}/R. \tag{B.23.3}$$

Thus require

$$(49/120)^2 R^2 \le D^2 V^3 < (227/500)^2 R^2. \tag{B.23.4}$$

For the two coordinates the actual deletion levels are respectively

$$H_1 = (1, q_0 + q_1, q_0), \qquad H_2 = (1, p_0 + p_1, p_0). \tag{B.23.5}$$

Write $L = 3D + z\phi(z)$. After multiplication by the positive factor $2VR$, the global dual contact equation at a raw centered support point u of coordinate i is

$$\boxed{2VD(|u|^3 - r_i) - LR(u^2 - v_i) - (6VDM_i + 2\sqrt{V}\phi(z)R)u + 2VR(F - H_i(u)) = 0.} \tag{B.23.6}$$

Indeed the unscaled dual price is D/R, its quadratic coefficient is $L/(2V)$, its linear coefficient is $3DM_i/R + \phi(z)/\sqrt{V}$, and averaging the majorant over the coordinate determines its constant. This proves (B.23.6) directly and preserves the actual centered moments.

Variance Hessian For the two centered signed mass vectors, use

$$w_1 = (1 - s, -1, s), \qquad w_2 = (s, -1, 1 - s), \qquad T = s(1 - s).$$

Both raw variance derivatives equal T. Their raw cubic derivatives are $\rho_i = \sum_j w_{i,j}|x_{i,j}|^3$, where $x_{1,j} = x_j$ and $x_{2,j} = y_j$ are the centered raw supports. The actual 321 mixed event coefficient is

$$M_0 = 1 - s + s^2 = 1 - T.$$

Put

$$\begin{aligned} W &= R\{15D + \phi'''(z)\}/4, \\ N_i &= WT^2 - 3DVT\rho_i, \\ H_{12} &= RV^2 M_0 + \frac{3}{2}DVT(\rho_1 + \rho_2) - WT^2. \end{aligned} \tag{B.23.7}$$

The necessary conditions from the original variance Hessian are

$$\boxed{N_1 \ge 0, \quad N_2 \ge 0, \quad N_1 N_2 - H_{12}^2 \ge 0.} \tag{B.23.8}$$

To verify the normalization, start with the independent unnormalized Hessian derived at the end of Section B.31, replace its G by R/D, and multiply all matrix entries by $D > 0$. The raw support scale need not equal K for that calculation. Formula (B.23.7) uses the actual $\phi'''(z) = (3z - z^3)\phi(z)$.

B.24 Three representations: the gap product

Use the unit-span scale of Section B.23.

In the three-representation 321 case, retain the notation

$$T = s(1-s), \quad L = 1 - T, \quad W = \frac{15D + \phi'''(z)}{4}, \quad C = \kappa V^{3/2}.$$

The original variance Hessian has diagonal necessary quantities

$$N_i = WT^2 - 3\kappa VT\rho_i,$$

and its determinant implies

$$N_1 + N_2 \geq V^2 L. \tag{B.24.1}$$

The pointwise third-moment slope bound Section B.22 gives $\rho_1 + \rho_2 \geq T$. Hence

$$V^2 L \leq N_1 + N_2 \leq (2W - 3\kappa V)T^2. \tag{B.24.2}$$

The bound (B.3) gives $W < 79/40$. Also $V \leq 1/2$ and $C \geq 49/120$ yield

$$\kappa V = \frac{C}{\sqrt{V}} > \frac{49\sqrt{2}}{120} > \frac{343}{600},$$

where the final step uses $\sqrt{2} > 7/5$. Thus

$$2W - 3\kappa V < 79/20 - 343/200 = 447/200.$$

Since $L \geq 3/4$, (B.24.2) proves

$$\boxed{T^2 > \frac{50}{149}V^2.} \tag{B.24.3}$$

The unit-span diameter bound is $V > 2401/24964$. Substituting it into (B.24.3) gives

$$T^2 > \frac{144120025}{46428496552} > \frac{1}{324}, \qquad \boxed{s(1-s) > 1/18.} \tag{B.24.4}$$

The exact final slack is $66597887/3760708220712 > 0$, saved in the computational supplement. Equivalently,

$$\frac{3 - \sqrt{7}}{6} < s < \frac{3 + \sqrt{7}}{6}.$$

In particular $1/18 < s < 17/18$.

B.25 Three representations: an upper variance bound

The scale is the unit-span scale of Section B.23.

At an attained violating fixed-two-coordinate maximum with two genuine three-point laws and all three representations of the closed 321 event,

$$\boxed{V < 3/10.} \tag{B.25.1}$$

A scalar lower bound Retain $D > 49/(120\sqrt{2})$ and $-5/4 < z < 3/5$ from Section B.3. Define

$$J = 3D + (z^3 + 3z)\phi(z).$$

Then

$$\boxed{J > -9/50.} \tag{B.25.2}$$

For $z \geq 0$ this is immediate. For $z = -t$, $0 < t < 5/4$, the derivative of $(t^3 + 3t)\phi(t)$ is $(3 - t^4)\phi(t) > 0$. Thus its maximum on this interval is less than $(365/64)\phi(5/4)$. The elementary bound $\phi(0) < 399/1000$, used in Section B.3, and the alternating exponential upper polynomial give

$$\phi(5/4) < \frac{399}{1000}\sum_{j=0}^{6}\frac{(-25/32)^j}{j!} = \frac{9415683872273}{51539607552000} < \frac{183}{1000}.$$

Also $\sqrt{2} < 99/70$, so $3D > 343/396$. Consequently

$$J > \frac{343}{396} - \frac{365}{64}\frac{183}{1000} = -\frac{9}{50} + \frac{631}{253440} > -\frac{9}{50}.$$

Eliminate the cubic slopes from the mass Hessian Write

$$T = s(1-s), \quad L = 1 - T, \quad u = (1-s)\nu, \quad v = s\mu, \quad w = m + n,$$
$$E_0 = \mu(1-\mu) + \nu(1-\nu), \quad V = E_0 - Tw,$$
$$A = \frac{3D + z\phi(z)}{2V}, \quad W = \frac{15D + \phi'''(z)}{4}, \quad \kappa = D/R.$$

Applying the centered mass direction $(1-s, -1, s)$ to the first coordinate's three actual contact equations cancels its constant and linear terms. The deletion levels $(1, q_0 + q_1, q_0)$ give

$$\kappa\rho_1 - AT = (1-s)q_2 - sq_1 = u - Ln =: \eta_1.$$

Similarly, the second coordinate's direction $(s, -1, 1-s)$ gives

$$\kappa\rho_2 - AT = sp_2 - (1-s)p_1 = v - Lm =: \eta_2. \tag{B.25.3}$$

The regularized diagonal Hessian quantities from Section B.23 are therefore

$$H_i = W - 3\kappa V\rho_i/T = -\frac{J}{4} - \frac{3V\eta_i}{T}, \tag{B.25.4}$$

since $W - 3VA = -J/4$. The determinant condition implies $H_1 + H_2 \geq V^2L/T^2$. Set

$$B_0 = LE_0 - T(u+v).$$

As $\eta_1 + \eta_2 = (LV - B_0)/T$, (B.25.4) yields the scalar necessary inequality

$$\boxed{3VB_0 - 4LV^2 - JT^2/2 \geq 0.} \tag{B.25.5}$$

This eliminates both third-moment slopes and their difference from this part of the Hessian.

Maximize the remaining quadratic in the two means For fixed s, independently maximize the two concave quadratics in μ, ν appearing in B_0. This gives

$$\frac{B_0}{L} \le \frac{(L-Ts)^2 + (L-T(1-s))^2}{4L^2} = \frac{(1-2T)(T^2-2T+2)}{4(1-T)^2} =: M(T). \tag{B.25.6}$$

The maximizing means lie in $[0,1]$; the displayed inequality also follows by completing the two squares and does not need this observation. The function M strictly decreases on $0 \le T \le 1/4$. Indeed, with $L = 1-T$, it is

$$M = L/2 - 1/4 + 1/(2L) - 1/(4L^2), \qquad \frac{dM}{dL} = \frac{L^3 - L + 1}{2L^3} > 0$$

on $3/4 \le L \le 1$. Its useful rational value is

$$M(19/100) = \frac{513391}{1312200} = \frac{47}{120} - \frac{277}{656100} < \frac{47}{120}. \tag{B.25.7}$$

Combining (B.25.2), (B.25.5), (B.25.6), and $T^2/L \le 1/12$ gives

$$4V^2 - 3M(T)V - 3/400 < 0. \tag{B.25.8}$$

In particular, whenever $T > 19/100$, (B.25.7) implies

$$f(V) := 4V^2 - (47/40)V - 3/400 < 0. \tag{B.25.9}$$

But $f(3/10) = 0$ exactly and f increases for $V \ge 3/10$.

Two finite bootstrap steps The gap bound is $T^2 > (50/149)V^2$. If $V \ge 1/3$, then

$$T^2 > \frac{50}{1341} > \frac{361}{10000},$$

so (B.25.9) contradicts $V \ge 1/3$. Thus first $V < 1/3$.
It follows that

$$\kappa V = \frac{C}{\sqrt{V}} > \frac{49\sqrt{3}}{120} > \frac{7}{10},$$

using $\sqrt{3} > 12/7$. The original gap-Hessian derivation now improves to

$$V^2 L \le (2W - 3\kappa V)T^2 < \frac{37}{20}T^2, \qquad T^2 > \frac{15}{37}V^2. \tag{B.25.10}$$

If $V \ge 3/10$, (B.25.10) gives

$$T^2 > \frac{27}{740} > \frac{361}{10000}.$$

Hence $T > 19/100$, so (B.25.9) contradicts $V \ge 3/10$. Therefore $V < 3/10$.

As a convenient weaker consequence of (B.25.1), $\kappa V > 37/50$, since $(37/50)^2\,(3/10) < (49/120)^2$. Repeating the same gap inequality gives $T^2 > (75/173)V^2$.

B.26 Three representations: joint mean and variance bounds

Keep the unit-span scale of Section B.23.

In the three-representation 321 case, coupling the two means to the threshold improves the variance bound to

$$\boxed{V < 11/40.} \tag{B.26.1}$$

Couple the quadratic mean bound with the scalar J Retain $T, L, B_0, M(T)$ from Section B.25 and $J = 3D + (z^3 + 3z)\phi(z)$, and write $g = T^2/L$ and $d_{\min} = 49/(120\sqrt{2})$. Set

$$y = z\sqrt{V} = 1 - \mu - \nu, \qquad d = \mu - \nu.$$

Completing the square in d while retaining y gives

$$\frac{B_0}{L} \le N_0(T) + \frac{T}{2L}y - \frac{y^2}{2},$$
$$N_0(T) = \frac{1}{2} - \frac{T}{2L} + \frac{T^2(1-4T)}{8L^2} = M(T) - \frac{T^2}{8L^2} \le M(T). \tag{B.26.2}$$

For $z \le 0$, monotonicity of $(t^2+3)\phi(t)$ for $t \ge 0$ gives

$$(z^3 + 3z)\phi(z) \ge (6/5)z, \qquad J > 3d_{\min} + (6/5)z. \tag{B.26.3}$$

Indeed the derivative of $(t^2+3)\phi(t)$ is $-t(t^2+1)\phi(t) \le 0$, and its value at zero is $3\phi(0) < 6/5$. In the actual contact/Hessian inequality

$$4V \le 3B_0/L - Jg/(2V),$$

substitute (B.26.2)–(B.26.3). The coefficient of $y \le 0$ is

$$\frac{3T}{2L} - \frac{3g}{5V^{3/2}} = \frac{3T}{2L}\left(1 - \frac{2T}{5V^{3/2}}\right) > 0$$

whenever $V \ge 1/4$, because then $T/V^{3/2} \le 2$. Hence that linear term and $-3y^2/2$ are nonpositive. It follows that

$$4V^2 - 3M(T)V + \frac{3d_{\min}}{2}g < 0. \tag{B.26.4}$$

For $z \ge 0$, the same inequality follows directly from $J \ge 3D > 3d_{\min}$ and $B_0/L \le M(T)$. Thus (B.26.4) holds for every candidate with $V \ge 1/4$. In particular, since $3d_{\min}/2 > 3/7$,

$$4V^2 - 3M(T)V + (3/7)g < 0. \tag{B.26.5}$$

Two rational bootstrap steps The already proved $V < 3/10$ gives $\kappa V > 37/50$. Therefore the gap-Hessian inequality gives

$$g > \frac{100}{173}V^2. \tag{B.26.6}$$

If $V \ge 7/25$, (B.26.6) implies $g > 196/4325 > 361/8100 = g(19/100)$. Since $g(T)$ increases, $T > 19/100$, and Section B.25 gives $M(T) < 47/120$. Equation (B.26.5) would then force

$$V < \frac{47/40}{4 + 300/1211} = \frac{56917}{205760} < \frac{7}{25}.$$

This contradiction proves $V < 7/25$. Consequently $\kappa V > 77/100$, so

$$g > \frac{25}{41} V^2. \tag{B.26.7}$$

If $V \geq 11/40$, (B.26.7) gives

$$g > \frac{121}{2624} > \frac{576}{12625} = g(24/125).$$

Thus $T > 24/125$, and exact evaluation gives

$$M(24/125) = \frac{994301}{2550250} = \frac{39}{100} - \frac{593}{5100500} < \frac{39}{100}.$$

Equation (B.26.5) now forces

$$V < \frac{117/100}{4 + 75/287} = \frac{33579}{122300} < \frac{11}{40},$$

again a contradiction. This proves (B.26.1).

B.27 Three representations: scalar exclusion

Keep the unit-span scale of Section B.23.

Use the notation $V, z, T = s(1-s), L = 1 - T, m, n, w = m + n$, and $v_0 = 2401/24964$. The range is

$$v_0 < V < 11/40, \qquad -5/4 < z < 3/5. \tag{B.27.1}$$

The actual bounded sum and its exact slack The raw sum U is in $[0, 2]$, has mean $1 - o$, variance V, and the selected event is $U \leq 1$, where $o = z\sqrt{V}$. Pointwise $U(U-1) \leq 2\mathbf{1}_{\{U>1\}}$. Therefore

$$F \leq 1 - \{V - o + o^2\}/2. \tag{B.27.2}$$

The precise nonnegative slack, using the actual independent 321 event, is

$$S := 2(1-F) - (V - o + o^2) = T(mq_0 + p_0 n) + (2 - s - s^2)mq_2 + (3s - s^2)p_2 n.$$

Both outside coefficients are at least T, so

$$S \geq T\{m(1-n) + n(1-m)\} \geq T\{w - w^2/2\}. \tag{B.27.3}$$

Let $H_\Sigma = H_1 + H_2$, with H_i defined in (B.25.4). The mass-Hessian condition gives

$$H_\Sigma \geq \frac{V^2 L}{T^2} \geq 12V^2, \qquad T = s(1-s) \leq \frac{1}{4}, \quad L = 1 - T.$$

The contact identities yield

$$\eta_{\text{sum}} = u + v - Lw = -\frac{T(H_\Sigma + J/2)}{3V}.$$

Since $J > -9/50$ and $12v_0^2 > 9/100$, this is negative. Therefore

$$w > \frac{u+v}{L}. \tag{B.27.4}$$

If $T > 1/5$, both s and $1-s$ belong to $(1/4, 3/4)$ and $L < 4/5$. Thus (B.27.4) gives $w > 5(\mu+\nu)/16$. The range (B.27.1) gives $\mu+\nu = 1 - z\sqrt{V} > 137/200$, using $\sqrt{11/40} < 21/40$. Hence

$$w > \frac{137}{640} =: a.$$

Let X_1, X_2 denote the raw marginals, and set $a_1 = s$, $a_2 = 1-s$, $m_1 = m$, $m_2 = n$. Since $a_i \in (1/4, 3/4)$, the raw marginal variances satisfy

$$v_i \le \mathbb{E}(X_i - a_i)^2 < 9(1-m_i)/16.$$

Summing gives

$$w \le 2 - 16V/9 < 2 - 16v_0/9 =: b.$$

The concave function $f(w) = w - w^2/2$ has its minimum on $[a, b]$ at an endpoint. Both exact endpoint values exceed $3/20$. Consequently

$$\boxed{T > 1/5 \quad \Longrightarrow \quad S > 3/100.} \tag{B.27.5}$$

Combine with the mass Hessian The mass Hessian and $\rho_1 + \rho_2 \ge T$ give

$$V^2 \le \left\{ \frac{15D + \phi'''(z)}{2} - \frac{3C}{\sqrt{V}} \right\} \frac{T^2}{L}. \tag{B.27.6}$$

If $T \le 1/5$, then $T^2/L \le 1/20$, so $15D + \phi'''(z) \ge 40V^2 + 6C/\sqrt{V}$. If $T > 1/5$, use $T^2/L \le 1/12$ instead, giving $15D + \phi'''(z) \ge 24V^2 + 6C/\sqrt{V}$, together with (B.27.5).

The two cases are therefore enclosed by $(A, S_0) = (40, 0)$ and $(A, S_0) = (24, 3/100)$. Put $c_0 = 49/120$ and

$$B(V, z; A, S_0) = 15(1-\Phi(z)) - \frac{15}{2}V(1+z^2) + \frac{15}{2}z\sqrt{V} - \frac{15}{2}S_0 + (3z - z^3)\phi(z) - AV^2 - \frac{6c_0}{\sqrt{V}}. \tag{B.27.7}$$

The plain probability bound also gives

$$P(V, z; A) = 15(1 - \Phi(z)) + (3z - z^3)\phi(z) - AV^2 - 6c_0/\sqrt{V}.$$

When $z < 0$, Cantelli gives the additional bound

$$Q(V, z; A) = 15\{(1+z^2)^{-1} - \Phi(z)\} + (3z - z^3)\phi(z) - AV^2 - 6c_0/\sqrt{V}. \tag{B.27.8}$$

Every genuine candidate requires each applicable upper expression to be strictly positive, including when $C = c_0$. The proper selected downset has probability less than one. The bounded-sum comparison has positive slack at an interior support point; when the prescribed slack is positive, (B.27.5) gives a strict improvement over it. For negative thresholds, equality in Cantelli's inequality would force a two-point sum law, whereas the sum of two genuine triples has at least five support points. At $z = 0$ the Cantelli expression equals the probability expression. Thus a nonpositive enclosure excludes the candidate also at $C = c_0$. It suffices to cover (B.27.1), separately for both (A, S_0) cases, by boxes on which at least one of B, P, or Q (when the whole box has $z \le 0$) is nonpositive.

Whole-box derivative enclosures For (B.27.7), the exact first derivatives are

$$B_V = -\frac{15}{2}(1+z^2) + \frac{15z}{4\sqrt{V}} - 2AV + 3c_0/V^{3/2},$$

$$B_z = (z^4 - 6z^2 - 12)\phi(z) - 15Vz + \frac{15}{2}\sqrt{V}.$$

For P, Q, their V derivative is $-2AV + 3c_0/V^{3/2}$; their z derivatives are respectively $(z^4 - 6z^2 - 12)\phi(z)$ and that quantity minus $30z/(1+z^2)^2$. These formulas apply throughout the original rectangle, independently of candidate constraints. A midpoint value plus these whole-box gradient enclosures gives a valid enclosure by the mean value integral. The numerical domain has $V \geq v_0 > 0$. The fixed certificate `three_owner321` covers both cases in (B.27.1) and proves that every point has a nonpositive applicable exclusion expression. Therefore no maximizing pair can have all three representations of the 321 threshold. Section 8 gives the enclosure and partition rules.

B.28 Two representations: contact geometry

Use the contact scale and the strict bounds in (6.1): $K = R_{\text{normalized}} + z\phi(z)/(3C)$, scaled support coordinates equal the normalized coordinates divided by K, and $\tau = CK^3$. Raw moments in this calculation refer to those scaled supports. Thus

$$\frac{1}{20} < \tau < 2, \qquad \frac{25}{64} < V < 4, \qquad C = \tau V^{3/2}. \tag{B.28.1}$$

The two coordinate variance fractions exceed 13/200. Define the piecewise cubic function

$$P_\beta(x) = |x|^3 - \frac{3}{2}x^2 - 3\beta x.$$

Every global majorant in this scale is a constant plus τP_β. A nonselected positive contact b has $\beta = b(b-1)$; a nonselected negative contact $-a$ has $\beta = a(1-a)$.

Adjacent representations Write the supports and the selected threshold as

$$X = (-t, c, b), \quad Y = (-a, d, e), \quad o = e - t = c + d, \qquad d = e - t - c. \tag{B.28.2}$$

Here the representations are $(0,2), (1,1)$. The unused maximal cell $(2,0)$ is strictly lower: $b - a < o$. The high point b and the low point $-a$ are nonselected smooth minima, so $b > 1/2$ and $a > 1/2$. In fact $b \geq 1$. If $1/2 < b < 1$, then $\beta = b(b-1) < 0$ and P_β has a negative minimum at $-a_*$ with $a_* > 1$. Its value is less than $-1/2$, whereas $P_\beta(b) = b^2(3/2 - 2b) > -1/2$. The deletion CDF at $-a_*$ is at least its value at b; global majorization therefore rules out this lower minimum.

Also $a < 1$. Otherwise $-a$ would be a global minimum of $P_{a(1-a)}$. Indeed, for $x \geq 0$,

$$P_{a(1-a)}(x) = x^3 - \frac{3}{2}x^2 + 3a(a-1)x \geq -\frac{1}{2} \geq P_{a(1-a)}(-a),$$

and for $x = -t \leq 0$ the difference from the value at $-a$ is $(t-a)^2(t + 2a - 3/2) \geq 0$. The contact level at $-a$ is one, so a global minimum there would preclude the smaller contact levels at the other support points.

The nonpositive derivative at the selected high point gives $e \leq B(a)$, where B is defined below. The unused-corner inequality now gives

$$t + b < a + e \leq a + B(a) \leq 2.$$

Since $b \geq 1$, we have $t < 1$. The nonpositive derivative at $-t$ gives $b(b-1) \geq t(1-t)$, hence $b \geq B(t)$. The function $x + B(x)$ is strictly increasing on $(0,1)$, because its derivative is $1 + (1-2x)/\sqrt{1+4x(1-x)} > 0$. Thus $t \geq a$ would contradict the strict span inequality. Finally $e > t + b - a > t$, so $o = e - t > 0$. We have proved

$$\frac{1}{2} < a < 1, \quad 0 < t < a, \quad B(t) \leq b < 2 - t, \quad 0 < e \leq B(a), \quad -t < c < b,$$

$$B(v) = \frac{1 + \sqrt{1 + 4v(1-v)}}{2}, \qquad A_2 := a + e > A_1 := t + b, \qquad o > 0. \tag{B.28.3}$$

The strict span ordering also implies $-a < d < e$; it need not be imposed by a separate chart. Set $h = t + c = e - d > 0$, $v = b - c > 0$.

Endpoint representations Now the representations are $(0,2), (2,0)$ and the unused middle-middle cell lies strictly below their common threshold. Both nonselected middle contacts are negative smooth minima. A positive smooth minimum would force a positive derivative at the larger contact, contrary to Lemma 6.2; zero cannot be a minimum because the quadratic coefficient is negative. Write

$$X = (-t_1, -a_1, b_1), \qquad Y = (-t_2, -a_2, b_2),$$

$$\frac{1}{2} < a_i < 1, \quad a_i < t_i < 4, \quad 0 < b_i \leq B(a_i), \quad A = t_1 + b_1 = t_2 + b_2,$$

$$o = b_2 - t_1 = b_1 - t_2 > -a_1 - a_2. \tag{B.28.4}$$

Variance Hessian For either threshold geometry, take arbitrary positive mass vectors (ℓ_i, m_i, r_i) on its centered supports. Let v_i and R_i be their variances and third absolute moments, and put

$$V = v_1 + v_2, \qquad R = R_1 + R_2, \qquad z = o/\sqrt{V}, \qquad F = 1 - m_1 r_2 - r_1(m_2 + r_2).$$

At the attained ratio, $\tau = C/V^{3/2}$ and $D = F - \Phi(z) = \tau R$.

The full variance Hessian can be evaluated in the original signed-mass basis. For each support vector (x_0, x_1, x_2), set

$$w_i = \left(\frac{x_2 - x_1}{x_2 - x_0}, -1, \frac{x_1 - x_0}{x_2 - x_0}\right), \quad T_i = (x_1 - x_0)(x_2 - x_1), \quad \rho_i = \sum_j w_{i,j} |x_j|^3.$$

Let $M_0 = \sum_{j+k \leq 2} w_{1,j} w_{2,k}$, the actual 321 mixed event coefficient, and put

$$W = \{15\tau R + \phi'''(z)\}/4, \quad N_i = W T_i^2 - 3\tau V T_i \rho_i,$$

$$Q = V^2 M_0 + \frac{3}{2}\tau V (T_1 \rho_2 + T_2 \rho_1) - W T_1 T_2.$$

Necessary second-order conditions are

$$N_1 \geq 0, \qquad N_2 \geq 0, \qquad N_1 N_2 - Q^2 \geq 0. \tag{B.28.5}$$

These are the already derived original Hessian entries multiplied by the positive scalar τ, with the actual normal derivative $\phi'''(z) = (3z - z^3)\phi(z)$.

B.29 Adjacent representations: scalar exclusion

Normalize by the larger span Translate each support to start at zero and divide by the second span. The supports and selected threshold become

$$X \in \{0, a, r\}, \quad Y \in \{0, 1-a, 1\}, \quad 0 < a < r < 1, \quad X + Y \le 1. \tag{B.29.1}$$

Let V denote total variance in this raw scale, z the standardized selected threshold, $D = F - \Phi(z)$, and $C = DV^{3/2}/R$. The uniform diameter and threshold bounds give

$$v_0 := 2401/24964 < V \le (1 + r^2)/4 \le 1/2, \qquad 0 < z < 3/5. \tag{B.29.2}$$

Put $s = a/r$, $T = s(1-s)$, and use the centered mass directions $(1-s, -1, s)$ and $(a, -1, 1-a)$. Their raw variance derivatives are $T_1 = a(r-a)$ and $T_2 = a(1-a)$. Summation over the actual 321 cells gives the mixed event derivative

$$M = 1 - a + a^2/r = 1 - rT. \tag{B.29.3}$$

The cubic third-moment derivatives divided by T_i are the centered second divided differences γ_i. The elementary divided-difference diameter bound gives $\gamma_1 \ge r/2$ and $\gamma_2 \ge 1/2$.

Retain the dependence between the mixed coefficient and span Write $W = (15D + \phi'''(z))/4$ and $H_i = W - 3C\gamma_i/\sqrt{V}$. The mass-Hessian condition (B.28.5), divided by the positive variance-direction factors, implies

$$H_1 + H_2 \ge V^2\Lambda, \qquad \Lambda = M/(T_1T_2). \tag{B.29.4}$$

This follows directly from its determinant: the off-diagonal entry is $V^2\Lambda - (H_1 + H_2)/2$, whose absolute value is at most $\sqrt{H_1H_2} \le (H_1 + H_2)/2$.

Now $a(1-a) \le 1/4$, and $T_1\,T_2 = r^2T\,a(1-a)$. Therefore

$$\Lambda \ge \frac{4(1-rT)}{r^2T} \ge \frac{16 - 4r}{r^2} \ge \frac{12}{r^2}. \tag{B.29.5}$$

Combining (B.29.4), (B.29.5), and the two divided-difference bounds proves

$$\boxed{15D + \phi'''(z) \ge \frac{32 - 8r}{r^2}V^2 + \frac{3C(1+r)}{\sqrt{V}}.} \tag{B.29.6}$$

The universal bound $15D + \phi'''(z) < 79/10$ also gives $V^2\Lambda < 79/20$. Hence $r^2 > (240/79)V^2 > (240/79)v_0^2 > 1/36$, proving the convenient strict bound $r > 1/6$. The final comparison is exact rational arithmetic.

The bounded sum supplies the scalar upper bounds Set $U = X + Y$. It belongs to $[0, 1+r]$, has mean $1 - z\sqrt{V}$, and its selected event is $U \le 1$. Pointwise,

$$U(U-1) \le r(1+r)\mathbf{1}_{\{U>1\}}.$$

Consequently

$$F \le 1 - \frac{V(1+z^2) - z\sqrt{V}}{r(1+r)}, \qquad F \le 1. \tag{B.29.7}$$

Put $c_0 = 49/120$, $A(r) = 32/r^2 - 8/r$, and

$$P(r,V,z) = 15(1-\Phi(z)) + (3z - z^3)\phi(z) - A(r)V^2 - \frac{3c_0(1+r)}{\sqrt{V}},$$

$$B(r,V,z) = P(r,V,z) - \frac{15\{V(1+z^2) - z\sqrt{V}\}}{r(1+r)}. \tag{B.29.8}$$

Every genuine candidate requires both $P > 0$ and $B > 0$, including when $C = c_0$. The selected downset is proper, so $F < 1$. The bounded-sum comparison is also strict: the sum has an interior support point at which its pointwise quadratic bound has positive slack. Together with (B.29.6), these strict comparisons show that a nonpositive enclosure excludes the candidate at the equality floor too. It suffices to cover the closed box

$$[1/6, 1] \times [v_0, 1/2] \times [0, 3/5] \tag{B.29.9}$$

by boxes on each of which one expression in (B.29.8) is nonpositive.

Whole-box derivatives for a fixed outward cover Let $N = V(1+z^2) - z\sqrt{V}$. On the whole domain (B.29.9), the exact derivatives are

$$P_r = (64/r^3 - 8/r^2)V^2 - 3c_0/\sqrt{V},$$

$$P_V = -2A(r)V + \frac{3c_0(1+r)}{2V^{3/2}}, \qquad P_z = (z^4 - 6z^2 - 12)\phi(z),$$

$$B_r = P_r + \frac{15N(1+2r)}{r^2(1+r)^2},$$

$$B_V = P_V - \frac{15\{1 + z^2 - z/(2\sqrt{V})\}}{r(1+r)}, \qquad B_z = P_z - \frac{15(2Vz - \sqrt{V})}{r(1+r)}.$$

Thus midpoint values plus whole-box derivative enclosures give a valid mean-value enclosure. Both denominators r and V are uniformly positive. The fixed records `adjacent_owner321` and `adjacent_owner321_completion` cover (B.29.9); at every point they prove $P \le 0$ or $B \le 0$. Thus the adjacent-representation case has no maximizing pair. Section 8 gives the enclosure and partition rules, including the subdivision of the four cells treated by the completion record.

B.30 Endpoint representations: scalar exclusion

Geometry, variance directions, and the first scalar inequality Translate both supports to start at zero and divide by their common span:

$$X \in \{0, a, 1\}, \quad Y \in \{0, b, 1\}, \quad a, b > 0, \quad s := a + b < 1, \qquad U = X + Y \le 1. \tag{B.30.1}$$

Let the respective means be μ, ν, middle masses m, n, and total raw variance and sum of third absolute moments be V, R. Write

$$D = F - \Phi(z), \quad C = DV^{3/2}/R, \quad T_1 = a(1-a), \quad T_2 = b(1-b), \quad M = a+b-ab, \quad g = T_1T_2/M.$$

The diameter and threshold results give

$$v_0 := 2401/24964 < V \le 1/2, \qquad -5/4 < z < 3/5, \qquad D > \frac{49}{120\sqrt{2}}, \quad C \ge c_0 := 49/120. \tag{B.30.2}$$

The actual mixed derivative of the 321 event in the centered mass directions $(1-a,-1,a)$ and $(1-b,-1,b)$ is M. The original mass Hessian, as in the two-representation reduction, implies

$$15D + \phi'''(z) \ge \frac{2V^2}{g} + \frac{6C}{\sqrt{V}}, \tag{B.30.3}$$

using the lower bound $1/2$ for each centered cubic divided difference.

We will use the uniform bound

$$\boxed{g < 1/11.} \tag{B.30.4}$$

Indeed put $p = ab$, $d = (1-a)(1-b) = 1-s+p$; then $g = pd/(s-p)$. For fixed s, its derivative in p has positive numerator $s(1-s+2p)-p^2$. Thus its maximum occurs at $p = s^2/4$, or $a = b = x$. The resulting function is $x(1-x)^2/(2-x)$, $0 < x \le 1/2$, whose derivative has the sign of x^2-3x+1. Its maximum is $(5\sqrt{5}-11)/2 < 1/11$; the last comparison follows from $\sqrt{5} < 123/55$, since $123^2 - 5\,55^2 = 4 > 0$.

At moderately negative thresholds the actual contacts give mass bounds For $z \ge -3/4$, define $J = 3D + (z^3+3z)\phi(z)$. Then

$$J > \frac{343}{396} - \frac{171}{64}\frac{31}{100} = \frac{24001}{633600} > 0. \tag{B.30.5}$$

Here $\sqrt{2} < 99/70$, and $\phi(3/4) < 31/100$. The latter follows from $\phi(0) < 399/1000$ and the degree-four upper Taylor polynomial for $\exp(-9/32)$. On $[0,3/4]$, both $t(t^2+3)\phi(t)$ and $t(t^2+2)\phi(t)$ increase; their derivatives are respectively $(3-t^4)\phi(t)$ and $(2+t^2-t^4)\phi(t)$.

Let $\kappa = D/R$, $A = (3D + z\phi(z))/(2V)$, and $W = (15D + \phi'''(z))/4$. Applying each centered mass direction to its three actual contact equations gives

$$\eta_1 := \kappa\rho_1 - AT_1 = (1-a)\nu - Mn, \qquad \eta_2 := \kappa\rho_2 - AT_2 = (1-b)\mu - Mm. \tag{B.30.6}$$

The diagonal Hessian quantities satisfy $H_i = W - 3\kappa V\rho_i/T_i = -J/4 - 3V\eta_i/T_i \ge 0$. Thus (B.30.5) gives $\eta_i < 0$. Both middle contacts are negative after centering, so $\mu > a$ and $\nu > b$. Consequently

$$\boxed{m > \frac{a(1-b)}{M}, \qquad n > \frac{b(1-a)}{M}.} \tag{B.30.7}$$

Define the cubic contact scale in the present unit-span coordinates by

$$k = \frac{\sqrt{V}\{D + z\phi(z)/3\}}{C}. \tag{B.30.8}$$

A nonselected negative minimum of the cubic majorant has distance strictly larger than $k/2$ from zero: its second derivative is positive, and the quadratic coefficient is $-3k/2$ relative to the cubic coefficient. Hence $\mu - a > k/2$ and $\nu - b > k/2$. Positivity of the low masses now gives

$$\boxed{1-m > \frac{k}{2(1-a)}, \qquad 1-n > \frac{k}{2(1-b)}.} \tag{B.30.9}$$

A uniform positive lower bound on this raw contact scale Combining (B.30.3), (B.30.4), and (B.30.8) yields

$$k \geq \frac{2}{5} + \frac{\sqrt{V}}{15C}\{22V^2 + (z^3 + 2z)\phi(z)\}. \tag{B.30.10}$$

For $z \geq 0$, the braces are positive. For $-3/4 \leq z \leq 0$, their value exceeds $1/5 - (123/64)(31/100) = -2533/6400$, because $22v_0^2 > 1/5$. Moreover $\sqrt{V}/(15C) < (5/7)/(49/8) = 40/343$. In either sign case,

$$\boxed{k > \frac{7}{20}.} \tag{B.30.11}$$

The exact margin in the negative sign case is $2/5 - (40/343)(2533/6400) - 7/20 = 211/54880 > 0$.

Exact event slack, retaining the independent probabilities Let p_0, m, p_2 and q_0, n, q_2 be the actual probabilities. The nonnegative bounded-sum slack is

$$\begin{aligned} S &:= 2(1-F) - \{V - z\sqrt{V} + z^2 V\} \\ &= T_1 m q_0 + T_2 p_0 n + s(1-s)mn + (1-a)(2+a)mq_2 + (1-b)(2+b)p_2 n. \end{aligned} \tag{B.30.12}$$

Since $\mu > a$, $\nu > b$, the high masses satisfy $p_2 > a(1-m)$ and $q_2 > b(1-n)$. After grouping each low/high pair in (B.30.12),

$$S > A_0 m(1-n) + B_0 n(1-m) + s(1-s)mn,$$

where $A_0 = (1-a)(a+2b)$ and $B_0 = (1-b)(b+2a)$. Inserting (B.30.7), (B.30.9) into the first two positive terms proves

$$S > \frac{k}{2M}\{aA_0 + bB_0\} = \frac{k}{2M}\{s^2(1-s) + p(s+2)\}. \tag{B.30.13}$$

The expression in braces is at least $4pd$. To see this, subtract $4pd$: $s^2(1-s) + p(5s-2) - 4p^2$ is concave in $p \in [0, s^2/4]$. Its endpoint values are $s^2(1-s) \geq 0$ and $s^2(2+s-s^2)/4 > 0$. Using $g = pd/M$, (B.30.13) therefore yields the simple exact inequality

$$\boxed{S > 2kg \qquad (z \geq -3/4).} \tag{B.30.14}$$

In particular, (B.30.11) proves

$$\boxed{g > 1/20,\ z \geq -3/4 \quad \Longrightarrow \quad S > 7/200.} \tag{B.30.15}$$

Three fixed scalar rectangles suffice The expressions P, B, Q below are upper bounds obtained respectively from $F \leq 1$, the bounded sum with slack S_0, and Cantelli for $z \leq 0$:

$$\begin{aligned} P &= 15(1 - \Phi(z)) + (3z - z^3)\phi(z) - A_0 V^2 - 6c_0/\sqrt{V}, \\ B &= P - \frac{15}{2}\{V(1+z^2) - z\sqrt{V} + S_0\}, \qquad Q = P + 15\{(1+z^2)^{-1} - 1\}. \end{aligned} \tag{B.30.16}$$

Every candidate requires each applicable expression to be strictly positive, including at $C = c_0$, by the strict probability and bounded-sum comparisons used in Section B.27. For the positive prescribed slack use (B.30.15); for negative thresholds Cantelli equality is impossible for a sum with at least five support points. The following three closed rectangles cover all possibilities:

Case	Coefficient A_0	Slack S_0	z interval
$g \le 1/20$	40	0	$[-5/4, 3/5]$
$g > 1/20$, negative tail	22	0	$[-5/4, -3/4]$
$g > 1/20$, remaining thresholds	22	7/200	$[-3/4, 3/5]$

Every row has the same V interval $[2401/24964, 1/2]$. The first row uses (B.30.3) directly; the others use (B.30.4), and the last also uses (B.30.15). Closed shared faces do not omit any genuine candidate.

The whole-box derivatives are the ones in Section B.27, with the displayed coefficient and slack. They hold independently of all candidate restrictions throughout these rectangles. The fixed certificate `endpoint_owner321` covers all three rectangles and proves that every point has a nonpositive applicable expression in (B.30.16). Hence the endpoint-representation case has no maximizing pair. Section 8 gives the enclosure and partition rules.

B.31 Two triples: the collided asymmetric event

$$K = R + z\phi(z)/(3C), \qquad \tau = CK^3, \qquad \sigma = \phi(z)/(CK^2).$$

The elementary fixed-two-coordinate bounds give $1/2 < K < 8/5$, $1/20 < \tau < 2$, and $49/120 \le C < 227/500$. All support quantities below are divided by K.

Supports, actual levels, and exact compact charts The maximal included cells are $(1,1)$ and $(2,0)$. A genuine selected collision therefore has these two representations. Write

$$T_1 = (-a, c, b), \qquad T_2 = (-t, d, e), \qquad h = b - c = t + d > 0,$$
$$u = a + c > 0, \qquad A_1 = a + b = u + h, \qquad v = e - d > 0, \qquad A_2 = t + e = h + v. \tag{B.31.1}$$

The unused corner $(0,2)$ is excluded, giving the strict geometric condition

$$A_2 > A_1, \qquad \text{equivalently } v > u. \tag{B.31.2}$$

Denote low/middle/high probabilities by ℓ_i, m_i, r_i, and put $q_i = \ell_i + m_i = 1 - r_i$. The first triple's deletion levels are (q_2, q_2, ℓ_2); the second triple's levels are $(1, q_1, 0)$. The first low support and the second high support are nonselected. They are smooth minima of their global cubic majorants.

In the first triple the negative minimum $-a$ and the selected middle contact c have equal height and nonpositive derivatives. The complete equal-height charts from the contact formulas above are

	$a(k)$	$c(k)$
negative	$\frac{1}{2} + \frac{k}{4}$	$\frac{k-1}{2}$
positive	$\frac{3(1+k)^2}{2(k^3+3k+2)}$	$ka(k)$

$0 \le k \le 1.$ (B.31.3)

The next positive minimum is

$$B(a) = \frac{1 + \sqrt{1 + 4a(1-a)}}{2}.$$

The selected high contact satisfies $0 < b \le B(a)$. Set $b = wB(a)$, $0 \le w \le 1$, retaining $b > c$ as a strict necessary condition. The genuine level drop below the negative minimum ensures $a < 1$; including the boundary $a = 1$ in (B.31.3) is harmless because the level drop is tested separately.

For the second triple its nonselected high contact has level zero. Its scaled cubic majorant is exactly

$$g_e(x) = |x|^3 - \frac{3}{2}x^2 - 3e(e-1)x + B_0, \qquad B_0 = e^2(2e - 3/2). \tag{B.31.4}$$

Necessarily $e \geq 1$. Indeed a smooth positive minimum has $e > 1/2$; if $1/2 < e < 1$, the corresponding negative minimum has smaller potential value. That contradicts global majorization of a nonnegative deletion CDF when $g_e(e) = 0$. More generally the same argument applies to a nonselected positive minimum at any deletion level, since the deletion CDF is nonincreasing.

At the low contact set

$$G = g_e(-t) = t^3 - \frac{3}{2}t^2 + 3e(e-1)t + B_0 = 1/\tau. \tag{B.31.5}$$

Thus $1/2 < G < 20$. It follows that $1 \leq e < 5/2$ and $0 < t < 7/2$. For $e \geq 5/2$, the expression in (B.31.5) increases with $t \geq 0$ and is at least $B_0 \geq 175/8 > 20$. For $t \geq 7/2$, it is at least $t^3 - (3/2)t^2 + 1/2 \geq 25 > 20$.

Use $e = 1 + (3/2)E$, $t = (7/2)T$, and $d = h - t$. Together with (B.31.3), the two parameter cubes are $(k, w, E, T, m_1) \in [0,1]^5$.

Contact values determine the remaining masses The first triple's scaled level difference is

$$U = a^2(3/2 - 2a) - b^3 + \frac{3}{2}b^2 + 3a(1-a)b > 0. \tag{B.31.6}$$

Actual deletion levels give

$$\boxed{m_2 = U/G.} \tag{B.31.7}$$

Centering determines all remaining probabilities:

$$\ell_1 = \frac{b - m_1 h}{A_1}, \quad r_1 = \frac{a - m_1 u}{A_1}, \qquad \ell_2 = \frac{e - m_2 v}{A_2}, \quad r_2 = \frac{t - m_2 h}{A_2}. \tag{B.31.8}$$

Require all six masses strictly positive. The second middle contact supplies the exact equation

$$\boxed{(1 - r_1)G - g_e(d) = 0.} \tag{B.31.9}$$

In the open region $u > 0$, equation (B.31.9) can eliminate the last free mass: $m_1 = (A_1 g_e(d)/G - b)/u$. The five-parameter box retains m_1 and (B.31.9), avoiding a division whose denominator vanishes on the coalescing-support face. This is a numerical choice, not an extra mathematical degree of freedom in an actual solution.

Collision-preserving gradients and actual moments Define the nonnegative contact slopes

$$\begin{aligned} H_c &= 3[a(1-a) + c - c|c|], \quad H_b = 3[a(1-a) + b - b^2], \\ H_t &= 3[t^2 - t + e(e-1)], \quad H_d = 3[e(e-1) + d - d|d|]. \end{aligned} \tag{B.31.10}$$

Moving the two support points in either selected representation by opposite amounts, with the usual recentering of the two marginals, preserves both selected raw sums. These two admissible variations give

$$\boxed{m_1H_c = m_2H_d, \qquad r_1H_b = \ell_2H_t.} \tag{B.31.11}$$

Consequently

$$\sigma = m_1H_c + r_1H_b = \ell_2H_t + m_2H_d > 0. \tag{B.31.12}$$

The scaled variances and third absolute moments are

$$\begin{gathered} V_1 = ab - m_1uh, \quad V_2 = te - m_2hv, \quad V = V_1 + V_2, \\ R_1 = \ell_1a^3 + m_1|c|^3 + r_1b^3, \quad R_2 = \ell_2t^3 + m_2|d|^3 + r_2e^3, \quad R_0 = R_1 + R_2. \end{gathered} \tag{B.31.13}$$

The selected scaled sum is $o = b - t = c + d$. Thus

$$\begin{gathered} K = V^{-1/2}, \quad z = o/\sqrt{V}, \quad C = V^{3/2}/G, \\ \boxed{3(V - R_0) - \sigma o = 0, \qquad \sigma\sqrt{V} - G\phi(o/\sqrt{V}) = 0.} \end{gathered} \tag{B.31.14}$$

The actual closed CDF is

$$F = q_2 - m_2r_1.$$

Its exact normal equation and a useful necessary positive expression are

$$\boxed{G[F - \Phi(o/\sqrt{V})] - R_0 = 0,} \tag{B.31.15}$$

$$\boxed{B_0 - \tfrac{3}{2}V_2 - R_1 > 0.} \tag{B.31.16}$$

Equation (B.31.16) follows by averaging the second dual (B.31.4), giving $GF = R_2 - (3/2)V_2 + B_0$, and using $\Phi(z) = F - R_0/G > 0$.

The fixed-two-coordinate and dominant-coordinate results impose

$$\begin{gathered} \frac{25}{64} < V < 4, \qquad V_i > \frac{13}{200}V \quad (i = 1, 2), \\ (49/120)^2G^2 \le V^3 < (227/500)^2G^2. \end{gathered} \tag{B.31.17}$$

The unconditional two-triple mixture-window result additionally requires

$$3(W_1 + W_2) > V, \tag{B.31.18}$$

where the actual canonical variance widths are $W_1 = \min(bu, ah)$ and $W_2 = \min(eh, tv)$. Equivalently, $W_1 = bu$ for $c \le 0$ and $W_1 = ah$ for $c \ge 0$, while $W_2 = (eh + tv - |d|A_2)/2$. The adjacent-gap products uh, hv only give upper bounds for these widths; they are not the canonical widths.

The full actual variance Hessian, with cleared denominators For the centered unit-variance directions of the two triples, the actual CDF first and mixed derivatives are

$$L_1 = -\frac{m_2 V}{hA_1}, \qquad L_2 = \frac{(r_1 A_2 - h)V}{hvA_2}, \qquad g = \frac{V^2}{A_1 h^2 v} > 0. \tag{B.31.19}$$

These follow directly from the deletion levels and signed weights in Proposition 6.4. In particular the mixed coefficient is that of the actual 221 event.

Use centered mass parameters whose individual variations have raw total-variance derivative V; these are unit-variance directions after standardization. Write

$$J = F - \Phi(o/\sqrt{V}) - \frac{CR}{V^{3/2}},$$

where C is fixed, the raw quantities F, V, R vary, and the raw threshold o stays fixed. At the attained maximum, both first derivatives vanish and the Hessian is negative semidefinite. If γ_i is the scaled second divided difference of $|x|^3$, put

$$B_i = \frac{15R_0 + G\phi'''(z)}{4} - 3V\gamma_i = -GJ_{ii}.$$

The full condition is

$$B_1 \geq 0, \quad B_2 \geq 0, \quad B_1 B_2 - \left[Gg - \frac{B_1 + B_2}{2}\right]^2 \geq 0. \tag{B.31.20}$$

Eliminating both divided differences and the normal derivative gives polynomial necessary inequalities. The variance identity in Section B.14 is

$$4J_{ii} = 12L_i + 3D + (z^3 + 3z)\phi(z), \qquad D = \frac{R_0}{G} = \frac{CR_0}{V^{3/2}}.$$

For completeness, the raw third-moment derivative in this direction is $V\gamma_i$. The first derivative gives $C\gamma_i/\sqrt{V} = L_i + 3D/2 + z\phi(z)/2$, while the second gives $J_{ii} = 3C\gamma_i/\sqrt{V} - (15D + \phi'''(z))/4$. Eliminating γ_i proves the displayed identity.

Using (B.31.14), define

$$\begin{gathered} M = 3R_0 V + \sigma o(o^2 + 3V), \qquad D_* = 4Vh^2 A_1 v A_2 > 0, \\ P_1 = 12V^2 UhvA_2 - Mh^2 A_1 v A_2, \\ P_2 = -12GV^2(r_1 A_2 - h)hA_1 - Mh^2 A_1 v A_2, \qquad Q = 4GV^3 A_2. \end{gathered} \tag{B.31.21}$$

Direct substitution gives $P_i = D_* B_i$ and $Q = D_* Gg$. Therefore the equivalent denominator-cleared necessary conditions are

$$\boxed{P_1 \geq 0, \qquad P_2 \geq 0, \qquad 4P_1 P_2 - (2Q - P_1 - P_2)^2 \geq 0.} \tag{B.31.22}$$

Only division by $G > 1/2$ and the positive outer spans is needed to evaluate the necessary system. The Hessian tests remain finite when one of the adjacent gaps approaches zero.

Additional global-dual and threshold-gap tests At a maximizing threshold, every excluded sum atom at $z+\delta$ with $\delta>0$ has mass at most $\Phi(z+\delta)-\Phi(z)\le\phi(0)\delta$. Since $\phi(0)<2/5$, a scaled gap $\Delta=\delta/K$ and atom mass w_0 must satisfy

$$25w_0^2V<4\Delta^2. \tag{B.31.23}$$

Four available excluded atoms give (w_0,Δ) respectively

$$(\ell_1r_2,A_2-A_1),\quad (m_1r_2,v),\quad (r_1m_2,h),\quad (r_1r_2,A_2).$$

For the first pair, the strict sum-gap (B.31.2) cannot approach zero with a fixed positive atom mass. This controls the transition from 221 to 321 at positive masses, rather than treating it as a missing boundary face.

Two further tests follow by evaluating the global majorants at unused jumps. For the first dual, the point $o-e=b-A_2<-a$ has deletion level one. The negative cubic factors at $-a$, giving

$$\boxed{(A_2-A_1)^2(A_2-b+2a-3/2)-r_2G\ge 0.} \tag{B.31.24}$$

For the second dual, $o+a=A_1-t$ lies strictly between d and e, and its deletion level is ℓ_1. Hence

$$\boxed{g_e(A_1-t)-\ell_1G\ge 0.} \tag{B.31.25}$$

These tests use only global majorization and the actual convolution levels.

B.32 Two triples: eliminating the masses

Keep the contact scale of Section B.31.

Use the notation and equal-height charts of Section B.31:

$$T_1=(-a,c,b),\quad T_2=(-t,d,e),\quad u=a+c,\ h=b-c=t+d,\ v=e-d,$$
$$A_1=a+b=u+h,\quad A_2=t+e=h+v,\quad A_2>A_1.$$

The supports depend on only $(k,w,E,T)\in[0,1]^4$ in each chart: a,c are the equal-height functions, $b=wB(a)$, $e=1+3E/2$, $t=7T/2$, and $d=b-c-t$. The eliminated variable was the first middle mass.

Contact height determines the useful mass products Let $G=g_e(-t)$, $g=g_e(d)$, and U be the first triple's positive level drop, as in the reduction. Put

$$N=A_1g-bG,\qquad N_0=bG-hg,\qquad Z=N/G. \tag{B.32.1}$$

The second middle deletion level gives $q_1=1-r_1=g/G$. Centering the first coordinate then yields

$$\boxed{r_1=1-g/G,\quad m_1u=Z,\quad m_1=\frac{N}{uG},\quad \ell_1=\frac{N_0}{uG}.} \tag{B.32.2}$$

Thus all three masses are positive exactly when $N>0$, $N_0>0$, $G-g>0$ (with $G,u>0$). They sum to one and have mean zero by direct substitution. Use $Z=m_1u$; the formulas below then require no division by u. The second coordinate is as follows:

$$m_2=U/G,\qquad \ell_2=(e-m_2v)/A_2,\qquad r_2=(t-m_2h)/A_2. \tag{B.32.3}$$

Equal height removes the apparent cubic-moment singularity The first low and middle contacts have equal cubic-majorant height. Subtracting those values gives the exact identity

$$|c|^3 - a^3 = (a + c)\left[\frac{3}{2}(c - a) + 3a(1 - a)\right].$$

Define its bracket as W_c. Since $\ell_1 + m_1 = q_1$, the first variance and sum of third absolute moments become

$$\boxed{V_1 = ab - hZ, \qquad R_1 = q_1 a^3 + ZW_c + r_1 b^3.} \tag{B.32.4}$$

These are exactly the centered raw moments. Neither contains a small-gap denominator. The same holds for the selected derivative. The quotient $J_c = H_c/u$ has the explicit closed-chart extensions

$$\boxed{J_c = \begin{cases} u = 3k/4, & \text{negative chart,} \\ \dfrac{3(1-k)(k^2+4k+1)}{2(k^3+3k+2)}, & \text{positive chart.} \end{cases}} \tag{B.32.5}$$

For the negative chart, $H_c = u^2$. For the positive chart, substitute $c = ka$ into $H_c/u = 3[1 - a(1 + k^2)/(1 + k)]$, and then the exact rational formula for $a(k)$. Hence

$$\boxed{\sigma = ZJ_c + r_1 H_b.} \tag{B.32.6}$$

The two retained representation-gradient equations can be written

$$\boxed{NJ_c - UH_d = 0, \qquad A_2(G - g)H_b - (eG - Uv)H_t = 0.} \tag{B.32.7}$$

They are just the original equations multiplied by candidate-positive factors. The second-middle height equation has been used exactly, not discarded.

Remaining equations, probability signs and curvature Set $V_2 = te - m_2 hv$ and compute R_2 from the second raw support and masses. Then $V = V_1 + V_2$, $R = R_1 + R_2$, $o = b - t$, $z = o/\sqrt{V}$. The actual event is

$$F = \ell_2 + m_2 q_1.$$

The remaining three zero equations are as follows:

$$3(V - R) - \sigma o = 0, \qquad \sigma^2 V - G^2 \phi(z)^2 = 0, \qquad G[F - \Phi(z)] - R = 0. \tag{B.32.8}$$

The positive slope requirement retains the sign of the unsquared density equation. All ratio, raw moment, variance-fraction, width, diameter and majorant tests remain necessary. In particular the useful CDF expression remains $e^2(2e - 3/2) - (3/2)V_2 - R_1 > 0$.

The full variance Hessian in (B.31.21)–(B.31.22) uses only U, V, R, σ, r_1, h, v, A_1, A_2, G and o. All are now functions of the four retained parameters with no division by u, so the identical three cleared Hessian inequalities can be retained. For an independent original-Hessian formulation, its first divided difference also has the regular expression

$$\gamma_1 = \frac{b^3 - a^3 - A_1 W_c}{hA_1}. \tag{B.32.9}$$

This follows by subtracting the two secant slopes and using the equal-height identity. It likewise has no u denominator.

Uniform bounds for the denominators that remain The canonical widths satisfy

$$W_1 = \min(bu, ah), \quad W_2 = \min(eh, tv), \quad W_1 + W_2 > V/3 > 25/192.$$

Since $a < 1$ and $e < 5/2$, their sum is at most $(a+e)h < (7/2)h$. Since $v > u$, $b < B(a) < 5/4$, and $t < 7/2$, it is also smaller than $(b+t)v < (19/4)v$. Therefore

$$\boxed{h > 25/672, \qquad v > 25/912.} \tag{B.32.10}$$

Also $V_1 > 13/512$ and $V_1 \le ab < b$ imply $b > 13/512$. Similarly $t > 13/1280$ follows from $V_2 > 13/512$ and $e < 5/2$. The outer spans obey $A_1 > 1/2$, $A_2 \ge 1$, and $G > 1/2$ was already proved. Thus every remaining rational moment/curvature denominator is uniformly bounded away from zero at a genuine candidate. The first gap u may tend to zero, but no evaluation denominator uses it.

Global-dual and atom-gap tests without individual first masses The first unused dual jump is the inequality already stated for g_a. For the second unused jump, multiply its necessary inequality by $u > 0$ and use (B.32.2):

$$\boxed{u\, g_e(A_1 - t) - N_0 \ge 0.} \tag{B.32.11}$$

For the excluded low/high and middle/high atoms, multiply the squared atom-gap inequalities by $(uG)^2 > 0$:

$$\begin{aligned} 4(A_2 - A_1)^2 u^2 G^2 - 25 N_0^2 r_2^2 V > 0, \\ 4v^2 u^2 G^2 - 25 N^2 r_2^2 V > 0. \end{aligned} \tag{B.32.12}$$

The other two atom-gap inequalities need only r_1, m_2, r_2, already available without u division. Two aggregate tests remain nontrivial on the coalescing face. At the middle/high sum, both first-low/second-high and first-middle/second-high atoms have entered the CDF, with total mass $q_1 r_2$ and maximal raw gap v. At the largest sum, the entire excluded mass $1 - F$ has entered, with raw gap A_2. Therefore

$$\boxed{4v^2 - 25(q_1 r_2)^2 V > 0, \qquad 4A_2^2 - 25(1-F)^2 V > 0.} \tag{B.32.13}$$

Both follow from threshold maximality and $\phi(0) < 2/5$, exactly as the individual atom-gap tests. Coincidences among later atoms cause no problem.

Each genuine collided maximum is covered by the support chart and maps into one of the two original closed four-cubes. Conversely, on $u, G > 0$, the retained probability signs reconstruct a normalized centered first triple through (B.32.2). The closed face $u = 0$ is included by continuity.

B.33 Support geometry inequalities

The following identities are used in the contact-scale charts of Sections B.16 and B.31.

If the candidate values satisfy an affine equation $\sum_j a_j x_j = b$, with fixed nonzero exact coefficients, then for every i they satisfy

$$x_i = \frac{b - \sum_{j \ne i} a_j x_j}{a_i}. \tag{B.33.1}$$

Evaluate the right side on current candidate intervals and intersect with the interval for x_i. This preserves every candidate. Any fixed sequence of such operations is valid by induction; stopping before a fixed point only weakens the resulting test. No derivative is involved. The coefficients used below are integers or exact half-integers.

For collided 210, use all support equations

$$A = t + b = h + v, \quad h = t + c = d + e, \quad v = b - c, \quad o = c - d = e - t,$$

and the probability, variance, and third-moment sums. For collided 221, use

$$A_1 = a + b = u + h, \quad A_2 = t + e = h + v, \quad u = a + c, \quad h = b - c = t + d, \quad v = e - d,$$

$$\delta = A_2 - A_1 = v - u, \qquad o = b - t = c + d.$$

On its chart with nonpositive first middle contact also use $3a - u = 3/2$, $3c - 2u = -3/2$, and $4u = 3k$, where k is that chart's parameter (the contact scale remains one). These are the exact chart identities, including the retained $u = 0$ face. On the positive chart $a > 0$, so $k = c/a$ may be intersected before re-evaluating its rational chart formulas.

After a support interval narrows, re-evaluate the original cubic duals and their slopes on those narrower candidate values. For example in 221, $G = g_e(-t)$, $g = g_e(d)$, and $m_2 = U/G$ remain exact candidate equalities. These are not ambient derivatives of the support-chart residuals.

Two existing unused-jump majorizations are also useful in inverse form:

$$r_2 \le \frac{\delta^2(A_2 - b + 2a - 3/2)}{G}, \qquad Y \le \frac{u\, g_e(A_1 - t)}{G}. \tag{B.33.2}$$

Here $Y = u\ell_1$; (B.33.2) never divides by u. The factor in the first bound is nonnegative at every candidate since $a \ge 1/2$ and $A_2 - b + 2a - 3/2 = \delta + 3a - 3/2$. On the negative chart it is exactly $\delta + u = v$, so the first bound simplifies to

$$\boxed{r_2 \le \delta^2 v/G.} \tag{B.33.3}$$

The potential $g_e(A_1 - t)$ in the second bound is nonnegative because the global cubic majorant dominates a nonnegative deletion CDF. Its factorized form is $\delta^2(3e - \delta - 3/2)$ when $A_1 - t \ge 0$, and its absolute-cubic form works without a sign restriction. Divisions by G, outer spans and a use their already proved positive lower bounds.

B.34 The support Hessian with fixed moments

Keep the contact scale of Section B.31.

Keep the scaled supports $(-a, c, b)$, $(-t, d, e)$, gaps $u = a + c$, $h = b - c = t + d$, $v = e - d$, spans $A_1 = u + h$, $A_2 = h + v$, and $A_2 > A_1$. Let G, the six masses, and the contact slopes have their previous meanings. Write $Z = um_1$, $J_c = H_c/u$, and $L(x) = 6|x| - 3$. The selected-support equations are

$$m_1 H_c = m_2 H_d, \qquad r_1 H_b = \ell_2 H_t. \tag{B.34.1}$$

An actual two-parameter family Vary the supports as

$$(-a, c+\epsilon, b+\eta), \qquad (-t-\eta, d-\epsilon, e). \tag{B.34.2}$$

For each marginal separately choose its three masses to keep its total mass, mean zero, and original variance fixed. The two 3×3 Vandermonde systems have unique smooth solutions near every genuine candidate. The base masses are positive, so both signs of small ϵ and η are admissible. Both selected raw sums remain exactly $o = c+d = b-t$ and all other event inequalities remain strict. Thus the event matrix is the actual 221 matrix throughout this family. Its CDF is $F = \ell_2 + m_2(1-r_1)$.

Both the centered threshold and the total variance are fixed. The normal CDF term is therefore constant, and the relevant objective is $F - R/G$. Let P_1, P_2 be the scaled cubic dual potentials. Their values at the three supports equal G times the actual deletion probabilities. Their first derivatives are respectively $(0, -H_c, -H_b)$ and $(-H_t, -H_d, 0)$, and their second derivatives are $L(x)$. Adding the fixed mean, variance and mass constraints to the objective changes it only by a constant on (B.34.2). The resulting Lagrangian has zero derivatives in all mass variables at the base point.

Consequently, for two tangent support motions s, t, its Hessian multiplied by G is exactly

$$G\sum_{ij} I_{ij}(w^s_{1i}w^t_{2j} + w^t_{1i}w^s_{2j}) - \sum_{ij} p_{ij}L(x_{ij})\dot{x}^s_{ij}\dot{x}^t_{ij} - \sum_{ij} P_i'(x_{ij})(w^s_{ij}\dot{x}^t_{ij} + w^t_{ij}\dot{x}^s_{ij}). \tag{B.34.3}$$

There are no second derivatives of the masses in (B.34.3): their coefficients are the zero first mass derivatives of this Lagrangian. The support paths are affine. Equation (B.34.1) makes both first support derivatives zero.

Exact weights and regular Hessian For a marginal with support points x_0, x_1, x_2, let L_i be its quadratic Lagrange basis, defined by $L_i(x_j) = \mathbf{1}_{\{i=j\}}$. If x_j moves with velocity s, preserving mass, mean, and variance requires the probability derivative $w_i = -p_j s L_i'(x_j)$. The entries needed in the actual event are

	ϵ	η
w_{1r}	$-Z/(A_1h)$	$-r_1(A_1+h)/(A_1h)$
w_{1m}	$-m_1(h-u)/(uh)$	$r_1A_1/(uh)$
w_{2m}	$m_2(v-h)/(hv)$	$\ell_2A_2/(hv)$
$w_{2\ell}$	$-m_2v/(hA_2)$	$-\ell_2(h+A_2)/(hA_2)$.

(B.34.4)

The mixed event form is exactly $-w_{1r}w_{2m}$. Substitution in (B.34.3), followed only by (B.34.1), gives the scaled Hessian

$$\begin{pmatrix} -C_m & B \\ B & -C_e \end{pmatrix} \preceq 0, \tag{B.34.5}$$

where

$$C_m = m_2L(d) + Z\left(\frac{L(c)+2J_c}{u} - \frac{2J_c}{v}\right) - \frac{2GZm_2(v-h)}{A_1h^2v}, \tag{B.34.6}$$

$$C_e = r_1L(b) + \ell_2L(-t) + 2r_1H_b\left(\frac{1}{A_1} - \frac{1}{A_2}\right) - \frac{2Gr_1\ell_2A_2(A_1+h)}{A_1h^2v}, \tag{B.34.7}$$

$$B = \frac{G\{Z\ell_2A_2 + r_1m_2(A_1+h)(v-h)\}}{A_1h^2v} - \frac{ZH_b}{A_1h} + \frac{r_1A_1J_c}{h} + \frac{m_2vH_t}{hA_2} - \frac{\ell_2A_2H_d}{hv}. \tag{B.34.8}$$

Thus every genuine candidate satisfies $C_m \geq 0$, $C_e \geq 0$, and $C_m C_e - B^2 \geq 0$. On the negative first-middle chart, $J_c = u$, $L(c) = -4u$, so (B.34.6) has the regular form

$$\boxed{C_m = m_2 L(d) - 2Z(1 + u/v) - \frac{2GZm_2(v-h)}{A_1 h^2 v}.} \tag{B.34.9}$$

Analytic restriction when $c \leq 0$ and $v \geq h$ In this genuine subcase $Z > 0$, $m_2 > 0$, and $0 < u \leq 3/4$. Equation (B.34.9) first gives $L(d) > 0$, hence $|d| > 1/2$. If $d > 1/2$, then $h = t + d > d$ and $e = d + v \geq d + h > 2d \geq 1$. Since $x(x-1)$ increases for $x \geq 1$,

$$H_d = 3\{e(e-1) + d - d^2\} > 3d(3d-1).$$

But (B.34.1) and (B.34.9) also give $L(d) \geq 2Z/m_2 = 2H_d/u$, so

$$6d - 3 > 8d(3d-1).$$

This is impossible because

$$24d^2 - 14d + 3 = 24(d - 7/24)^2 + 23/24 > 0.$$

Therefore

$$\boxed{c \leq 0, \quad v \geq h \quad \Longrightarrow \quad d < -1/2.} \tag{B.34.10}$$

There is also a useful upper-support bound. Write $w = -d > 1/2$. Equation (B.34.9) is strict in the weaker form $m_2 L(d) > 2Z$, since $u/v > 0$. Using (B.34.1) yields

$$e(e-1) + w^2 - w < u(w - 1/2).$$

Complete the square in w to obtain

$$\boxed{e(e-1) < \frac{1+u^2}{4}, \qquad e < \frac{1+\sqrt{2+u^2}}{2} \leq \frac{1}{2} + \frac{\sqrt{41}}{8}.} \tag{B.34.11}$$

The last two inequalities hold when $v \geq h$; the next calculation proves this condition on the negative chart.

B.35 The negative middle contact

Keep the contact scale of Section B.31.

On the chart with nonpositive first middle contact,

$$c = 2a - 3/2, \quad u = a + c \in (0, 3/4], \quad h = b - c \leq 7/4, \quad v = e - d > u, \quad e \geq 1.$$

The actual first dual height and selected-flow identity give

$$U = A_1^2 h - 2b^3 < A_1^2 h, \qquad (A_1 g - bG)u = UH_d, \quad g = g_e(d), \quad G > 0. \tag{B.35.1}$$

In particular $H_d > 0$ and $U/A_1 < ug/H_d$. The middle support curvature, after using $Zu = m_2 H_d$, is

$$\frac{C_m}{m_2} = L(d) - \frac{2H_d}{u} Q, \quad Q = 1 + \frac{u}{v} - \frac{U(h-v)}{A_1 h^2 v}, \quad L(d) = 6|d| - 3. \tag{B.35.2}$$

A maximum requires $C_m \geq 0$.

First suppose $-1/2 \leq d \leq 1/8$. If $v \geq h$, then $Q \geq 1 + u/v > 0$. If $v < h$, use the first upper bound in (B.35.1):

$$Q > 2 - h/v + u/h.$$

Since $v = e - d \geq 7/8$ and $h \leq 7/4$, the right side is at least $u/h > 0$. But $L(d) \leq 0$, so (B.35.2) is strictly negative. This interval is impossible.

Now suppose $d \geq 1/8$. If $v \geq h$, (B.35.2) and $v > u$ give

$$C_m/m_2 < L(d) - 2H_d/v = 3 - 6e < 0.$$

If $v < h$, write $r = v/h$ and $\ell = e + d - 1 \geq d \geq 1/8$. The positive-side cubic potential factors exactly as

$$\frac{g}{H_d} = \frac{v}{2} + \frac{v^2}{6\ell}.$$

Hence, using $0 < r < 1$, $h \leq 7/4$,

$$\frac{g(h-v)}{H_d h^2} = \frac{r(1-r)}{2} + \frac{h}{6\ell} r^2(1-r) \leq \frac{1}{8} + \frac{7}{3}\frac{4}{27} = \frac{305}{648} < 1. \tag{B.35.3}$$

The second upper bound in (B.35.1) now implies

$$Q > 1 + \frac{u}{v} - \frac{u}{v}\frac{305}{648} > 1 > \frac{u}{v}.$$

Again (B.35.2) is strictly below $L(d) - 2H_d/v = 3 - 6e < 0$. This excludes every $d \geq 1/8$. Combining the two ranges proves

$$\boxed{d < -1/2 \quad \text{throughout the chart with nonpositive first middle contact.}} \tag{B.35.4}$$

In fact the opposite-gap sector is impossible. Put $w = -d > 1/2$. The first high support satisfies the exact bound

$$b < B(a) = \frac{1 + \sqrt{1 + 4a(1-a)}}{2} \leq \frac{1 + \sqrt{2}}{2}.$$

Consequently

$$h = b - c < 1 + 1/\sqrt{2} - 2u/3 < 1 + 1/\sqrt{2}. \tag{B.35.5}$$

If $v \leq h$, then $v = e + w$ lies in $(3/2, 1 + 1/\sqrt{2})$. For fixed v, convexity in w on $[1/2, v-1]$ gives

$$\frac{H_d}{3} = e^2 - e + w^2 - w \leq \max\{v^2 - 2v + 1/2,\ (v-1)(v-2)\} < 0.$$

Both endpoint expressions are strictly negative on the stated interval. This contradicts $H_d > 0$ from the selected flow. Thus

$$\boxed{v > h \quad \text{throughout the chart with nonpositive first middle contact.}} \tag{B.35.6}$$

The conditional negative-chart support bound therefore also holds unconditionally on this chart: $e(e-1) < (1+u^2)/4$, hence $e < 1/2 + \sqrt{41}/8$.

B.36 The signed divided-difference inequality

In the two-triple 221 contact scale let $r = \sqrt{V}$. The identities $C = r^3/G$, $\phi(z) = \sigma r/G$, $z = o/r$, and $3(V - R) = \sigma o$ give

$$\frac{C}{r} = D + \frac{z\phi(z)}{3}, \qquad L = \frac{1}{4} + \frac{\sigma o(o^2 + 2V)}{12V^2} = \frac{1}{4} + \frac{rz\phi(z)(z^2+2)}{12C} = \frac{1}{4} + \frac{z\phi(z)(z^2+2)}{12(D + z\phi(z)/3)}.$$

The own-variance curvature inequality is

$$\gamma_i \le \frac{r\{15D + (3z - z^3)\phi(z)\}}{12C} = \frac{3}{2} - L.$$

For $z \ge 0$ one has $L \ge 1/4$. For $z = -y \in [-5/4, 0]$, the function $y(y^2 + 2)\phi(y)$ increases: its derivative is $(2 - y^2)(1 + y^2)\phi(y) > 0$. Since $\phi(0) < 2/5$ and $\sum_{j=0}^{5}(25/32)^j/j! > 24/11$, $\phi(5/4) < 11/60$. Using $r < 2$ and $C \ge 49/120$ gives

$$y(y^2 + 2)\phi(y) < \frac{285}{64}\frac{11}{60} = \frac{209}{256}, \qquad L > \frac{1}{4} - \frac{20}{49}\frac{209}{256} = -\frac{261}{3136} > -\frac{1}{12}.$$

The height identities, with q allowed either sign, are

$$p = \frac{3}{2} - \gamma_1 = \frac{U}{A_1 h} > 0, \qquad q = \frac{3}{2} - \gamma_2 = \frac{G(h - A_2 r_1)}{A_2 h v}, \qquad p, q \ge L.$$

Thus $\gamma_2 < 4965/3136$ and $A_2 < 4965/1568$ by (B.2). The exact mass and event identities read

$$m_2 = \frac{pA_1 h}{G}, \quad r_1 = \frac{h}{A_2} - \frac{qhv}{G}, \quad q_1 = \frac{v}{A_2} + \frac{qhv}{G}, \quad F = \frac{e}{A_2} + \frac{m_2 qhv}{G}.$$

Put $x = p - L$, $y = q - L$, and $b_0 = GV/(3A_1 h^2 v)$. The original probability Hessian is negative semidefinite only if

$$x \ge 0, \quad y \ge 0, \quad x + y \ge b_0, \qquad 4xy - (2b_0 - x - y)^2 \ge 0.$$

These are the signed inequalities used in the supplied positive-chart calculation. They impose no positivity assumption on q or $F - e/A_2$.

B.37 A parameter bound for a positive middle contact

Keep the contact scale of Section B.31.

In the positive chart for the collided two-triple 221 event, $k < 8/9$. Indeed, with $u = a(1 + k)$ and $A_1 = u + h$, the centered marginal and selected contact equations give

$$\beta_1 = 1 - \frac{2a^2}{A_1 u}, \quad \sigma = \frac{aH_b}{A_1} + \left(J_c - \frac{H_b}{A_1}\right) Z, \quad J_c - \frac{H_b}{A_1} = 3h\beta_1, \quad Z = a - A_1 r_1.$$

Thus $0 < Z < a$. If $k \ge 8/9$, then $A_1 u > u^2 > 2a^2$, so $\beta_1 > 0$ and $\sigma \le aJ_c \le J_c$. But

$$J_c = \frac{3(1 - k)(k^2 + 4k + 1)}{2(k^3 + 3k + 2)}$$

decreases on $[1/2, 1]$: its derivative has the sign of $k^4 - 4k^3 - 6k^2 - 4k + 1 < 0$, as follows from $k^4 \le k^3$ and evaluation of the resulting decreasing polynomial at $1/2$. Consequently $\sigma \le J_c(8/9) =$

1299/7828. On the other hand $\sigma = V\phi(z)/C$, and the bounds $V > 2401/5476$, $\phi(z) > 9/50$, $C < 227/500$ imply

$$\sigma > \frac{108045}{621526} = \frac{1299}{7828} + \frac{19206993}{2432652764}.$$

This contradiction proves the parameter bound used in the positive-chart certificate.

The fixed certificates `tt221_negative` and `tt221_positive` prove that neither compact support chart contains a point satisfying the necessary conditions in Sections B.31–B.37. Thus the collided 221 event and its transpose 320 cannot occur at a maximizing pair. Section 8 gives the enclosure and partition rules.

References

[1] Logan Bell. An upper bound on the Berry–Esseen constant for independent summands. 6 October 2026; companion manuscript in this collection, 2026. URL https://github.com/epicycloids/independent-berry-esseen-bounds/blob/main/paper/independent-berry-esseen.pdf.

[2] V. Bentkus and K. Kirsa. Estimates of the proximity of a distribution to the normal law. *Lithuanian Math. J.*, 29(4):321–332, 1989. doi: 10.1007/BF00971947.

[3] Dimitris Bertsimas and Ioana Popescu. Optimal inequalities in probability theory: A convex optimization approach. *SIAM Journal on Optimization*, 15(3):780–804, 2005. doi: 10.1137/S1052623401399903.

[4] C.-G. Esseen. A moment inequality with an application to the central limit theorem. *Skand. Aktuarietidskr.*, 39:160–170, 1956. doi: 10.1080/03461238.1956.10414946.

[5] Larry Goldstein. Bounds on the constant in the mean central limit theorem. *The Annals of Probability*, 38(4):1672–1689, 2010. doi: 10.1214/10-AOP527. URL https://arxiv.org/abs/0906.5145.

[6] Larry Goldstein and Gesine Reinert. Stein's method and the zero bias transformation with application to simple random sampling. *The Annals of Applied Probability*, 7(4):935–952, 1997. doi: 10.1214/aoap/1043862419.

[7] Fredrik Johansson. Arb: Efficient arbitrary-precision midpoint-radius interval arithmetic. *IEEE Transactions on Computers*, 66(8):1281–1292, 2017. doi: 10.1109/TC.2017.2690633. URL https://arxiv.org/abs/1611.02831.

[8] Lutz Mattner. Extreme expectations of Bernoulli convolutions given their first few moments are attained at shifted convolutions of as few binomials, 2022. URL https://arxiv.org/abs/2204.06040v1. Version 1.

[9] Aaron Meurer, Christopher P. Smith, Mateusz Paprocki, et al. SymPy: Symbolic computing in Python. *PeerJ Computer Science*, 3:e103, 2017. doi: 10.7717/peerj-cs.103. URL https://peerj.com/articles/cs-103/.

[10] Jona Schulz. *The optimal Berry–Esseen constant in the binomial case*. Dissertation, Universität Trier, 2016. URL https://d-nb.info/1197702695/34. 152 pp.

[11] Ilya Tyurin. New estimates of the convergence rate in the Lyapunov theorem, 2009. URL https://arxiv.org/abs/0912.0726v1. Version 1.

[12] Gerhard Winkler. Extreme points of moment sets. *Mathematics of Operations Research*, 13 (4):581–587, 1988. doi: 10.1287/moor.13.4.581. URL https://www.jstor.org/stable/3689944.